Vertex algebras of chiral differential operators on a reductive group and representation theory

Algèbres vertex des opérateurs différentiels chiraux sur un groupe réductif et théorie des représentations

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Résumé: Cette thèse s'inscrit dans le programme de Langlands géométrique quantique, un vaste réseau de conjectures reliant théorie des représentations et géométrie algébrique. Notre approche emploie le langage des algèbres vertex, un cadre algébrique adapté pour l'étude des différents objets apparaissant dans ce contexte.

Un objet central pour notre étude est l'algèbre vertex des opérateurs différentiels chiraux $\mathcal{D}_G^\kappa$ sur un groupe algébrique réductif G. On s'intéresse à sa théorie des représentations pour des niveaux génériques. Dans ce contexte, on montre que l'équivalence de Satake géométrique dégénère. Ce phénomène nous mène à formuler un analogue vertex de l'équivalence locale fondamentale de Gaitsgory et Lurie.

Cette perspective nous conduit naturellement à la $\mathcal{W}$-algèbre affine principale équivariante $\mathcal{W}_G^\kappa$, introduite par Arakawa comme la réduction hamiltonienne principale de $\mathcal{D}_G^\kappa$. On étudie sa théorie des représentations et construit une famille de $\mathcal{W}_G^\kappa$-modules simples dont la combinatoire coïncide avec celle de la théorie des représentations du groupe de Langlands dual $G^\vee$. On prouve ensuite l'équivalence attendue dans les cas où G est un tore algébrique ou un groupe classique simple adjoint simplement lacé.

Un ingrédient clé de notre preuve est une famille d'algèbres vertex d'opérateurs différentiels chiraux décalés. Même dans les cas les plus simples, on sait très peu de ces algèbres. Quand G est un tore algébrique, on calcule et étudie leurs schémas de Poisson associés.

Title: Vertex algebras of chiral differential operators on a reductive group and representation theory

Keywords: Vertex algebras, Chiral differential operators, Representation theory, Reductive group, W-algebra.

Abstract: This thesis is situated in the framework of the quantum geometric Langlands program, a vast network of conjectures relating representation theory and algebraic geometry. We approach this program through the language of vertex algebras, which provide a suitable algebraic framework for studying the objects that arise in this setting.

A central object of our work is the vertex algebra of chiral differential operators $\mathcal{D}_G^\kappa$ on a reductive algebraic group G. We investigate its representation theory at generic levels. In this setting, we show that the geometric Satake equivalence degenerates. This phenomenon leads us to formulate a vertex-algebraic analogue of the fundamental local equivalence of Gaitsgory and Lurie.

This perspective naturally brings us to the principal equivariant affine $\mathcal{W}$-algebra $\mathcal{W}_G^\kappa$, introduced by Arakawa as the principal quantum Hamiltonian reduction of $\mathcal{D}_G^\kappa$. We study its representation theory and construct a family of simple $\mathcal{W}_G^\kappa$-modules whose combinatorics matches that of the representation theory of the Langlands dual group $G^\vee$. We then prove the expected equivalence when G is an algebraic torus or a simple adjoint group of classical simply laced type.

A key ingredient in the proof is a family of vertex algebras, the shifted chiral differential operators. Very little is known about these algebras, even in the simplest cases. When G is an algebraic torus, we compute and study their associated Poisson schemes.

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Contents

Chapter 1

Introduction (english version)

Let G be a reductive algebraic group over the field $\mathbb{C}$ of complex numbers with Lie algebra $\mathfrak{g}$. Denote by $V^\kappa(\mathfrak{g})$ the associated universal affine vertex algebra at the level κ. It can be thought of as a vertex algebraic version of the enveloping algebra $\mathcal{U}(\mathfrak{g})$ of $\mathfrak{g}$. Let $\mathcal{D}_G^\kappa$ be the vertex algebra of chiral differential operators on the group G. Similarly, this is a Kac–Moody version of the algebra $\mathcal{D}_G$ of differential operators on G. We have a natural inclusion of $\mathcal{U}(\mathfrak{g})$ in $\mathcal{D}_G$ which follows from thinking of $\mathfrak{g}$ as the Lie algebra of left-invariant vector fields on G. This classical setting chiralizes (that is, extends to the vertex algebraic setting) and we have an embedding of $V^\kappa(\mathfrak{g})$ in $\mathcal{D}_G^\kappa$. Let $\mathcal{W}_G^\kappa$ be the principal equivariant $\mathcal{W}$-algebra on the group G at level κ. It is defined as the Drinfeld–Sokolov reduction of $\mathcal{D}_G^\kappa$ with respect to the previous embedding. One contribution of this thesis is to prove:

Theorem A (Theorem 7.2.5)**.** *For generic levels, there exists a family of pairwise nonequivalent simple $\mathcal{W}_G^\kappa$-modules labeled by the dominant cocharacters of G. They are obtained as quantum Hamiltonian reduction of spectral flow twists of the adjoint module $\mathcal{D}_G^\kappa$.*

In other words, we give a vertex algebraic construction that, out of the group G, produces simple objects whose combinatorics match that of the representation theory of the Langlands dual group $\check{G}$. The appearance of the dual group hints that something deeper is at play here. Before giving a more satisfying interpretation, we need some context.

Let T be a maximal torus of G and B be a Borel containing it. The study of the representation theory of G naturally leads one to consider various geometric spaces built out of the group. An example is the flag variety G/B, a smooth projective algebraic variety of finite type. For instance, the study of (twisted) algebraic $\mathcal{D}_{G/B}$-modules *i.e.*, the theory of modules over the sheaf of (twisted) differential operators on G/B leads to the theory of Beilinson–Bernstein localization ([BB93]).

The theory of $\mathcal{D}$-modules on a space X can be thought of as the study of systems of linear partial differential equations on X ([Kas03], [Bor87], [HTT08]). When solving those, one produces topological objects, typically local systems of finite dimensional vector spaces on X. It sometimes happens that a $\mathcal{D}$-module and the associated topological object are equivalent, and this kind of phenomenon goes under the name of Riemann–Hilbert correspondence. A large class of $\mathcal{D}$-modules for which that happens has been identified. These are the so-called regular holonomic $\mathcal{D}$-module and their topological counterparts are called perverse sheaves ([Del70], [Meb84], [Kas84], [GM80], [BBDG82]). They are also important because of their interactions with the sheaves-functions dictionary of Grothendieck ([Del77]).

An example is the geometric Satake equivalence ([Sat63], [Gin90], [MV00], [BR18]) and is a first step in the geometric version of the Langlands program, as envisioned by Drinfeld and Laumon in the 1980s ([Dri83], [Lau87]). In order to state it, let $\mathcal{J}_\infty G$ be the jet space and $\mathcal{L}G$ be the loop space of the group G. These are algebraic groups such that $\mathcal{J}_\infty G(\mathbb{C}) = G(\mathbb{C}[[t]])$ and $\mathcal{L}G(\mathbb{C}) = G(\mathbb{C}((t)))$. Define the affine Grassmannian $\mathrm{Gr}_G = \mathcal{L}G/\mathcal{J}_\infty G$ to be the quotient of the loop group of G by its jet space, it is an ind-scheme of infinite type. The geometric Satake equivalence states that the category of $\mathcal{J}_\infty G$-equivariant perverse sheaves on Gr_G is equivalent to the category of representations of $\check{G}$.

This raises the following question: is there a way to state the geometric Satake equivalence in terms of $\mathcal{D}$-modules on the affine Grassmannian Gr_G? To answer that, one may hope for a formalism of $\mathcal{D}$-modules which is robust enough to make sense on Gr_G as well as a suitable Riemann–Hilbert correspondence. One step in that direction is suggested by the work of Arkhipov and Gaitsgory ([AG02]) in which they build a chiral algebra whose category of modules is a substitute for a category of $\mathcal{D}$-modules on the loop group $\mathcal{L}G$. This kind of chiral algebras are known as chiral algebras of differential operators and were introduced by Beilinson and Drinfeld ([BD04]). They were also independently discovered in the language of vertex algebras by Gorbounov, Malikov, and Schechtman and later called vertex algebras of chiral differential operators ([MSV99], [GMS04]). In terms of vertex algebras, the construction of Arkhipov and Gaitsgory reads as follows: there exists a vertex algebra $\mathcal{D}_G^\kappa$ whose category of modules is an appropriate category $\mathcal{D}_\kappa(\mathcal{L}G)$−Mod of κ-twisted $\mathcal{D}$-modules on the loop group. In particular for a certain level κ_c, called the critical level, the category $\mathcal{D}_G^{\kappa_c}$−Mod behaves as an appropriate category of untwisted $\mathcal{D}$-modules on the loop group.

In this thesis, we will use the latter language. Vertex algebras are relatively recent algebraic structures that were axiomatized by Borcherds ([Bor86]). Their emergence is closely tied to the monster group through the celebrated moonshine conjectures of Conway and Norton ([CN79]) and their proof by Borcherds ([Bor92]). A crucial step is the construction of the moonshine module by Frenkel, Lepowsky, and Meurman ([FLM89]). The genesis of vertex algebras is inseparable from ideas arising in physics; see the beautifully written preface and introduction of *loc. cit.*. The mathematician learning about vertex algebras quickly faces a paradox as their definition is deceptively simple. A vertex algebra is a vector space endowed with countably many products satisfying certain relations (see Definition 3.2.1). Yet constructing (noncommutative) examples is far from easy, as both experience and history show (see for instance [FLM89], [FZ92a], [Zhu08]). The theory of vertex algebras appears to exhibit an intrinsic rigidity. Strikingly, the scarcity of examples leads one to formulate uniqueness results that would be overly ambitious in the classical setting. This led us to prove Theorems 4.4.3 and 9.3.9, which are the technical backbone of our work.

Let us now say a word on the classical setting of the theory of algebraic $\mathcal{D}$-modules. Let X be a smooth affine scheme of finite type with structure sheaf $\mathcal{O}_X$. It carries a sheaf $\mathcal{D}_X$ of differential operators built out of $\mathcal{O}_X$ and the tangent sheaf Θ_X. It is a sheaf of noncommutative rings that contains $\mathcal{O}_X$ as a subring and Θ_X as a Lie subalgebra interacting via the Leibniz rule. The category of algebraic $\mathcal{D}_X$-module on X is then the category of sheaves of, say left, modules over $\mathcal{D}_X$ that are quasi-coherent as $\mathcal{O}_X$-modules. The quasicoherence assumption ensures that a $\mathcal{D}_X$-module is controlled by its global sections, which is a left module over the noncommutative ring $\mathcal{D}_X(X)$ of global differential operators on X. Let us from now on use the same notation for a sheaf and its global sections.

A key example for us is the case where $X = G$ is the group itself. The theory of $\mathcal{D}$-modules is then nothing but the representation theory of the ring $\mathcal{D}_G$. This ring has a nice description in terms of functions on G and its Lie algebra $\mathfrak{g}$. In fact, we can think of $\mathfrak{g}$ as global left-invariant vector fields of G and they generate over $\mathcal{O}_G$ the Lie algebra of global vector fields Θ_G. As a vector space we have

$$\mathcal{D}_G = \mathcal{U}(\mathfrak{g}) \otimes \mathcal{O}_G$$

and the ring structure is determined by the Leibniz rule. It naturally contains $\mathcal{U}(\mathfrak{g})$ and $\mathcal{O}_G$ as subalgebras. Note that we made the choice of considering $\mathfrak{g}$ as left-invariant global vector fields on G. Thinking of $\mathfrak{g}$ as right-invariant global vector fields show that $\mathcal{D}_G$ contains another subalgebra that is a copy of $\mathcal{U}(\mathfrak{g})$. By construction, these two copies commute with one another.

The construction of $\mathcal{D}_G^\kappa$ is a variation on that of $\mathcal{D}_G$ where G is replaced by $\mathcal{L}G$ and rings are replaced by vertex algebras. Introduce the affine Kac–Moody algebra $\hat{\mathfrak{g}}^\kappa = \mathfrak{g}((t)) \oplus \mathbb{C}\mathbb{1}$ ([Kac90]). It is a central extension of $\mathrm{Lie}(\mathcal{L}G) = \mathfrak{g}((t))$ given by the level κ. The universal affine vertex algebra $V^\kappa(\mathfrak{g})$ plays an analogous role to the universal enveloping algebra for $\hat{\mathfrak{g}}^\kappa$. The Lie subalgebra $\mathfrak{g}[[t]] = \mathrm{Lie}(\mathcal{J}_\infty G)$ of $\hat{\mathfrak{g}}^\kappa$ acts on $\mathcal{O}_{\mathcal{J}_\infty G}$ as left-invariant vector fields and so by analogy we define, as vector spaces

$$\mathcal{D}_G^\kappa = V^\kappa(\mathfrak{g}) \otimes \mathcal{O}_{\mathcal{J}_\infty G}.$$

One shows that there exists a vertex algebra structure on $\mathcal{D}_G^\kappa$ such that $V^\kappa(\mathfrak{g})$ and $\mathcal{O}_{\mathcal{J}_\infty G}$ are vertex subalgebras of $\mathcal{D}_G^\kappa$. Again, we made the choice to consider left-invariant vector fields on $\mathcal{J}_\infty G$ over right-invariant ones. This time, corresponding to the latter, there is a copy of $V^{\kappa^*}(\mathfrak{g})$ in $\mathcal{D}_G^\kappa$ but the level gets shifted (see Theorem 4.1.4). Again, these two copies commute in an appropriate sense.

This vertex algebra is going to play a role analogous to (the global sections of) the sheaf of (κ-twisted) differential operators on the loop group $\mathcal{L}G$. It is thus natural to define

$$\mathcal{D}\text{–Mod}_\kappa(\mathcal{L}G) := \mathcal{D}^\kappa_G\text{–Mod}.$$

In particular, when $\kappa = \kappa_c$ is the critical level, then we get a definition of untwisted $\mathcal{D}$-modules on the loop group

$$\mathcal{D}\text{–Mod}(\mathcal{L}G) := \mathcal{D}^{\kappa_c}_G\text{–Mod}.$$

To take into account objects on the affine Grassmannian, and the equivariance of these objects themselves, one needs to further develop the formalism. In the classical setting where X is a smooth scheme of finite type and is acted on by a connected subgroup K of G, it is remarkable that the category of K-equivariant $\mathcal{D}$-modules on X is a full subcategory of the category of $\mathcal{D}$-modules on X. That is, equivariance of $\mathcal{D}$-modules is a property and not an additional structure (see for instance Remark 30.10 of [Eti24]). Precisely, given a $\mathcal{D}_X$-module $\mathcal{M}$, then $\mathcal{M}$ is K-equivariant if the induced action of $\mathrm{Lie}(K)$ integrates to an action of the group K (see Proposition 30.18 of *loc.cit.*).

Back to the infinite dimensional setting, let $K = \mathcal{J}_\infty G \subset \mathcal{L}G$ act on $\mathcal{L}G$ by left multiplication. First, because Gr_G is a quotient, it is natural to define the category of $\mathcal{D}$-modules on Gr_G to be the category of K-equivariant $\mathcal{D}$-modules on the loop group. In terms of the vertex algebra $\mathcal{D}^\kappa_G$, given a $\mathcal{D}^\kappa_G$-module M this translates to asking that when seen as a $V^{\kappa^*}(\mathfrak{g})$-module via restriction, the action of $\mathfrak{g}[[t]]$ integrates to an action of $\mathcal{J}_\infty G$. This subcategory of $V^{\kappa^*}(\mathfrak{g})\text{–Mod}$ is known as the Kazhdan–Lusztig category ([KL93a], [KL93b], [KL94a], [KL94b]), we denote it by $V^{\kappa^*}(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$. Let

$$\mathcal{D}^\kappa_G\text{–Mod}^{1\times\mathcal{J}_\infty G}$$

be the full subcategory of $\mathcal{D}^\kappa_G\text{–Mod}$ consisting of objects satisfying this condition. That defines (see Theorem A.3 of [AG02]) the category of κ-twisted $\mathcal{D}$-modules on the affine Grassmannian

$$\mathcal{D}\text{–Mod}_\kappa(\mathrm{Gr}_G) := \mathcal{D}^\kappa_G\text{–Mod}^{1\times\mathcal{J}_\infty G}.$$

In the same fashion, define

$$\mathcal{D}^\kappa_G\text{–Mod}^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$$

to be the full subcategory of modules consisting of objects such that, when seen as $V^{\kappa\oplus\kappa^*}(\mathfrak{g}\oplus\mathfrak{g})$-modules, belong to $V^{\kappa\oplus\kappa^*}(\mathfrak{g}\oplus\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$. Let

$$\mathcal{D}\text{–Mod}_\kappa(\mathrm{Gr}_G)^{\mathcal{J}_\infty G} := \mathcal{D}^\kappa_G\text{–Mod}^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$$

be the category of κ-twisted $\mathcal{J}_\infty G$-equivariant $\mathcal{D}$-modules on the affine Grassmannian. In this language, it is reasonable to expect that the geometric Satake equivalence is something that happens at the critical level and takes the form of an equivalence of categories

$$\mathcal{D}_G^{\kappa_c}\text{–Mod}^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G} \simeq \check{G}\text{–Mod}$$

where $\check{G}$–Mod is the category of representation of the algebraic group $\check{G}$. It is natural to ask if it is possible to describe the category

$$\mathcal{D}_G^{\kappa}\text{–Mod}^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$$

for levels κ that are not critical. When moving away from the critical level, we enter the paradigm of the quantum geometric Langlands program ([Sto06], [Gai16]). We prove the following in the case where G is a torus or a simple group:

Theorem B (Theorem 4.5.3, Corollary 8.1.8)**.** *For generic enough levels κ the adjoint module $\mathcal{D}_G^{\kappa}$ is the only simple object in the category $\mathcal{D}_G^{\kappa}\text{–Mod}^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$. Moreover, the category $\mathcal{D}_G^{\kappa}\text{–Mod}^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$ is equivalent to the category of $\mathbb{C}$-vector spaces.*

This can be interpreted as a quantum degeneration of the geometric Satake correspondence. This fact was well-known to experts and a category of substitution was introduced by Gaitsgory following an idea of Lurie ([Gai08]). It is the appropriately defined category of κ-twisted Whittaker $\mathcal{D}$-modules on Gr_G. We now introduce a conjecturally equivalent category in our language. This guess was influenced by the affine Skryabin equivalence proven by Raskin ([Ras21]).

Let H^*_{DS} be the (principal) quantum Hamiltonian reduction functor ([FF90]). This functor allows one to define the (principal) $\mathcal{W}$-algebra of $\mathfrak{g}$ at level κ from the universal affine algebras, $W^\kappa(\mathfrak{g}) := H^*_{DS}(V^\kappa(\mathfrak{g}))$. Moreover $W^\kappa(\mathfrak{g})$ is a vertex algebra, and we get a well-defined functor

$$H^*_{DS} : V^\kappa(\mathfrak{g})\text{–Mod} \longrightarrow W^\kappa(\mathfrak{g})\text{–Mod}.$$

It also works in a relative setting, namely if V is a vertex algebra equipped with a morphism

$$V^\kappa(\mathfrak{g}) \longrightarrow V$$

then $H^*_{DS}(V)$ is a vertex algebra that comes equipped with a morphism

$$W^\kappa(\mathfrak{g}) \longrightarrow H^*_{DS}(V).$$

Moreover we have a well-defined functor

$$H^*_{DS} : V\text{–Mod} \longrightarrow H^*_{DS}(V)\text{–Mod}.$$

When this construction is applied to $\mathcal{D}_G^{\kappa}$ with respect to the morphisms of vertex algebras

$$V^\kappa(\mathfrak{g}) \longrightarrow V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g}) \longrightarrow \mathcal{D}_G^{\kappa}$$

that defines the (principal) equivariant affine $\mathcal{W}$-algebra $\mathcal{W}^{\kappa}_{G}$ on the group G at level κ ([Ara18]). It comes equipped with a morphism

$$W^{\kappa}(\mathfrak{g}) \longrightarrow W^{\kappa}(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g}) \longrightarrow \mathcal{W}^{\kappa}_{G}.$$

We define

$$\mathcal{W}^{\kappa}_{G}\text{–Mod}^{\mathcal{J}_\infty G}$$

to be the full subcategory of $\mathcal{W}^{\kappa}_{G}$–Mod consisting of modules that are integrable with respect to $\mathcal{J}_\infty G$, when seen as $V^{\kappa^*}(\mathfrak{g})$-modules by restriction. This category should deform nicely and be related for generic κ to the category of representations of the Langlands dual group $\check{G}$.

We make the following conjecture which is an abelian category version of the fundamental local equivalence (FLE) of the quantum geometric Langlands program ([CDR21]) in our language:

Conjecture A (Conjecture 7.2.4)**.** *For generic levels, the category $\mathcal{W}^{\kappa}_{G}$–Mod$^{\mathcal{J}_\infty G}$ is semisimple with simple objects given by the $H^*_{DS}(\gamma\cdot\mathcal{D}^{\kappa}_{G})$ for $\gamma \in X_*(T)_+$. In particular, there exists an equivalence of abelian categories*

$$\mathcal{W}^{\kappa}_{G}\text{–Mod}^{\mathcal{J}_\infty G} \simeq \check{G}\text{–Mod}$$

that sends for all $\gamma \in X_*(T)_+$ the simple object $H^*_{DS}(\gamma \cdot \mathcal{D}^{\kappa}_{G})$ to V_γ.

We explain a general approach to solve this conjecture. In the case of an algebraic torus, we can prove it by direct computations. In the general case, this approach relies on, among other things, certain vertex algebras whose existence is still conjectural. As they have been shown to exist with some restrictions on the group G, we are able to prove:

Theorem C (Theorem 8.1.6 and Theorem 9.4.1)**.** *Conjecture 7.2.4 holds when G is a torus or when G is a simple adjoint group of classical simply laced type.*

The main idea of this thesis is that, for generic levels, the chiral Peter–Weyl Theorem 4.4.1 allows us to think of $\mathcal{D}^{\kappa}_{G}$ as an analogue of a lattice vertex algebra. In this analogy, the underlying lattice is the semigroup of dominant characters and Fock spaces are replaced by Weyl modules. A guiding principle in the study of lattice vertex algebras is that their representation theory is controlled by the dual lattice. This principle applied here is what gives birth to the Langlands dual combinatorics.

This work raises many possible future research directions that we discuss in Chapter 11. The one that seems the most interesting to us is to understand what this whole picture, *e.g.*, the fundamental local equivalence, becomes for an arbitrary choice of nilpotent element. To the best of our knowledge there are no clear expectations coming from the quantum geometric Langlands program. We have high hopes that certain techniques displayed here will be useful to explore these questions.

Outline

Chapter 3

We set up our conventions and notations about algebraic groups, their Lie algebras and vertex algebras.

Chapter 4

We define the central object $\mathcal{D}_G^\kappa$ of this thesis together with the Kazhdan–Lusztig category and reprove some of their basic properties in view of recalling the proof of the chiral Peter–Weyl Theorem 4.4.1. We also prove a unicity statement that will be crucial later (Theorem 4.4.3). Our proof uses the theory of spherical varieties ([LV83], [Kno91]) in an essential way. Then we give a proof of the degeneration of the geometric Satake equivalence (Theorem 4.5.3).

Chapter 5

We start to make our key observation more precise by introducing Li's delta operator ([Li97]) to define what we call the spectral flow group and compute it in the case of $\mathcal{D}_G^\kappa$ (Proposition 5.3.1). Because this group fully controls the representation theory in the case of lattice vertex algebra (see *e.g.*, Proposition 3.4 of [Li97]), our analogy dictates that it might be relevant in our situation. In general, the spectral flow group provides a large supply of simple modules on $\mathcal{D}_G^\kappa$.

Chapter 6

As explained in the introduction, we need to go through the quantum Hamiltonian reduction procedure to produce meaningful objects. This chapter is a technical interlude where we recall some definitions and explain the relation between taking the quantum Hamiltonian reduction of the spectral flow of a module and the twisted quantum Hamiltonian reduction functor. More precisely, we prove that these two operations are equal up to cohomological shift (Theorem 6.3.1).

Chapter 7

We recall the definition of $\mathcal{W}_G^\kappa$, the equivariant $\mathcal{W}$-algebra on the group G, which is the other main character of this thesis. We then state our main conjecture (Conjecture 7.2.4). We are then able to describe the objects appearing in it and prove their simplicity (Theorem 7.2.5) by using the results of Arakawa and Frenkel ([AF19]) in a crucial way.

Chapter 8

To test our conjecture, we restrict our attention the case of an algebraic torus. In fact, in this case $\mathcal{D}^{\kappa}_{G}$ is isomorphic to what is called in the literature a half-lattice vertex algebra (see *e.g.*, [BDT02]). We then show how the standard techniques apply and prove Conjecture 7.2.4 in this setting. In fact, we even get a more precise statement about how the geometric Satake equivalence behaves with respect to the level (Corollary 8.1.8).

Chapter 9

We propose a general strategy to prove the conjecture. Once again, we are guided by a somewhat informal principle: vertex algebras tend to be more rigid than ordinary algebras. This is what led us in Chapter 4 to prove a unicity statement on simple vertex algebras having the chiral Peter–Weyl decomposition (see Theorem 4.4.3). In turn, this unicity statement is the key for us to prove properties of some shifted versions of $\mathcal{D}^{\kappa}_{G}$ (called the quantum geometric Langlands kernels vertex algebras and introduced in [CG20]) and to study their behaviour under the quantized Moore–Tachikawa convolution operation ([Ara18]). In particular, when G is a simple adjoint group of type A or D, *e.g.* $G = \mathrm{PSL}_n$, we are able to prove the conjecture. Our approach is, among other things, related to the Goddard–Kent–Olive coset realization of $\mathcal{W}$-algebras (see [GKO86] for $\mathfrak{sl}_2$, [ACL19] and [CN23] for type ADE, [CL22a] for type A, [CL22b] for type D).

Chapter 10

We return to the case of an algebraic torus to study the quantum geometric Langlands kernels vertex algebras in this setting. In particular, we compute their associated schemes (Proposition 10.2.5) and study their behaviour under the Moore–Tachikawa convolution operation (Proposition 10.3.2).

Chapter 11

We discuss possible extensions of the ideas developed in this work and further research directions.

Chapter 2

Introduction (version française)

Soit G un groupe algébrique réductif sur le corps $\mathbb{C}$ des nombres complexes, d'algèbre de Lie $\mathfrak{g}$. Soit $V^\kappa(\mathfrak{g})$ l'algèbre vertex affine universelle associée au niveau κ. On peut y penser comme une version vertex de l'algèbre enveloppante $\mathcal{U}(\mathfrak{g})$ de $\mathfrak{g}$. Soit $\mathcal{D}_G^\kappa$ l'algèbre vertex des opérateurs différentiels chiraux sur le groupe G. De même, c'est une version Kac–Moody de l'algèbre $\mathcal{D}_G$ des opérateurs différentiels sur G. On dispose d'une inclusion naturelle de $\mathcal{U}(\mathfrak{g})$ dans $\mathcal{D}_G$ obtenue en pensant $\mathfrak{g}$ comme l'algèbre de Lie des champs de vecteurs invariants à gauche sur G. Ce cadre classique se chiralise (c'est-à-dire s'étend au cadre des algèbres vertex) et on a un plongement de $V^\kappa(\mathfrak{g})$ dans $\mathcal{D}_G^\kappa$. Soit $\mathcal{W}_G^\kappa$ la $\mathcal{W}$-algèbre équivariante principale sur le groupe G au niveau κ. Elle est définie comme la réduction de Drinfeld–Sokolov de $\mathcal{D}_G^\kappa$ par rapport au plongement précédent. Une contribution de cette thèse est de prouver

Théorème A (Theorem 7.2.5)**.** *Pour des niveaux génériques, il existe une famille de $\mathcal{W}_G^\kappa$-modules simples, deux à deux non équivalents, indexés par les cocaractères dominants de G. Ils sont obtenus par réduction hamiltonienne quantique de flots spectraux du module adjoint $\mathcal{D}_G^\kappa$.*

En d'autres termes, nous donnons une construction dans le monde des algèbres vertex qui, à partir du groupe G, produit des objets simples dont la combinatoire correspond à celle de la théorie des représentations du groupe dual de Langlands $\check{G}$. L'apparition du groupe dual suggère qu'un phénomène plus profond est à l'œuvre ici. Afin de donner une interprétation plus satisfaisante, nous avons besoin d'un peu de contexte.

Soit T un tore maximal de G et B un Borel le contenant. L'étude de la théorie des représentations de G conduit naturellement à considérer divers espaces géométriques construits à partir du groupe. Un exemple est la variété de drapeaux G/B, une variété algébrique projective lisse de type fini. Par exemple, l'étude des $\mathcal{D}_{G/B}$-modules algébriques (tordus), *i.e.*, la théorie des modules sur le faisceau des opérateurs différentiels (tordus) sur G/B, conduit à la théorie de la localisation de Beilinson–Bernstein ([BB93]).

La théorie des $\mathcal{D}$-modules sur un espace X peut être vue comme l'étude des systèmes d'équations aux dérivées partielles linéaires sur X ([Kas03], [Bor87], [HTT08]). Lorsqu'on les résout, on produit des objets topologiques, typiquement des systèmes locaux d'espaces vectoriels de dimension finie sur X. Il arrive parfois qu'un $\mathcal{D}$-module et l'objet topologique associé soient équivalents, et ce type de phénomène porte le nom de correspondance de Riemann–Hilbert. Une large classe de $\mathcal{D}$-modules pour lesquels cela se produit a été identifiée. Ce sont les $\mathcal{D}$-modules holonomes réguliers et leurs pendants topologiques sont appelés faisceaux pervers ([Del70], [Meb84], [Kas84], [GM80], [BBDG82]). Ils sont également importants en raison de leurs interactions avec le dictionnaire faisceaux-fonctions de Grothendieck ([Del77]).

Un exemple est l'équivalence de Satake géométrique ([Sat63], [Gin90], [MV00], [BR18]) qui constitue un premier pas dans la version géométrique du programme de Langlands, tel qu'envisagé par Drinfeld et Laumon dans les années 1980 ([Dri83], [Lau87]). Pour l'énoncer, soit $\mathcal{J}_\infty G$ l'espace des jets et $\mathcal{L}G$ l'espace des lacets du groupe G. Ce sont des groupes algébriques tels que $\mathcal{J}_\infty G(\mathbb{C}) = G(\mathbb{C}[[t]])$ et $\mathcal{L}G(\mathbb{C}) = G(\mathbb{C}((t)))$. On définit la Grassmannienne affine $\mathrm{Gr}_G = \mathcal{L}G/\mathcal{J}_\infty G$ comme le quotient du groupe des lacets de G par son espace des jets ; c'est un ind-schéma de type infini. L'équivalence de Satake géométrique affirme que la catégorie des faisceaux pervers $\mathcal{J}_\infty G$-équivariants sur Gr_G est équivalente à la catégorie des représentations de $\check{G}$.

Cela soulève la question suivante : existe-t-il un moyen d'énoncer l'équivalence de Satake géométrique en termes de $\mathcal{D}$-modules sur la Grassmannienne affine Gr_G ? Pour y répondre, on peut espérer un formalisme de $\mathcal{D}$-modules suffisamment robuste pour avoir un sens sur Gr_G, ainsi qu'une correspondance de Riemann–Hilbert adaptée. Un pas dans cette direction est suggéré par les travaux d'Arkhipov et Gaitsgory ([AG02]) dans lesquels ils construisent une algèbre chirale dont la catégorie de modules joue le rôle de la catégorie des $\mathcal{D}$-modules sur le groupe des lacets $\mathcal{L}G$. Ce type d'algèbres chirales est connu sous le nom d'algèbres chirales d'opérateurs différentiels et a été introduit par Beilinson et Drinfeld ([BD04]). Elles ont également été découvertes indépendamment dans le langage des algèbres vertex par Gorbounov, Malikov et Schechtman, et appelées par la suite algèbres vertex d'opérateurs différentiels chiraux ([MSV99], [GMS04]). En termes d'algèbres vertex, la construction d'Arkhipov et Gaitsgory s'énonce ainsi : il existe une algèbre vertex $\mathcal{D}_G^\kappa$ dont la catégorie de modules est une catégorie appropriée $\mathcal{D}_\kappa(\mathcal{L}G)-\mathrm{Mod}$ de $\mathcal{D}$-modules κ-tordus sur le groupe des lacets. En particulier, pour un certain niveau κ_c, appelé le niveau critique, la catégorie $\mathcal{D}_G^{\kappa_c}-\mathrm{Mod}$ se comporte comme une catégorie appropriée de $\mathcal{D}$-modules non tordus sur le groupe des lacets.

Dans cette thèse, nous utiliserons ce dernier langage. Les algèbres vertex sont des structures algébriques relativement récentes qui ont été axiomatisées par Borcherds ([Bor86]). Leur émergence est étroitement liée au groupe Monstre à travers les célèbres conjectures du Moonshine de Conway et Norton ([CN79]) et leur démonstration par Borcherds ([Bor92]). Une étape cruciale est la construction du module de Moonshine par Frenkel, Lepowsky et Meurman ([FLM89]). La genèse des algèbres vertex est indissociable d'idées provenant de la physique ; voir la préface et l'introduction magnifiquement écrites de *loc. cit.*. Le mathématicien qui apprend la théorie des algèbres vertex fait rapidement face à un paradoxe car leur définition est d'une simplicité trompeuse. Une algèbre vertex est un espace vectoriel muni d'une infinité dénombrable de produits satisfaisant certaines relations (voir la Définition 3.2.1). Pourtant, construire des exemples (non commutatifs) est loin d'être facile, comme le montrent à la fois l'expérience et l'histoire (voir par exemple [FLM89], [FZ92a], [Zhu08]). La théorie des algèbres vertex semble intrinsèquement rigide. De manière frappante, la rareté des exemples conduit à formuler des résultats d'unicité qui sembleraient trop ambitieux dans le cadre classique. Cela nous a amené à démontrer les Théorèmes 4.4.3 et 9.3.9, qui jouent un rôle technique central dans ce travail.

Disons maintenant un mot sur le cadre classique de la théorie des $\mathcal{D}$-modules algébriques. Soit X un schéma affine lisse de type fini, de faisceau structural $\mathcal{O}_X$. Il vient équipé d'un faisceau $\mathcal{D}_X$ d'opérateurs différentiels construit à partir de $\mathcal{O}_X$ et du faisceau tangent Θ_X. C'est un faisceau d'anneaux non commutatifs qui contient $\mathcal{O}_X$ comme sous-anneau et Θ_X comme sous-algèbre de Lie, interagissant via la règle de Leibniz. La catégorie des $\mathcal{D}_X$-modules algébriques sur X est alors la catégorie des faisceaux de modules, disons à gauche, sur $\mathcal{D}_X$ qui sont quasi-cohérents en tant que $\mathcal{O}_X$-modules. L'hypothèse de quasi-cohérence assure qu'un $\mathcal{D}_X$-module est contrôlé par ses sections globales, qui forment un module à gauche sur l'anneau non commutatif $\mathcal{D}_X(X)$ des opérateurs différentiels globaux sur X. On identifie dès à présent un faisceau et ses sections globales.

Un exemple clé pour nous est le cas où $X = G$ est le groupe lui-même. La théorie des $\mathcal{D}$-modules n'est alors rien d'autre que la théorie des représentations de l'anneau $\mathcal{D}_G$. Cet anneau admet une description agréable en termes de fonctions sur le groupe G et de son algèbre de Lie $\mathfrak{g}$. En effet, on peut voir $\mathfrak{g}$ comme les champs de vecteurs globaux invariants à gauche de G et ils engendrent sur $\mathcal{O}_G$ l'algèbre de Lie des champs de vecteurs globaux Θ_G. En tant qu'espace vectoriel, on a

$$\mathcal{D}_G = \mathcal{U}(\mathfrak{g}) \otimes \mathcal{O}_G$$

et la structure d'anneau est déterminée par la règle de Leibniz. $\mathcal{D}_G$ contient naturellement $\mathcal{U}(\mathfrak{g})$ et $\mathcal{O}_G$ comme sous-algèbres. Notons que nous avons fait le choix de considérer $\mathfrak{g}$ comme des champs de vecteurs globaux invariants à gauche sur G. En considérant $\mathfrak{g}$ comme des champs de vecteurs globaux invariants à droite, on montre que $\mathcal{D}_G$ contient une autre sous-algèbre qui est une copie de $\mathcal{U}(\mathfrak{g})$. Par construction, ces deux copies commutent l'une avec l'autre.

La construction de $\mathcal{D}_G^\kappa$ est une variante de celle de $\mathcal{D}_G$ où G est remplacé par $\mathcal{L}G$ et les anneaux sont remplacés par des algèbres vertex. Introduisons l'algèbre de Kac–Moody affine $\hat{\mathfrak{g}}^\kappa = \mathfrak{g}((t)) \oplus \mathbb{C}\mathbb{1}$ ([Kac90]). C'est une extension centrale de $\mathrm{Lie}(\mathcal{L}G) = \mathfrak{g}((t))$ donnée par le niveau κ. L'algèbre vertex affine universelle $V^\kappa(\mathfrak{g})$ joue un rôle analogue à celui de l'algèbre enveloppante universelle pour $\hat{\mathfrak{g}}^\kappa$. La sous-algèbre de

Lie $\mathfrak{g}[[t]] = \mathrm{Lie}(\mathcal{J}_\infty G)$ de $\hat{\mathfrak{g}}^\kappa$ agit sur $\mathcal{O}_{\mathcal{J}_\infty G}$ comme des champs de vecteurs invariants à gauche et ainsi, par analogie, on définit, en tant qu'espaces vectoriels

$$\mathcal{D}_G^\kappa = V^\kappa(\mathfrak{g}) \otimes \mathcal{O}_{\mathcal{J}_\infty G}.$$

On peut alors démontrer qu'il existe une structure d'algèbre vertex sur $\mathcal{D}_G^\kappa$ telle que $V^\kappa(\mathfrak{g})$ et $\mathcal{O}_{\mathcal{J}_\infty G}$ sont des sous-algèbres vertex de $\mathcal{D}_G^\kappa$. De nouveau, nous avons fait le choix de considérer les champs de vecteurs invariants à gauche sur $\mathcal{J}_\infty G$ plutôt que les invariants à droite. Cette fois, correspondant à ces derniers, il y a une copie de $V^{\kappa^*}(\mathfrak{g})$ dans $\mathcal{D}_G^\kappa$ mais le niveau est décalé (voir le Théorème 4.1.4). Là encore, ces deux copies commutent en un sens approprié.

Cette algèbre vertex va jouer un rôle analogue à celui (des sections globales) du faisceau des opérateurs différentiels (κ-tordus) sur le groupe des lacets $\mathcal{L}G$. Il est donc naturel de définir

$$\mathcal{D}\mathrm{-Mod}_\kappa(\mathcal{L}G) := \mathcal{D}_G^\kappa\mathrm{-Mod}.$$

En particulier, lorsque $\kappa = \kappa_c$ est le niveau critique, on obtient une définition des $\mathcal{D}$-modules non tordus sur le groupe des lacets

$$\mathcal{D}\mathrm{-Mod}(\mathcal{L}G) := \mathcal{D}_G^{\kappa_c}\mathrm{-Mod}.$$

Pour prendre en compte les objets sur la Grassmannienne affine, et l'équivariance de ces objets eux-mêmes, il faut développer davantage le formalisme. Dans le cadre classique où X est un schéma lisse de type fini et est muni d'une action d'un sous-groupe connexe K de G, il est remarquable que la catégorie des $\mathcal{D}$-modules K-équivariants sur X soit une sous-catégorie pleine de la catégorie des $\mathcal{D}$-modules sur X. Autrement dit, l'équivariance des $\mathcal{D}$-modules est une propriété et non une structure supplémentaire (voir par exemple la Remarque 30.10 de [Eti24]). Précisément, étant donné un $\mathcal{D}_X$-module $\mathcal{M}$, alors $\mathcal{M}$ est K-équivariant si l'action induite de $\mathrm{Lie}(K)$ s'intègre en une action du groupe K (voir la Proposition 30.18 de *loc.cit.*).

Revenons au cadre de la dimension infinie, soit $K = \mathcal{J}_\infty G \subset \mathcal{L}G$ agissant sur $\mathcal{L}G$ par multiplication à gauche. Premièrement, puisque Gr_G est un quotient, il est naturel de définir la catégorie des $\mathcal{D}$-modules sur Gr_G comme la catégorie des $\mathcal{D}$-modules K-équivariants sur le groupe des lacets. En termes de l'algèbre vertex $\mathcal{D}_G^\kappa$, étant donné un $\mathcal{D}_G^\kappa$-module M, cela revient à demander que, vu comme $V^{\kappa^*}(\mathfrak{g})$-module par restriction, l'action de $\mathfrak{g}[[t]]$ s'intègre en une action de $\mathcal{J}_\infty G$. Cette sous-catégorie de $V^{\kappa^*}(\mathfrak{g})\mathrm{-Mod}$ est connue sous le nom de catégorie de Kazhdan–Lusztig ([KL93a], [KL93b], [KL94a], [KL94b]), nous la notons $V^{\kappa^*}(\mathfrak{g})\mathrm{-Mod}^{\mathcal{J}_\infty G}$. Soit

$$\mathcal{D}_G^\kappa\mathrm{-Mod}^{1\times\mathcal{J}_\infty G}$$

la sous-catégorie pleine de $\mathcal{D}_G^\kappa\mathrm{-Mod}$ constituée des objets satisfaisant cette condition. Cela définit (voir le Théorème A.3 de [AG02]) la catégorie des $\mathcal{D}$-modules κ-tordus sur la Grassmannienne affine

$$\mathcal{D}\mathrm{-Mod}_\kappa(\mathrm{Gr}_G) := \mathcal{D}_G^\kappa\mathrm{-Mod}^{1\times\mathcal{J}_\infty G}.$$

De la même manière, on définit

$$\mathcal{D}_G^\kappa\text{−Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}$$

comme la sous-catégorie pleine de modules constituée des objets qui, vus comme $V^{\kappa \oplus \kappa^*}(\mathfrak{g} \oplus \mathfrak{g})$-modules, appartiennent à $V^{\kappa \oplus \kappa^*}(\mathfrak{g} \oplus \mathfrak{g})\text{−Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}$. Soit

$$\mathcal{D}\text{−Mod}_\kappa(\mathrm{Gr}_G)^{\mathcal{J}_\infty G} := \mathcal{D}_G^\kappa\text{−Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}$$

la catégorie des $\mathcal{D}$-modules κ-tordus $\mathcal{J}_\infty G$-équivariants sur la Grassmannienne affine. Dans ce langage, il est raisonnable de s'attendre à ce que l'équivalence de Satake géométrique se produise au niveau critique et prenne la forme d'une équivalence de catégories

$$\mathcal{D}_G^{\kappa_c}\text{−Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G} \simeq \check{G}\text{−Mod}$$

où $\check{G}$−Mod est la catégorie des représentations du groupe algébrique $\check{G}$. Il est naturel de se demander s'il est possible de décrire la catégorie

$$\mathcal{D}_G^\kappa\text{−Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}$$

pour des niveaux κ qui ne sont pas critiques. Lorsqu'on s'éloigne du niveau critique, on entre dans le paradigme du programme de Langlands géométrique quantique ([Sto06], [Gai16]). Nous démontrons le résultat suivant dans le cas où G est un tore ou un groupe simple :

Théorème B (Theorem 4.5.3, Corollary 8.1.8)**.** *Pour des niveaux κ suffisamment génériques, le module adjoint $\mathcal{D}_G^\kappa$ est le seul objet simple de la catégorie $\mathcal{D}_G^\kappa\text{−Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}$. De plus, la catégorie $\mathcal{D}_G^\kappa\text{−Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}$ est équivalente à la catégorie des $\mathbb{C}$-espaces vectoriels.*

Cela peut être interprété comme une dégénérescence quantique de la correspondance de Satake géométrique. Ce fait était bien connu des experts et une catégorie de substitution a été introduite par Gaitsgory en suivant une idée de Lurie ([Gai08]). C'est la catégorie, convenablement définie, des $\mathcal{D}$-modules de Whittaker κ-tordus sur Gr_G. Influencé par l'équivalence de Skryabin affine démontrée par Raskin ([Ras21]), nous introduisons maintenant une catégorie conjecturalement équivalente dans notre langage.

Soit H_{DS}^* le foncteur de réduction hamiltonienne quantique (principale) ([FF90]). Ce foncteur permet de définir la $\mathcal{W}$-algèbre (principale) de $\mathfrak{g}$ au niveau κ à partir des algèbres affines universelles, $W^\kappa(\mathfrak{g}) := H_{DS}^*(V^\kappa(\mathfrak{g}))$. De plus, $W^\kappa(\mathfrak{g})$ est une algèbre vertex, et on obtient un foncteur bien défini

$$H_{DS}^* : V^\kappa(\mathfrak{g})\text{−Mod} \longrightarrow W^\kappa(\mathfrak{g})\text{−Mod}.$$

Cela fonctionne également dans un cadre relatif, à savoir si V est une algèbre vertex munie d'un morphisme

$$V^\kappa(\mathfrak{g}) \longrightarrow V$$

alors $H^*_{DS}(V)$ est une algèbre vertex qui vient avec un morphisme

$$W^\kappa(\mathfrak{g}) \longrightarrow H^*_{DS}(V).$$

De plus, on a un foncteur bien défini

$$H^*_{DS} : V\text{−Mod} \longrightarrow H^*_{DS}(V)\text{−Mod}.$$

Lorsque cette construction est appliquée à $\mathcal{D}^\kappa_G$ par rapport au morphisme d'algèbres vertex

$$V^\kappa(\mathfrak{g}) \longrightarrow V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g}) \longrightarrow \mathcal{D}^\kappa_G$$

cela définit la $\mathcal{W}$-algèbre affine équivariante (principale) $\mathcal{W}^\kappa_G$ sur le groupe G au niveau κ ([Ara18]). Elle vient équipée d'un morphisme

$$W^\kappa(\mathfrak{g}) \longrightarrow W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g}) \longrightarrow \mathcal{W}^\kappa_G.$$

On définit

$$\mathcal{W}^\kappa_G\text{−Mod}^{\mathcal{J}_\infty G}$$

comme la sous-catégorie pleine de $\mathcal{W}^\kappa_G$−Mod constituée des modules qui sont intégrables par rapport à $\mathcal{J}_\infty G$ lorsqu'ils sont vus comme $V^{\kappa^*}(\mathfrak{g})$-modules par restriction. Cette catégorie devrait se déformer de manière satisfaisante et être reliée, pour κ générique, à la catégorie des représentations du groupe dual de Langlands $\check{G}$.

Nous formulons la conjecture suivante, qui est une version en catégories abéliennes de l'équivalence locale fondamentale du programme de Langlands géométrique quantique ([CDR21]) dans notre langage :

Conjecture A (Conjecture 7.2.4)**.** *Pour des niveaux génériques, la catégorie $\mathcal{W}^\kappa_G\text{−Mod}^{\mathcal{J}_\infty G}$ est semi-simple, avec pour objets simples les $H^*_{DS}(\gamma \cdot \mathcal{D}^\kappa_G)$ pour $\gamma \in X_*(T)_+$. En particulier, il existe une équivalence de catégories abéliennes*

$$\mathcal{W}^\kappa_G\text{−Mod}^{\mathcal{J}_\infty G} \simeq \check{G}\text{−Mod}$$

qui envoie pour tout $\gamma \in X_*(T)_+$ l'objet simple $H^*_{DS}(\gamma \cdot \mathcal{D}^\kappa_G)$ sur V_γ.

Nous expliquons une approche générale pour résoudre cette conjecture. Dans le cas d'un tore algébrique, nous pouvons la démontrer par des calculs directs. Dans le cas général, notre approche repose, entre autres, sur certaines algèbres vertex dont l'existence est encore conjecturale. Comme leur existence a été établie avec certaines restrictions sur le groupe G, nous sommes en mesure de prouver :

Théorème C (Theorem 8.1.6 and Theorem 9.4.1)**.** *La Conjecture 7.2.4 est vérifiée lorsque G est un tore ou lorsque G est un groupe simple adjoint de type classique simplement lacé.*

L'idée principale de cette thèse est que, pour des niveaux génériques, le Théorème de Peter–Weyl chiral 4.4.1 permet de voir $\mathcal{D}_G^\kappa$ comme un analogue d'une algèbre vertex réseau. Dans cette analogie, le réseau sous-jacent est le semi-groupe des caractères dominants et les espaces de Fock sont remplacés par les modules de Weyl. Un principe directeur dans l'étude des algèbres vertex de réseau est que leur théorie des représentations est contrôlée par le réseau dual. Ce principe appliqué ici est ce qui donne naissance à la combinatoire duale de Langlands.

Ce travail ouvre de nombreuses directions de recherche possibles que nous discutons au Chapitre 11. Celle qui nous semble la plus intéressante est de comprendre ce que l'ensemble de ce tableau, *e.g.*, l'équivalence locale fondamentale, devient pour un choix arbitraire d'élément nilpotent. À notre connaissance, il n'y a pas d'attentes claires provenant du programme de Langlands géométrique quantique. Nous avons bon espoir que certaines techniques présentées ici seront utiles pour explorer ces questions.

Plan

Chapitre 3

Nous fixons nos conventions et notations concernant les groupes algébriques, leurs algèbres de Lie et les algèbres vertex.

Chapitre 4

Nous définissons $\mathcal{D}_G^\kappa$ qui est l'objet central de cette thèse ainsi que la catégorie de Kazhdan–Lusztig et redémontrons certaines de leurs propriétés de base en vue de rappeler la preuve du Théorème de Peter–Weyl chiral 4.4.1. Nous démontrons également un énoncé d'unicité qui sera crucial par la suite (Théorème 4.4.3). Notre preuve utilise la théorie des variétés sphériques ([LV83], [Kno91]) de manière essentielle. Nous donnons ensuite une preuve de la dégénérescence de l'équivalence de Satake géométrique (Théorème 4.5.3).

Chapitre 5

Nous commençons à préciser notre observation clé en introduisant l'opérateur delta de Li ([Li97]) pour définir ce que nous appelons le groupe de flot spectral et le calculer dans le cas de $\mathcal{D}_G^\kappa$ (Proposition 5.3.1). Puisque ce groupe contrôle entièrement la théorie des représentations dans le cas des algèbres vertex de réseau (voir *e.g.*, la Proposition 3.4 de [Li97]), notre analogie indique qu'il pourrait être pertinent dans notre situation. En général, le groupe de flot spectral fournit une grande quantité de modules simples sur $\mathcal{D}_G^\kappa$.

Chapitre 6

Comme expliqué dans l'introduction, nous devons passer par la procédure de réduction hamiltonienne quantique pour produire des objets significatifs. Ce chapitre est un interlude technique où nous rappelons certaines définitions et expliquons la relation entre la réduction hamiltonienne quantique du flot spectral d'un module et le foncteur de réduction hamiltonienne quantique tordue. Plus précisément, nous démontrons que ces deux opérations sont égales à un décalage cohomologique près (Théorème 6.3.1).

Chapitre 7

Nous rappelons la définition de $\mathcal{W}^{\kappa}_{G}$, la $\mathcal{W}$-algèbre équivariante sur le groupe G, qui est l'autre protagoniste de cette thèse. Nous énonçons ensuite notre conjecture principale (Conjecture 7.2.4). Nous sommes alors en mesure de décrire les objets qui y apparaissent et de démontrer leur simplicité (Théorème 7.2.5) en utilisant les résultats d'Arakawa et Frenkel ([AF19]) de manière cruciale.

Chapitre 8

Pour mettre notre conjecture à l'épreuve, nous restreignons notre attention au cas d'un tore algébrique. En fait, dans ce cas, $\mathcal{D}^{\kappa}_{G}$ est isomorphe à ce que l'on appelle dans la littérature une algèbre vertex demi-réseau (voir *e.g.*, [BDT02]). Nous montrons ensuite comment les techniques standard s'appliquent et démontrons la Conjecture 7.2.4 pour un tore. En fait, nous obtenons même un énoncé plus précis sur le comportement de l'équivalence de Satake géométrique par rapport au niveau (Corollaire 8.1.8).

Chapitre 9

Nous proposons une stratégie générale pour démontrer la conjecture. Encore une fois, nous sommes guidés par un principe quelque peu informel : les algèbres vertex tendent à être plus rigides que les algèbres ordinaires. C'est ce qui nous a conduit au Chapitre 4 à démontrer un énoncé d'unicité sur les algèbres vertex simples possédant la décomposition chirale de Peter–Weyl (voir le Théorème 4.4.3). À son tour, cet énoncé d'unicité est la clé pour démontrer des propriétés de certaines versions décalées de $\mathcal{D}^{\kappa}_{G}$ (appelées algèbres vertex noyaux de Langlands géométrique quantique et introduites dans [CG20]) et pour étudier leur comportement sous l'opération de convolution quantifiée de Moore–Tachikawa ([Ara18]). En particulier, lorsque G est un groupe simple adjoint de type A ou D, *e.g.* $G = \mathrm{PSL}_n$, nous sommes en mesure de démontrer la conjecture. Notre approche est, entre autres, liée à la réalisation par commutant de Goddard–Kent–Olive des $\mathcal{W}$-algèbres (voir [GKO86] pour $\mathfrak{sl}_2$, [ACL19] et [CN23] pour le type ADE, [CL22a] pour le type A, [CL22b] pour le type D).

Chapitre 10

Nous revenons au cas d'un tore algébrique pour étudier les algèbres vertex noyaux de Langlands géométrique quantique dans ce cadre. En particulier, nous calculons leurs schémas associés (Proposition 10.2.5) et étudions leur comportement sous l'opération de convolution de Moore–Tachikawa (Proposition 10.3.2).

Chapitre 11

Nous discutons des extensions possibles des idées développées dans ce travail et des directions de recherche futures.

Chapter 3

Conventions and notations

All constructions are over the field $\mathbb{C}$ of complex numbers and unlabeled tensor products are understood to be taken over $\mathbb{C}$.

For us, algebraic groups are assumed to be affine and connected, we denote the multiplicative group by $\mathbb{G}_m = \operatorname{Spec}(\mathbb{C}[t, t^{-1}])$. By an algebraic torus we mean a direct product of finitely many copies of $\mathbb{G}_m$.

If $\mathfrak{g}$ is a Lie algebra and M is a $\mathfrak{g}$-module, we denote by $M^{\mathfrak{g}}$ the invariant subspace consisting of elements of M annihilated by every element of $\mathfrak{g}$.

If V is a vector space then we denote by $\mathcal{F}(V)$ the vector space of fields on V. It consists of series $a(z) \in \operatorname{End}(V)[[z, z^{-1}]]$ such that for all $v \in V$, $a(z)v \in V((z))$.

If I is a set and $i, j \in I$ we denote by $\delta_{i,j}$ the Dirac delta function equals to 1 if $i = j$ and 0 else.

3.1 Reductive groups and their Lie algebras

Some good general references on algebraic groups are [Bor91], [DG11], [TY05], and [Mil17]. We denote by

$$G-\mathrm{Mod}$$

the category of representations of G. Let $Z(G)$ be the center of G and choose a maximal torus T, a Borel subgroup B containing T, and denote by N the unipotent radical of B. Let $N_G(T)$ be the normalizer of the maximal torus T, $W = N_G(T)/T$ be the Weyl group and ω_0 be its longest element.

Let $\mathfrak{g} = \operatorname{Lie}(G)$ be the Lie algebra of G, $\mathfrak{z}(\mathfrak{g}) = \operatorname{Lie}(Z(G))$, $\mathfrak{h} = \operatorname{Lie}(T)$ (it is a Cartan subalgebra of $\mathfrak{g}$), $\mathfrak{b} = \operatorname{Lie}(B)$, $\mathfrak{n} = \operatorname{Lie}(N)$ the corresponding Lie algebras. We denote by $\mathfrak{g}-\mathrm{Mod}$ the category of representations of $\mathfrak{g}$. Since $\mathfrak{g}$ is reductive, it decomposes as the direct sum of its center $\mathfrak{z}(\mathfrak{g})$ and its derived Lie algebra $[\mathfrak{g}, \mathfrak{g}]$. Write

$$\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}] = \mathfrak{z}(\mathfrak{g}) \oplus \bigoplus_{s=1}^{r} \mathfrak{g}_s, \tag{3.1}$$

with $\mathfrak{g}_s$ simple. For each $s \in \{1, \ldots, r\}$, let $\mathfrak{h}_s := \mathfrak{h} \cap \mathfrak{g}_s$, it is a Cartan subalgebra of $\mathfrak{g}_s$ and we have the equality

$$\mathfrak{h} = \mathfrak{z}(\mathfrak{g}) \oplus \bigoplus_{s=1}^{r} \mathfrak{h}_s.$$

Let $X^*(T)$ be the group of characters and $X_*(T)$ the group of cocharacters of the maximal torus T. They both are free abelian groups of rank equal to the dimension of T. Via the functor Lie we identify $X^*(T)$ (resp. $X_*(T)$) with subgroups of $\mathfrak{h}^*$ (resp. $\mathfrak{h}$). These are full rank lattices in the sense that the following canonical maps are isomorphisms of vector spaces

$$X_*(T) \otimes_{\mathbb{Z}} \mathbb{C} \longrightarrow \mathfrak{h}, \qquad X^*(T) \otimes_{\mathbb{Z}} \mathbb{C} \longrightarrow \mathfrak{h}^*.$$

Recall that each element of $X^*(\mathbb{G}_m)$ is of the form $t \in \mathbb{G}_m \mapsto t^n \in \mathbb{G}_m$ for a unique $n \in \mathbb{Z}$. This yields a group isomorphism between $X^*(\mathbb{G}_m)$ and $\mathbb{Z}$. If $\gamma \in X_*(T)$ is a cocharacter and $\alpha \in X^*(T)$ is a character, their composition $\alpha \circ \gamma$ is, via the previous identification, an element of $\mathbb{Z}$. That defines a $\mathbb{Z}$-bilinear map

$$X^*(T) \otimes_{\mathbb{Z}} X_*(T) \longrightarrow \mathbb{Z} \tag{3.2}$$

that happens to be a perfect pairing of free $\mathbb{Z}$-modules in a compatible way with the perfect pairing between $\mathfrak{h}$ and $\mathfrak{h}^*$.

The group G acts on its Lie algebra $\mathfrak{g}$ via the so-called adjoint representation $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$. By restricting this action to the maximal torus T we obtain the decomposition

$$\mathfrak{g} = \bigoplus_{\alpha \in X^*(T)} \mathfrak{g}^{\alpha}$$

where $\mathfrak{g}^{\alpha} = \{x \in \mathfrak{g} \mid \mathrm{Ad}(t)(x) = \alpha(t)x \text{ for all } t \in T\}$. Recall that a nonzero $\alpha \in X^*(T)$ such that $\mathfrak{g}^{\alpha}$ is nonzero is called a root, they form a finite set denoted $\Phi(G, T)$. We have $\mathfrak{g}^0 = \mathfrak{h}$ and we get the following root space decomposition of $\mathfrak{g}$:

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi(G,T)} \mathfrak{g}^{\alpha}. \tag{3.3}$$

If we differentiate the adjoint representation of the group G we obtain the adjoint representation of its Lie algebra $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ and, after restricting to the Cartan subalgebra, a similar story holds. We denote by $\Phi(\mathfrak{g}, \mathfrak{h})$ the set of roots with respect to $\mathfrak{h}$.

Now, because $\mathfrak{h} = \mathrm{Lie}(T)$, it is easy to check that $\Phi(G, T) = \Phi(\mathfrak{g}, \mathfrak{h})$ under the inclusion $X^*(T) \subset \mathfrak{h}$, we denote this set by Φ. The story for coroots is parallel, let $\Phi(G, T)^{\vee} \subset X_*(T)$ be the finite subset of coroots for the group and $\Phi(\mathfrak{g}, \mathfrak{h})^{\vee} \subset \mathfrak{h}$ be the finite subset of coroots for the Lie algebra. We have that $\Phi(G, T)^{\vee} = \Phi(\mathfrak{g}, \mathfrak{h})^{\vee}$ under the inclusion $X_*(T) \subset \mathfrak{h}$. We denote this set by $\Phi^{\vee}$. To each root is naturally associated a coroot, and we denote by $\alpha \in \Phi \mapsto \alpha^{\vee} \in \Phi^{\vee}$ this bijection.

Let Q (resp. $Q^\vee$) be the subgroup of $\mathfrak{h}^*$ (resp. $\mathfrak{h}$) generated by roots (resp. coroots). By the theory of semisimple Lie algebras, the $\mathbb{C}$-span of the roots (resp. coroots) is

$$\bigoplus_{s=1}^{r} \mathfrak{h}_s^* \text{ (resp. } \bigoplus_{s=1}^{r} \mathfrak{h}_s).$$

The choice we made of a Borel B containing T defines a subset $\Phi_+ \subset \Phi$ of positive roots by saying that $\alpha \in \Phi$ is positive if $\mathfrak{g}^\alpha \subset \mathfrak{b}$. The following definitions are important to classify the representations of the algebraic group G. Let

$$X^*(T)_+ = \{\lambda \in X^*(T) \mid \lambda(\alpha^\vee) \geq 0 \text{ for all } \alpha \in \Phi_+\}$$

be the semigroup of dominant characters,

$$X_*(T)_+ = \{\gamma \in X_*(T) \mid \gamma(\alpha) \geq 0 \text{ for all } \alpha \in \Phi_+\}$$

be the semigroup of dominant cocharacters,

$$P = \{\lambda \in \mathfrak{h}^* \mid \lambda(\alpha^\vee) \in \mathbb{Z} \text{ for all } \alpha \in \Phi\},$$

be the group of weights,

$$P_+ = \{\lambda \in \mathfrak{h}^* \mid \lambda(\alpha^\vee) \in \mathbb{Z}_+ \text{ for all } \alpha \in \Phi_+\}.$$

be the semigroup of dominant weights,

$$P^\vee = \{\gamma \in \mathfrak{h} \mid \alpha(\gamma) \in \mathbb{Z} \text{ for all } \alpha \in \Phi\},$$

be the group of coweights,

$$P_+^\vee = \{\gamma \in \mathfrak{h} \mid \alpha(\gamma) \in \mathbb{Z}_+ \text{ for all } \alpha \in \Phi_+\}.$$

be the semigroup of dominant coweights. The two terminologies are compatible in the sense that

$$\begin{aligned} X^*(T)_+ &= P_+ \cap X^*(T), \\ X_*(T)_+ &= P_+^\vee \cap X_*(T). \end{aligned}$$

The following inclusions of abelian groups hold

$$\begin{aligned} Q \subset X^*(T) \subset P \subset \mathfrak{h}^*, \\ Q^\vee \subset X_*(T) \subset P^\vee \subset \mathfrak{h}. \end{aligned}$$

For $\lambda \in X^*(T)_+$, denote by V_λ a simple representation of G or $\mathfrak{g}$ of highest weight λ.

Theorem 3.1.1 (Theorems 22.2 and 22.42 of [Mil17])**.** *Any simple, hence finite dimensional, G-module is isomorphic to V_λ for a unique $\lambda \in X^*(T)_+$. Moreover the category $G-\mathrm{Mod}$ is semisimple.*

Any finite dimensional simple $\mathfrak{g}$-module is isomorphic to V_λ for a unique $\lambda \in P_+$. Moreover the category of locally finite $\mathfrak{g}$-modules on which $\mathfrak{z}(\mathfrak{g})$ acts semisimply is semisimple.

The following numerical criterion for integrability will be useful.

Corollary 3.1.2. *Let M be a locally finite $\mathfrak{g}$-module on which the center $\mathfrak{z}(\mathfrak{g})$ of $\mathfrak{g}$ acts semisimply. Then the action of $\mathfrak{g}$ on M integrates to a representation of the algebraic group G if and only if the set of weights of M with respect to $\mathfrak{h}$ is contained in $X^*(T)$.*

3.2 Vertex algebras

Some good general references on vertex algebras are [Kac98], [FB04], [LL04], and [AM25].

Definition 3.2.1. *A vertex algebra consists of a vector space V equipped with a linear map*

$$Y(\cdot, z) : v \in V \longmapsto Y(v, z) = v(z) = \sum_{n\in\mathbb{Z}} v_{(n)} z^{-n-1} \in \mathcal{F}(V) \subset \mathrm{End}(V)[[z, z^{-1}]],$$

a distinguished vector $|0\rangle \in V$, and an endomorphism $\partial : V \longrightarrow V$. We call $Y(\cdot, z)$ the vertex operator map, $|0\rangle$ the vacuum vector and ∂ the translation operator. They are subject to the following axioms for all $v \in V$:

$$\begin{aligned} Y(|0\rangle, z) &= \mathrm{id}, \\ Y(v, z)|0\rangle &= v, \\ Y(\partial v, z) &= \partial_z Y(v, z). \end{aligned}$$

And, most importantly, such that for all $x, y \in V$ and $m, n \in \mathbb{Z}$ we have the Borcherds identities:

$$[x_{(m)}, y_{(n)}] = \sum_{i\geqslant 0} \binom{m}{i} (x_{(i)} y)_{(m+n-i)}, \tag{3.4}$$

$$(x_{(m)} y)_{(n)} = \sum_{i\geqslant 0} (-1)^i \binom{m}{i} (x_{(m-i)} y_{(n+i)} - (-1)^m y_{(m+n-i)} x_{(i)}). \tag{3.5}$$

If $x, y \in V$ we will sometimes denote the element $x_{(-1)}y$ by xy. One easily defines the notion of morphism between vertex algebras, of vertex subalgebra and of tensor product of two vertex algebras. To describe the relations between the generators, we will make use of the λ-bracket notation (see *e.g.* §2.3 of [Kac98]).

Definition 3.2.2. *For $x, y \in V$ and set*

$$[x_\lambda y] = \sum_{n\geqslant 0} x_{(n)}y\frac{\lambda^n}{n!} \in V[\lambda].$$

Remark 3.2.3. *In view of equation (3.4), we see that the λ-bracket measures the noncommutativity of the vertex algebra. For instance if x, y are such that $[x_\lambda y] = 0$ then for all $m \in \mathbb{Z}$ we have $x_{(m)}y = y_{(m)}x$. Such a vertex algebra is called commutative, and can be regarded as a differential algebra with respect to the multiplication given by $xy = x_{(-1)}y$ and the derivation ∂. It is well-known that a commutative vertex algebra is the same thing as a commutative differential algebra (see §4 of [Bor86]).*

A natural source of commutative vertex algebras are coordinate rings of jet spaces of affine schemes:

Example 3.2.4. *Let $X = \mathrm{Spec}(R)$ be an affine scheme of finite type and let $\mathcal{J}_\infty X = \mathrm{Spec}(\mathcal{J}_\infty R)$ be the jet scheme of X (see e.g. Chapter 1 of [AM25]). Concretely if $R = \mathbb{C}[x_1, ..., x_n]/(f_1, ..., f_r)$ then its jet algebra*

$$\mathcal{J}_\infty R = \mathbb{C}[\partial^j x_i, i = 1, ..., n, j \geqslant 0]/(\partial^j f_i, i = 1, ..., r, j \geqslant 0)$$

is naturally a commutative differential algebra with the derivation ∂, and hence a commutative vertex algebra.

Definition 3.2.5. *A module for a vertex algebra V consists of a vector space M together with a map*

$$Y^M(\cdot, z) : v \in V \longmapsto Y^M(v, z) = \sum_{n\in\mathbb{Z}} v^M_{(n)}z^{-n-1} \in \mathcal{F}(M)$$

such that

$$Y^M(|0\rangle, z) = \mathrm{id},$$
$$Y^M(\partial v, z) = \partial_z Y^M(v, z).$$

And such that for all $x, y \in V$ and $m, n \in \mathbb{Z}$, the following Borcherds identities hold in $\mathrm{End}(M)$:

$$[x^M_{(m)}, y^M_{(n)}] = \sum_{i\geqslant 0}\binom{m}{i}(x_{(i)}y)^M_{(m+n-i)},$$

$$(x_{(m)}y)^M_{(n)} = \sum_{i\geqslant 0}(-1)^i\binom{m}{i}(x^M_{(m-i)}y^M_{(n+i)} - (-1)^m y^M_{(m+n-i)}x^M_{(i)}).$$

When it is clear from the context, we will drop the superscript M from the notations. One easily defines the notion of a submodule, submodule generated by a subset and of simple module. Any vertex algebra is naturally a module over itself, this structure is called the adjoint module structure. Therefore, we have the notion of ideal and of quotient of vertex algebras. A vertex algebra is said to be simple if it is simple for its adjoint module structure.

Remark 3.2.6. *Note that while a commutative vertex algebra is the same thing as a commutative differential algebra, their module categories are different. In fact, with the notation introduced above and setting $m = n = -1$, the second Borcherds identity expresses the failure of associativity of the (-1)-operation:*

$$(x_{(-1)}y)^M_{(-1)} - x^M_{(-1)}y^M_{(-1)} = \sum_{i \geqslant 0} x^M_{(-2-i)}y^M_{(i)} + y^M_{(-2-i)}x^M_{(i)}.$$

Therefore, in general, M need not be a module, in the sense of commutative algebra, for the structure induced by the (-1)-product. This is closely related to the notion of loop module introduced in [Bor01]. We will encounter such an instance in this thesis: $\mathcal{D}^\kappa_G$ is naturally a vertex $\mathcal{O}_{\mathcal{J}_\infty G}$-module but not in the sense of commutative algebra.

Definition 3.2.7. *A conformal vertex algebra consists of a vertex algebra V that is $\mathbb{Z}$-graded as a vector space*

$$V = \bigoplus_{n \in \mathbb{Z}} V_n$$

further equipped with a distinguished vector $\omega \in V_2$ called a conformal vector. This vector is subject to the following axioms: we require that $\omega_{(0)} = \partial$ and that for all $n \in \mathbb{Z}$, we have the equality

$$V_n = \mathrm{Ker}(\omega_{(1)} - n\mathrm{id}).$$

Moreover, we require that there exists a complex number c called the central charge, such that:

$$[\omega_\lambda \omega] = \partial(\omega) + 2\omega\lambda + \frac{c}{12}|0\rangle\lambda^3.$$

If (V, ω) is a vertex subalgebra of a vertex algebra W, we say that W is a conformal extension of (V, ω) if (W, ω) is a conformal vertex algebra.

Remark 3.2.8. *A very natural generalization of Definitions 3.2.1 and 3.2.7 is the notion of a vertex superalgebra. The underlying vector space becomes a vector superspace (that is a $\mathbb{Z}/2\mathbb{Z}$-graded vector space) and the Borcherds identities must be modified accordingly (see e.g. [Kac98]). There is also a natural notion of conformal vertex superalgebra and it is usual to allow the grading to take half-integral values (but note that the $\mathbb{Z}/2\mathbb{Z}$-grading need not to come from the conformal weight grading). Yet, to make the statements and formulas a little less heavy to the unfamiliar reader, we made the choice in this chapter to state them in the purely even, that is, vertex algebra case.*

Let us collect some useful facts on the representation theory of conformal vertex algebras. Let V be a conformal vertex algebra with conformal vector ω, M be a V-module and $m \in M$.

Lemma 3.2.9 (Discussion following Definition 4.1.6 of [LL04])**.** *Assume that the operator $\omega^M_{(1)}$ acts semisimply on M and that M is a monogenous V-module. Then the eigenvalues of $\omega^M_{(1)}$ all belong to the same congruence class modulo $\mathbb{Z}$.*

Proposition 3.2.10 (Proposition 4.5.11 and Corollary 4.5.15 of [LL04]). *Let*

$$\mathrm{Ann}(m) = \{v \in V \mid v^M(z)m = 0\} = \{v \in V \mid v^M_{(n)}m = 0 \text{ for all } n \in \mathbb{Z}\}.$$

It is an ideal of V. In particular, if V is simple and $m \in M$ is nonzero then for all nonzero $v \in V$, $v^M(z)m \neq 0$ i.e., there exists $n \in \mathbb{Z}$ such that

$$v^M_{(n)}m \neq 0.$$

Lemma 3.2.11 (Definition 4.7.1 and Corollary 4.7.8 of [LL04]). *We say that m is a vacuum-like vector if for all $v \in V$ and $n \geqslant 0$*

$$v^M_{(n)}m = 0.$$

In that case there exists a unique morphism of V-modules

$$V \longrightarrow M$$

mapping the vacuum vector $|0\rangle$ to m. In particular, if V is simple and M is a simple V-module, it is isomorphic to the adjoint module V if and only if it contains a nonzero vacuum-like vector.

Lemma 3.2.12 (Corollary 4.7.6 of [LL04]). *The vector m is vacuum-like if and only if $\omega^M_{(0)}m = 0$.*

Let κ be a level, *i.e.*, a bilinear symmetric form on $\mathfrak{g}$ such that for all $x, y, z \in \mathfrak{g}$ the following equality of complex numbers holds $\kappa([x, y], z) = \kappa(x, [y, z])$. An important example of level is the Killing form, denoted by $\kappa_{\mathfrak{g}}$.

We now introduce the affine Kac–Moody algebra associated to our data. Let $\hat{\mathfrak{g}}^\kappa$ be the central extension of $\mathfrak{g}((t))$ by $\mathbb{C}\mathbb{1}$ given by κ. By definition, $\hat{\mathfrak{g}}^\kappa$ fits in the exact sequence of Lie algebras

$$0 \longrightarrow \mathbb{C}\mathbb{1} \longrightarrow \hat{\mathfrak{g}}^\kappa \longrightarrow \mathfrak{g}((t)) \longrightarrow 0 \tag{3.6}$$

where $\mathbb{1}$ is central in $\hat{\mathfrak{g}}^\kappa$, and for all $x, y \in \mathfrak{g}$, $m, n \in \mathbb{Z}$, we have

$$[xt^m, yt^n] = [x, y]t^{m+n} + m\delta_{m+n,0}\kappa(x, y)\mathbb{1}.$$

Note that the exact sequence (3.6) splits over the Lie subalgebra $\mathfrak{g}[[t]] \subset \mathfrak{g}((t))$, hence $\mathfrak{g}[[t]] \oplus \mathbb{C}\mathbb{1}$ is a Lie subalgebra of $\hat{\mathfrak{g}}^\kappa$.

A $\hat{\mathfrak{g}}^\kappa$-module M is said to be smooth if for all $m \in M$ there exists $n \geqslant 0$ such that $t^n\mathfrak{g}[[t]]m = 0$ and $\mathbb{1}$ acts as the identity. As every module we will encounter will be smooth, we denote again by $\hat{\mathfrak{g}}^\kappa$−Mod the category of smooth $\hat{\mathfrak{g}}^\kappa$-modules.

There is a natural induction procedure that produces a smooth $\hat{\mathfrak{g}}^\kappa$-module out of a $\mathfrak{g}$-module V. We first see V as a $\mathfrak{g}[[t]] \oplus \mathbb{C}\mathbb{1}$-module by having $\mathbb{1}$ act as the identity and $\mathfrak{g}[[t]]$ act through the projection $\mathfrak{g}[[t]] \to \mathfrak{g}$ that maps t to zero. We then induce the resulting module to $\hat{\mathfrak{g}}^\kappa$. This defines a functor

$$\operatorname*{Ind}_{\hat{\mathfrak{g}}^\kappa \leftarrow \mathfrak{g}[[t]] \to \mathfrak{g}} : \mathfrak{g}\text{−Mod} \longrightarrow \hat{\mathfrak{g}}^\kappa\text{−Mod}.$$

If $\lambda \in P_+$ and V_λ denotes the simple highest weight $\mathfrak{g}$-module of weight λ, let $|\lambda\rangle \in V_\lambda$ be a highest weight vector, we call the resulting module

$$V_\lambda^\kappa = \operatorname*{Ind}_{\hat{\mathfrak{g}}^\kappa \leftarrow \mathfrak{g}[[t]] \to \mathfrak{g}} V_\lambda \tag{3.7}$$

the Weyl module of highest weight λ. The $\hat{\mathfrak{g}}^\kappa$-module $V^\kappa(\mathfrak{g}) := V_0^\kappa$ has the structure of a vertex algebra and is generated, as a vertex algebra, by elements of $\mathfrak{g}$ ([FZ92b]). For all $x, y \in \mathfrak{g}$, we have

$$[x_\lambda y] = [x, y] + \kappa(x, y)\lambda. \tag{3.8}$$

The vertex algebra $V^\kappa(\mathfrak{g})$ is called the universal affine vertex algebra associated with $\mathfrak{g}$ at level κ.

The categories $\hat{\mathfrak{g}}^\kappa$−Mod and $V^\kappa(\mathfrak{g})$−Mod are isomorphic (see *e.g.* Theorem 6.2.13 of [LL04]). Explicitly, given a smooth $\hat{\mathfrak{g}}^\kappa$-module M, one sets for all $x \in \mathfrak{g}$

$$Y^M(x, z) = \sum_{n \in \mathbb{Z}} (xt^n) z^{-n-1}.$$

Weyl modules satisfy the following universal property:

Proposition 3.2.13. *Let $\lambda \in P_+$, M be a $V^\kappa(\mathfrak{g})$-module and $m \in M$ be a highest weight vector for $\mathfrak{g}$ of weight λ such that $t\mathfrak{g}[[t]]m = 0$ and $\dim(\mathcal{U}(\mathfrak{g})m) < \infty$. Then there exists a unique morphism of $V^\kappa(\mathfrak{g})$-modules $V_\lambda^\kappa \longrightarrow M$ mapping $|\lambda\rangle$ to m.*

Proof. This follows from the definition of Weyl modules and the fact that $\mathrm{Hom}_{\mathfrak{g}}(V_\lambda, M)$ is naturally in bijection with the highest weight vector for $\mathfrak{g}$ of weight λ in M that span a finite dimensional $\mathfrak{g}$-module. □

Definition 3.2.14. *We denote by*

$$\mathfrak{g}\text{−Mod}^G$$

the full subcategory of $\mathfrak{g}$−Mod *consisting of locally finite $\mathfrak{g}$-modules for which the action integrates to a representation of the algebraic group G. Define the Kazhdan–Lusztig category*

$$V^\kappa(\mathfrak{g})\text{−Mod}^{\mathcal{J}_\infty G}$$

to be the full subcategory of the category of $V^\kappa(\mathfrak{g})$-modules consisting of objects M such that

- *as a $\mathfrak{g}$-module, M is locally finite and the action of $\mathfrak{g}$ integrates to a representation of the algebraic group G i.e., $M \in \mathfrak{g}\text{−Mod}^G$,*
- *the Lie subalgebra $t\mathfrak{g}[[t]]$ of $\hat{\mathfrak{g}}^\kappa$ acts locally nilpotently.*

Remark 3.2.15. *Because G is connected the functor* Lie *defines an isomorphism of categories*

$$\mathrm{Lie} : G\text{−Mod} \simeq \mathfrak{g}\text{−Mod}^{G}$$

which we use to identity them from now on. When the group G is a connected, simple and simply connected group, say for instance $G = \mathrm{SL}_2$, then for any locally finite $\mathfrak{g}$-module the action automatically integrates to a representation of the algebraic group G. As alluded to in the introduction, the Kazhdan–Lusztig category $V^{\kappa}(\mathfrak{g})\text{−Mod}^{\mathfrak{J}_\infty G}$ consists precisely of $\hat{\mathfrak{g}}^{\kappa}$-modules such that the action of $\mathfrak{g}[[t]]$ integrates to an algebraic representation of $\mathfrak{J}_\infty G$. To make sense of this statement and prove it, the formalism of pro-algebraic groups and their pro-Lie algebras is useful (see e.g. §4.4 of [Kum02]).

Example 3.2.16. *When $\mathfrak{g} = \mathfrak{h} = \mathfrak{z}(\mathfrak{g})$ is abelian and κ is generic i.e., nondegenerate (see Definition 4.3.1 below), then $V^{\kappa}(\mathfrak{h})$ is called a Heisenberg vertex algebra and Weyl modules are more commonly referred to as Fock spaces.*

An important feature of Fock spaces for us will be the following well-known realization

Proposition 3.2.17 (Proposition 6.3.4 of [LL04])**.** *Let $\mathfrak{h}$ be a finite dimensional abelian Lie algebra and κ a nondegenerate form on $\mathfrak{h}$ and $V^{\kappa}(\mathfrak{h})$ the associated Heisenberg vertex algebra. Let $(h^i)_{1\leqslant i\leqslant \dim\mathfrak{h}}$ be an orthonormal basis of $\mathfrak{h}$ with respect to κ. There exists on the $\mathbb{C}$-vector space*

$$\mathcal{F}^{\kappa}_{\lambda} = \mathbb{C}[(h^i t^{-n})_{1\leqslant i\leqslant \dim\mathfrak{h}, n>0}]|\lambda\rangle$$

a structure of smooth $\hat{\mathfrak{h}}^{\kappa}$-module such that for all $i \in \{1, \dots, \dim\mathfrak{h}\}$ and all $n > 0$

- *$h^i t^0$ act as the multiplication by $\lambda(h^i)$,*
- *$h^i t^{-n}$ act as the multiplication by $h^i t^{-n}$,*
- *$h^i t^n$ act as $n\frac{\partial}{\partial(h^i t^{-n})}$,*
- *$\mathbb{1}$ acts as the identity.*

Endowing $\mathcal{F}^{\kappa}_{\lambda}$ with this structure of $\hat{\mathfrak{h}}^{\kappa}$-module, there exists a unique isomorphism of $\hat{\mathfrak{h}}^{\kappa}$-modules

$$V^{\kappa}_{\lambda} \longrightarrow \mathcal{F}^{\kappa}_{\lambda}$$

mapping $|\lambda\rangle \in V^{\kappa}_{\lambda}$ to $1|\lambda\rangle \in \mathcal{F}^{\kappa}_{\lambda}$.

Chapter 4

Chiral differential operators on an algebraic group

4.1 Preliminary results

There are essentially two ways of thinking of $\mathcal{O}_G$, the coordinate ring of G, as a G-module. For $\alpha \in G$, $f \in \mathcal{O}_G$ and $g \in G$ set

$$\lambda_\alpha(f)(g) = f(\alpha^{-1}g),$$
$$\rho_\alpha(f)(g) = f(g\alpha).$$

That defines two group morphisms

$$\lambda : \alpha \in G \longmapsto \lambda_\alpha \in \mathrm{GL}(\mathcal{O}_G),$$
$$\rho : \alpha \in G \longmapsto \rho_\alpha \in \mathrm{GL}(\mathcal{O}_G).$$

They each make $\mathcal{O}_G$ into a G-module and commute with one another so they make $\mathcal{O}_G$ into a $G \times G$-module. We will refer to this $G \times G$-module structure on $\mathcal{O}_G$ as the regular representation of the group G. Explicitly, if $\alpha, \beta \in G$ and $f \in \mathcal{O}_G$ then define $(\alpha, \beta) \cdot f \in \mathcal{O}_G$ to take on each $g \in G$ the value

$$((\alpha, \beta) \cdot f)(g) = f(\alpha^{-1}g\beta).$$

As a vector space, $\mathfrak{g}$ is the tangent space of G at e, the identity element. Recall that it consists of the linear maps $x : \mathcal{O}_G \longrightarrow \mathbb{C}$ satisfying for all $f, g \in \mathcal{O}_G$ the equality of complex numbers

$$x(fg) = f(e)x(g) + x(f)g(e).$$

Let $\mathrm{Der}(\mathcal{O}_G)$ denote the Lie algebra of global vector fields on the group G *i.e.*, the derivations of the algebra $\mathcal{O}_G$. We say that $X \in \mathrm{Der}(\mathcal{O}_G)$ is a left-invariant derivation if for all $\alpha \in G$, we have the equality $\lambda_\alpha \circ X = X \circ \lambda_\alpha$. They form a Lie subalgebra of

$\mathrm{Der}(\mathcal{O}_G)$ that we denote by $\mathrm{Der}(\mathcal{O}_G)^\lambda$. We define in a similar fashion right-invariant derivations and denote their set by $\mathrm{Der}(\mathcal{O}_G)^\rho$. Any derivation $X \in \mathrm{Der}(\mathcal{O}_G)$ can be made into an element of the tangent space at the identity so we have two linear maps

$$\mathrm{ev}_e \circ \cdot : \mathrm{Der}(\mathcal{O}_G)^\lambda \longrightarrow T_e(G) \longleftarrow \mathrm{Der}(\mathcal{O}_G)^\rho : -\mathrm{ev}_e \circ \cdot$$

where $\mathrm{ev}_e : f \in \mathcal{O}_G \mapsto f(e) \in \mathbb{C}$ is the evaluation at the identity. Both these maps admit inverses that are defined for $x \in \mathfrak{g}$ by

$$x_L = (\mathrm{id} \circ x) \circ \Delta \in \mathrm{Der}(\mathcal{O}_G)^\lambda, \tag{4.1}$$
$$x_R = (-x \otimes \mathrm{id}) \circ \Delta \in \mathrm{Der}(\mathcal{O}_G)^\rho, \tag{4.2}$$

where $\Delta : \mathcal{O}_G \longrightarrow \mathcal{O}_G \otimes \mathcal{O}_G$ is the comultiplication map. More explicitly, we have for all $x \in \mathfrak{g}$, $f \in \mathcal{O}_G$ and $g \in G$ the equalities of complex numbers

$$x_L(f)(g) = x(\lambda_{g^{-1}} f), \tag{4.3}$$
$$x_R(f)(g) = -x(\rho_g f). \tag{4.4}$$

Recall that by definition the Lie algebra structure of $\mathfrak{g}$ is such that for all $x, y \in \mathfrak{g}$ the following equality holds in $\mathrm{Der}(\mathcal{O}_G)$

$$[x_L, y_L] = [x, y]_L. \tag{4.5}$$

Because we have put a minus sign in formula (4.2) we have the equality in $\mathrm{Der}(\mathcal{O}_G)$

$$[x_R, y_R] = [x, y]_R. \tag{4.6}$$

This shows that we have defined two Lie algebra maps

$$x \in \mathfrak{g} \longmapsto x_L \in \mathrm{Der}(\mathcal{O}_G) \subset \mathfrak{gl}(\mathcal{O}_G),$$
$$x \in \mathfrak{g} \longmapsto x_R \in \mathrm{Der}(\mathcal{O}_G) \subset \mathfrak{gl}(\mathcal{O}_G),$$

whose images commute, that is for all $x, y \in \mathfrak{g}$ we have in $\mathrm{Der}(\mathcal{O}_G)$ the equality

$$[x_L, y_R] = 0. \tag{4.7}$$

This endows $\mathcal{O}_G$ with the structure of a $\mathfrak{g} \oplus \mathfrak{g}$-module, explicitly, for all $x, y \in \mathfrak{g}$ and $f \in \mathcal{O}_G$ define

$$(x, y) \cdot f = x_L(y_R(f)) = y_R(x_L(f)).$$

This is, up to exchanging the factors, the differential of the regular representation, since for all $x, y \in \mathfrak{g}$ and $f \in \mathcal{O}_G$ we have

$$\mathrm{d}\lambda(y)(f) = y_R(f), \tag{4.8}$$
$$\mathrm{d}\rho(x)(f) = x_L(f), \tag{4.9}$$

where $\mathrm{d}\lambda : \mathfrak{g} \longrightarrow \mathfrak{gl}(\mathcal{O}_G)$ and $\mathrm{d}\rho : \mathfrak{g} \longrightarrow \mathfrak{gl}(\mathcal{O}_G)$ denote the differential at the identity of the morphism of groups λ and ρ (see *e.g.* §23.4 of [TY05]).

The coordinate ring $\mathcal{O}_{\mathcal{J}_\infty G}$ of the algebraic group $\mathcal{J}_\infty G$ is a differential graded commutative algebra *i.e.*, a commutative vertex algebra (recall Example 3.2.4). Its Lie algebra is $\mathfrak{g}[[t]]$ and the bracket is given for all $x, y \in \mathfrak{g}$ and $m, n \geqslant 0$ by

$$[xt^m, yt^n] = [x, y]t^{m+n}.$$

By the same construction as before, there are two Lie algebra maps from $\mathfrak{g}[[t]]$ to $\mathrm{Der}(\mathcal{O}_{\mathcal{J}_\infty G})$ corresponding to left-invariant and right-invariant vector fields. In particular, $\mathcal{O}_{\mathcal{J}_\infty G}$ is a $\mathfrak{g}[[t]] \oplus \mathfrak{g}[[t]]$-module. Let us view elements of $\mathfrak{g}[[t]]$ as left-invariant vector fields of $\mathcal{J}_\infty G$. This makes $\mathcal{O}_{\mathcal{J}_\infty G}$ into a $\mathfrak{g}[[t]]$-module. By having $\mathbb{1}$ act as the identity we make it into a $\mathfrak{g}[[t]] \oplus \mathbb{C}\mathbb{1}$-module.

Define the vertex algebra of chiral differential operators on the algebraic group G to be the induced module

$$\mathcal{D}_G^\kappa = \mathrm{Ind}_{\mathfrak{g}[[t]]\oplus\mathbb{C}\mathbb{1}}^{\hat{\mathfrak{g}}^\kappa} \mathcal{O}_{\mathcal{J}_\infty G} = \mathcal{U}(\hat{\mathfrak{g}}^\kappa) \otimes_{\mathcal{U}(\mathfrak{g}[[t]])\oplus\mathbb{C}\mathbb{1}} \mathcal{O}_{\mathcal{J}_\infty G}. \tag{4.10}$$

Note that, as vector spaces,

$$\mathcal{D}_G^\kappa = V^\kappa(\mathfrak{g}) \otimes \mathcal{O}_{\mathcal{J}_\infty G}.$$

Since both $V^\kappa(\mathfrak{g})$ and $\mathcal{O}_{\mathcal{J}_\infty G}$ are $\mathbb{Z}_{\geqslant 0}$-graded vector spaces, then so is $\mathcal{D}_G^\kappa$ by setting

$$(\mathcal{D}_G^\kappa)_n = \bigoplus_{k+k'=n} V^\kappa(\mathfrak{g})_k \otimes (\mathcal{O}_{\mathcal{J}_\infty G})_{k'}$$

for each $n \in \mathbb{Z}_{\geqslant 0}$.

According to [AG02] and [GMS01] (see also §3.4 of [AM25]) there exists a unique vertex algebra structure on $\mathcal{D}_G^\kappa$ such that the injective maps

$$\pi_L : x \in V^\kappa(\mathfrak{g}) \longmapsto x \otimes 1 \in \mathcal{D}_G^\kappa$$
$$\iota : f \in \mathcal{O}_{\mathcal{J}_\infty G} \longmapsto 1 \otimes f \in \mathcal{D}_G^\kappa$$

are vertex algebra morphisms satisfying for all $x \in \mathfrak{g}$ and $f \in \mathcal{O}_G$ the following Leibniz rule

$$[\pi_L(x)_\lambda \iota(f)] = \iota(x_L(f)).$$

As it is harmless we will forget the ι in the notations so that the Leibniz rule reads

$$[\pi_L(x)_\lambda f] = x_L(f).$$

To summarize, $\mathcal{D}_G^\kappa$ is generated as a vertex algebra by elements of the form $\pi_L(x)$ for $x \in \mathfrak{g}$ and regular functions on the group G in such a way that for all $x, y \in \mathfrak{g}$ and $f, g \in \mathcal{O}_G$ we have

$$[\pi_L(x)_\lambda \pi_L(y)] = \pi_L([x, y]) + \kappa(x, y)\lambda, \tag{4.11}$$
$$[\pi_L(x)_\lambda f] = x_L(f), \tag{4.12}$$
$$[f_\lambda g] = 0. \tag{4.13}$$

Any element of $\mathcal{D}_G^\kappa$ can be written as a linear combination of elements of the form

$$\pi_L(x^{i_1})_{(-m_1)} \dots \pi_L(x^{i_r})_{(-m_r)} f^{j_1}_{(-n_1)} \cdots f^{j_s}_{(-n_s)} |0\rangle \tag{4.14}$$

where $r, s \in \mathbb{Z}_{\geqslant 0}, m_1, \dots, m_r, n_1, \dots, n_s > 0, x^{i_1}, \dots, x^{i_r} \in \mathfrak{g}, f^{j_1}, \dots, f^{j_s} \in \mathcal{O}_G$.

Remark 4.1.1. *Note that since $\mathcal{O}_G$ is of finite type, $\mathcal{D}_G^\kappa$ is strongly finitely generated. Moreover, if G is simple, then κ can be identified with a complex number (see §4.2.2). Thinking of it as a formal parameter leads to an interesting point of view on $\mathcal{D}_G^\kappa$ as coming from a deformable family of vertex algebras (see §3.1 of [CL19] for the terminology). Notice that all the relations* (4.11)-(4.13) *between the generators of $\mathcal{D}_G^\kappa$ are polynomial functions of the variable κ. In fact, this is unsurprising as one can define by generators and relations a certain* 1*-parameter vertex algebra $\mathcal{D}_G^\kappa$ over the commutative ring $\mathbb{C}[\kappa]$ and recover $\mathcal{D}_G^{\kappa_0}$ at any given level κ_0 by specialization.*

We can describe the degree one part of $\mathcal{D}_G^\kappa$, as vector spaces we have

$$(\mathcal{D}_G^\kappa)_1 = \mathcal{O}_G \otimes \mathfrak{g} \oplus (\mathcal{O}_{\mathcal{J}_\infty G})_1.$$

Let Ω_G^1 be the $\mathcal{O}_G$-module of global differential 1-forms on the algebraic group G. The maps

$$f \otimes x \in \mathcal{O}_G \otimes \mathfrak{g} \longmapsto f x_L \in \mathrm{Der}(\mathcal{O}_G),$$

and

$$f \partial(g) \in (\mathcal{O}_{\mathcal{J}_\infty G})_1 \longmapsto f \mathrm{d}g \in \Omega_G^1$$

are isomorphisms of $\mathcal{O}_G$-modules. So, in fact, as vector spaces

$$(\mathcal{D}_G^\kappa)_1 = \mathrm{Der}(\mathcal{O}_G) \oplus \Omega_G^1.$$

Beware that this is not a decomposition as $\mathcal{O}_G$-modules since $(\mathcal{D}_G^\kappa)_1$ is not endowed with a $\mathcal{O}_G$-module structure by the natural map

$$f \in \mathcal{O}_G \longmapsto f_{(-1)} \in \mathrm{End}(\mathcal{D}_G^\kappa).$$

In fact, while this is a perfectly well-defined map, it is not a morphism of algebras (recall Remark 3.2.6).

Finally, recall that $\mathrm{Der}(\mathcal{O}_G)$ acts on Ω_G^1 by the Lie derivative as follows. Given $\omega \in \Omega_G^1$ and $\theta, \theta_1 \in \mathrm{Der}(\mathcal{O}_G)$ we set

$$(\mathrm{Lie}(\theta) \cdot \omega)(\theta_1) = \theta(\omega(\theta_1)) - \omega([\theta, \theta_1]).$$

In [AM21], Arakawa and Moreau introduced a very convenient criterion to show that a vertex algebra V is simple in terms of the geometry of its associated Poisson scheme $\tilde{X}_V$ (we refer the unfamiliar reader to §10.2 for the definition).

Proposition 4.1.2 (Corollary 9.3 of *loc.cit.*)**.** *If $\tilde{X}_V$ is smooth, reduced and symplectic then V is simple.*

One easily sees from the generating family (4.14) of $\mathcal{D}_G^\kappa$ that its associated Poisson scheme is the cotangent bundle T^*G of G. Because this is a smooth, reduced and symplectic scheme, we have:

Proposition 4.1.3. *For any level κ, the vertex algebra $\mathcal{D}_G^\kappa$ is simple.*

As mentioned in the introduction, the right action chiralizes, but the level for which it does develops an anomaly.

Theorem 4.1.4. *There exists a unique level κ^* such that there exists a $\mathbb{Z}$-graded injective vertex algebra map*

$$\pi_R : V^{\kappa^*}(\mathfrak{g}) \longrightarrow \mathcal{D}_G^\kappa$$

satisfying for all $x, y \in \mathfrak{g}$ and $f \in \mathcal{O}_G$

$$[\pi_R(x)_\lambda \pi_L(y)] = 0, \tag{4.15}$$

$$[\pi_R(x)_\lambda f] = x_R(f). \tag{4.16}$$

The level κ^* is related to κ via the formula

$$\kappa^* = -\kappa_\mathfrak{g} - \kappa,$$

in particular

$$(\kappa^*)^* = \kappa.$$

Explicitly, let $(x^i)_{1\leqslant i\leqslant \dim\mathfrak{g}}$ be a basis of $\mathfrak{g}$ and $(\omega^i)_{1\leqslant i\leqslant \dim\mathfrak{g}}$ the $\mathcal{O}_G$-basis of Ω_G^1 dual to the $\mathcal{O}_G$-basis of $\mathrm{Der}(\mathcal{O}_G)$ given by the $(x_L^i)_{1\leqslant i\leqslant \dim\mathfrak{g}}$. Then, for all $1 \leqslant i \leqslant \dim\mathfrak{g}$, letting

$$x_R^i = \sum_p f_p^i x_L^p,$$

where $(f_p^i) \in \mathrm{GL}(\mathcal{O}_G)$, we have the formula

$$\pi_R(x^i) = x_R^i + \sum_q \left(\sum_p \kappa^*(x^p, x^q) f_p^i \right) \omega^q. \tag{4.17}$$

Proof. To see why formula (4.17) satisfies all the requirements, see §4 of [GMS01], Theorem 4.4 of [AG02] or Theorem 3.3 of [AM25].

Conversely, assume that such a level κ^* exists. Fix $(x^i)_{1\leqslant i\leqslant \dim\mathfrak{g}}$ a basis of $\mathfrak{g}$, then $(x_L^i)_{1\leqslant i\leqslant \dim\mathfrak{g}}$ is a $\mathcal{O}_G$-basis of $\mathrm{Der}(\mathcal{O}_G)$, denote by $(\omega^i)_{1\leqslant i\leqslant \dim\mathfrak{g}} \in \Omega_G^1$ its $\mathcal{O}_G$-dual basis. Plainly this means we have for all $1 \leqslant i, j \leqslant \dim\mathfrak{g}$ the equality

$$\omega^i(x_L^j) = \delta_{i,j}.$$

We now try to find a formula for the embedding $\pi_R : V^{\kappa^*}(\mathfrak{g}) \longrightarrow \mathcal{D}_G^\kappa$. Because it has to be a morphism of $\mathbb{Z}$-graded vertex algebras, we can write for all i

$$\pi_R(x^i) = \sum_j f_j^i \pi_L(x^j) + \sum_j g_j^i \omega^j.$$

where $f^i_j, g^i_j \in \mathcal{O}_G$. In all the following computations, we repeatedly use the Borcherds identities (3.4) and (3.5) and equations (4.11), (4.12), (4.13).

Let $f \in \mathcal{O}_G$, we must have by assumption on the one hand

$$[\pi_R(x^i)_\lambda f] = x^i_R(f).$$

On the other hand

$$\begin{aligned}
[\pi_R(x^i)_\lambda f] &= \left[\sum_j f^i_j \pi_L(x^j) + \sum_j g^i_j \omega^j_\lambda f\right] \\
&= \left[\sum_j f^i_j \pi_L(x^j)_\lambda f\right] \\
&= \sum_j [f^i_j \pi_L(x^j)_\lambda f] \\
&= \sum_j f^i_j x^j_L(f).
\end{aligned}$$

Hence, the f^i_j are uniquely determined by the equality in $\mathrm{Der}(\mathcal{O}_G)$:

$$x^i_R = \sum_j f^i_j x^j_L.$$

By assumption, on the one hand

$$[\pi_R(x^\alpha)_\lambda \pi_L(x^\beta)] = 0$$

and on the other hand

$$\begin{aligned}
[\pi_R(x^\alpha)_\lambda \pi_L(x^\beta)] &= \left[\sum_j f^\alpha_j \pi_L(x^j) + \sum_j g^\alpha_j \omega^j_\lambda \pi_L(x^\beta)\right] \\
&= \sum_j [f^\alpha_j \pi_L(x^j)_\lambda \pi_L(x^\beta)] + [g^\alpha_j \omega^j_\lambda \pi_L(x^\beta)] \\
&= \sum_j f^\alpha_j \pi_L([x^j, x^\beta]) - (x^\beta)_L(f^\alpha_j)\pi_L(x^j) - \partial((x^j)_L(x^\beta_L)(f^\alpha_j)) \\
&\quad - \mathrm{Lie}(x^\beta_L)(g^\alpha_j \omega^j) + \partial(g^\alpha_\beta)\delta_{j,\beta} + \left(\sum_j f^\alpha_j \kappa(x^j, x^\beta) - (x^j)_L(x^\beta)_L(f^\alpha_j) + \delta_{j,\beta} g^\alpha_\beta\right)\lambda.
\end{aligned}$$

Which implies the two equalities

$$\sum_j f^\alpha_j \pi_L([x^j, x^\beta]) - (x^\beta)_L(f^\alpha_j)\pi_L(x^j) - \partial((x^j)_L(x^\beta_L)(f^\alpha_j)) - \mathrm{Lie}(x^\beta_L)(g^\alpha_j \omega^j) + \partial(g^\alpha_\beta)\delta_{j,\beta} = 0,$$
$$\sum_j f^\alpha_j \kappa(x^j, x^\beta) - (x^j)_L(x^\beta)_L(f^\alpha_j) + \delta_{j,\beta} g^\alpha_\beta = 0,$$

better written as

$$\partial(g^\alpha_\beta) = -\left(\sum_j f^\alpha_j \pi_L([x^j, x^\beta]) - (x^\beta)_L(f^\alpha_j)\pi_L(x^j) - \partial((x^j)_L(x^\beta_L)(f^\alpha_j)) - \mathrm{Lie}(x^\beta_L)(g^\alpha_j \omega^j)\right),$$
$$g^\alpha_\beta = \sum_j (x^j_L(x^\beta_L(f^\alpha_j)) - \kappa(x^j, x^\beta) f^\alpha_j). \tag{4.18}$$

The second equality shows that the coefficient g^α_β is uniquely determined. To summarize, we have no choice for the formula of π_R, for each i we have the equality

$$\pi_R(x^i) = \sum_j f^i_j \pi_L(x^j) + \sum_j \left(\sum_k (x^k)_L(x^j)_L(f^i_k) - \kappa(x^k, x^j) f^i_k\right)\omega^j.$$

Now we need to see if that determines κ^* uniquely. By assumption for all α,β we have, on the one hand

$$[\pi_R(x^\alpha)_\lambda \pi_R(x^\beta)] = \pi_R([x^\alpha, x^\beta]) + \kappa^*(x^\alpha, x^\beta)\lambda,$$

and on the other hand

$$\begin{aligned}\pi_R(x^\alpha)_{(1)}\pi_R(x^\beta) &= \left(\sum_j f^\alpha_j \pi_L(x^j) + g^\alpha_j \omega^j\right)_{(1)} \left(\sum_k f^\beta_k \pi_L(x^k) + g^\beta_k \omega^k\right)\\
&= \left(\sum_j f^\alpha_j \pi_L(x^j)\right)_{(1)} \left(\sum_k f^\beta_k \pi_L(x^k)\right) + \left(\sum_j f^\alpha_j \pi_L(x^j)\right)_{(1)} \left(\sum_k g^\beta_k \omega^k\right)\\
&+ \left(\sum_j g^\alpha_j \omega^j\right)_{(1)} \left(\sum_k f^\beta_k \pi_L(x^k)\right) + \left(\sum_j g^\alpha_j \omega^j\right)_{(1)} \left(\sum_k g^\beta_k \omega^k\right).\end{aligned}$$

By commutativity of $\mathcal{O}_{\mathcal{J}_\infty G}$, the last sum is equals zero. We compute each of the remaining sum separately, for each j, k we have

$$\begin{aligned}(f^\alpha_j \pi_L(x^j))_{(1)}(f^\beta_k \pi_L(x^k)) &= f^\alpha_j f^\beta_k \kappa(x^j, x^k) - f^\alpha_j x^k_L(x^j_L(f^\beta_k)) - x^j_L(f^\beta_k) x^k_L(f^\alpha_j) - f^\beta_k x^j_L(x^k_L(f^\alpha_j)),\\
(f^\alpha_j \pi_L(x^j))_{(1)}(g^\beta_k \omega^k) &= f^\alpha_j g^\beta_k \delta_{j,k},\\
(g^\alpha_j \omega^j)_{(1)}(f^\beta_k \pi_L(x^k)) &= g^\alpha_j f^\beta_k \delta_{j,k}.\end{aligned}$$

So we have the equalities

$$\begin{aligned}\kappa^*(x^\alpha, x^\beta) &= 2\left(\sum_j f^\alpha_j g^\beta_j\right) + \sum_{j,k} f^\alpha_j f^\beta_k \kappa(x^j, x^k) - f^\alpha_j x^k_L(x^j_L(f^\beta_k)) - x^j_L(f^\beta_k) x^k_L(f^\alpha_j) - f^\beta_k x^j_L(x^k_L(f^\alpha_j))\\
&= 2\left(\sum_{j,k} f^\alpha_j x^k_L(x^j_L(f^\beta_k)) - f^\alpha_j f^\beta_k \kappa(x^k, x^j)\right)\\
&+ \sum_{j,k} f^\alpha_j f^\beta_k \kappa(x^j, x^k) - f^\alpha_j x^k_L(x^j_L(f^\beta_k)) - x^j_L(f^\beta_k) x^k_L(f^\alpha_j) - f^\beta_k x^j_L(x^k_L(f^\alpha_j))\\
&= \sum_{j,k} -f^\alpha_j f^\beta_k \kappa(x^j, x^k) + f^\alpha_j x^k_L(x^j_L(f^\beta_k)) - x^j_L(f^\beta_k) x^k_L(f^\alpha_j) - f^\beta_k x^j_L(x^k_L(f^\alpha_j)).\end{aligned}$$

Better written as

$$\kappa^*(x^\alpha, x^\beta) = -\left(\sum_{j,k} f_j^\alpha f_k^\beta \kappa(x^j, x^k)\right) + \left(\sum_{j,k} f_j^\alpha x_L^k(x_L^j(f_k^\beta)) - x_L^j(f_k^\beta) x_L^k(f_j^\alpha) - f_k^\beta x_L^j(x_L^k(f_j^\alpha))\right) \tag{4.19}$$

Introduce the structure constants

$$[x^i, x^j] = \sum_k c_{i,j}^k x^k,$$

we have that

$$\kappa_{\mathfrak{g}}(x^i, x^j) = \sum_{p,q} c_{j,q}^p c_{i,p}^q = \sum_{p,q} c_{i,q}^p c_{j,p}^q. \tag{4.20}$$

The following equation holds for all i, j, k:

$$x_L^i(f_k^j) = -\sum_l c_{i,l}^k f_l^j. \tag{4.21}$$

It follows from the commutation relation in $\mathrm{Der}(\mathcal{O}_G)$

$$[x_L^i, x_R^j] = 0.$$

In fact, writing x_R^j in terms of the x_L^k yields

$$\begin{aligned} 0 = \sum_k [x_L^i, f_k^j x_L^k] = \sum_k [x_L^i, f_k^j] x_L^k + f_k^j [x_L^i, x_L^k] &= \sum_k \left(x_L^i(f_k^j) x_L^k + f_k^j \sum_p c_{i,k}^p x_L^p \right) \\ &= \sum_k (x_L^i(f_k^j) x_L^k) + \sum_p \left(\sum_k c_{i,k}^p f_k^j \right) x_L^p. \end{aligned}$$

Looking at the coefficient in front of x_L^k for each k we have proven equality (4.21). As a direct consequence of this equality, we have

$$\begin{aligned} f_j^\alpha x_L^k(x_L^j(f_k^\beta)) &= \sum_{l,l'} c_{j,l}^k c_{k,l'}^l f_j^\alpha f_{l'}^\beta, \\ x_L^j(f_k^\beta) x_L^k(f_j^\alpha) &= \sum_{l,l'} c_{j,l}^k c_{k,l'}^j f_{l'}^\alpha f_l^\beta, \\ f_k^\beta x_L^j(x_L^k(f_j^\alpha)) &= \sum_{l,l'} c_{k,l}^j c_{j,l'}^l f_{l'}^\alpha f_k^\beta. \end{aligned}$$

Hence

$$\begin{aligned} &\sum_{j,k} f_j^\alpha x_L^k(x_L^j(f_k^\beta)) - x_L^j(f_k^\beta) x_L^k(f_j^\alpha) - f_k^\beta x_L^j(x_L^k(f_j^\alpha)) \\ &\qquad = \sum_{j,k,l,l'} c_{j,l}^k c_{k,l'}^l f_j^\alpha f_{l'}^\beta - c_{j,l}^k c_{k,l'}^j f_{l'}^\alpha f_l^\beta - c_{k,l}^j c_{j,l'}^l f_{l'}^\alpha f_k^\beta. \end{aligned}$$

Now the key point is that for all i, j, because of the minus sign in equation (4.2) we have

$$f^i_j(e) = -\delta_{i,j}. \tag{4.22}$$

Because the left-hand side of equation (4.19) is a constant function we can evaluate everything at the identity element e to compute $\kappa^*(x^\alpha, x^\beta)$. By using repeatedly equations (4.20) and (4.22) we have

$$-\sum_{j,k} f^\alpha_j(e) f^\beta_k(e)\kappa(x^j, x^k) = -\kappa(x^\alpha, x^\beta)$$

and

$$\begin{aligned}
&\left(\sum_{j,k,l,l'} c^k_{j,l}c^l_{k,l'}f^\alpha_j f^\beta_{l'} - c^k_{j,l}c^j_{k,l'}f^\alpha_{l'}f^\beta_l - c^j_{k,l}c^l_{j,l'}f^\alpha_{l'}f^\beta_k\right)(e)\\
&\qquad = \sum_{j,k,l,l'} c^k_{j,l}c^l_{k,l'}\delta_{\alpha,j}\delta_{\beta,l'} - c^k_{j,l}c^j_{k,l'}\delta_{\alpha,l'}\delta_{\beta,l} - c^j_{k,l}c^l_{j,l'}\delta_{\alpha,l'}\delta_{\beta,k}\\
&\qquad = \sum_{k,l} c^k_{\alpha,l}c^l_{k,\beta} - \sum_{j,k} c^k_{j,\beta}c^j_{k,\alpha} - \sum_{j,l} c^j_{\beta,l}c^l_{j,\alpha}\\
&\qquad = -\sum_{k,l} c^k_{\alpha,l}c^l_{\beta,k} - \sum_{j,k} c^j_{k,\alpha}c^k_{j,\beta} + \sum_{j,l} c^l_{\alpha,j}c^j_{\beta,l}\\
&\qquad = -\kappa_{\mathfrak{g}}(x^\alpha, x^\beta) - \kappa_{\mathfrak{g}}(x^\alpha, x^\beta) + \kappa_{\mathfrak{g}}(x^\alpha, x^\beta) = -\kappa_{\mathfrak{g}}(x^\alpha, x^\beta).
\end{aligned}$$

That finally proves

$$\kappa^* = -\kappa - \kappa_{\mathfrak{g}}.$$

Now notice we can give a better formula for the g^α_β, using formula (4.21) twice we get

$$\begin{aligned}
\sum_j (x^j_L(x^\beta_L(f^\alpha_j)) &= \sum_j \left(\sum_{l,l'} c^j_{\beta,l}c^l_{j,l'}f^\alpha_{l'}\right)\\
&= \sum_{j,l,l'} c^j_{\beta,l}c^l_{j,l'}f^\alpha_{l'}\\
&= \sum_{l'} \left(\sum_{j,l} c^j_{\beta,l}c^l_{j,l'}\right) f^\alpha_{l'}\\
&= \sum_{l'} \left(-\sum_{j,l} c^j_{\beta,l}c^l_{l',j}\right) f^\alpha_{l'}\\
&= \sum_{l'} (-\kappa_{\mathfrak{g}}(x^\beta, x^{l'}))f^\alpha_{l'}\\
&= \sum_j (-\kappa_{\mathfrak{g}}(x^\beta, x^j))f^\alpha_j
\end{aligned}$$

and plugging this in the formula (4.18) yields

$$g_\beta^\alpha = \sum_j (-\kappa(x^j, x^\beta) - \kappa_{\mathfrak{g}}(x^j, x^\beta)) f_j^\alpha = \sum_j \kappa^*(x^j, x^\beta) f_j^\alpha.$$

Finally, putting everything together yields the formula

$$\pi_R(x^i) = x_R^i + \sum_{p,q} \kappa^*(x^p, x^q) f_p^i \omega^q = x_R^i + \sum_q \left(\sum_p \kappa^*(x^p, x^q) f_p^i \right) \omega^q.$$

□

Remark 4.1.5. *This beautiful fact is attributed to Feigin, Frenkel and Gaitsgory in [GMS01]. To the best of our knowledge, its earliest appearance in the case of special linear groups can be found in [FP96]. We also note that we did not need the group G to be reductive for all the results of this paragraph to hold. This assumption will be essential from now on.*

As we have just seen, $\mathcal{D}_G^\kappa$ comes equipped with two vertex algebra morphisms

$$\pi_L : V^\kappa(\mathfrak{g}) \longrightarrow \mathcal{D}_G^\kappa \longleftarrow V^{\kappa^*}(\mathfrak{g}) : \pi_R$$

whose images commute (see equation (4.15)). Equivalently this defines a morphism of vertex algebras

$$\pi_L \otimes \pi_R : V^{\kappa \oplus \kappa^*}(\mathfrak{g} \oplus \mathfrak{g}) \longrightarrow \mathcal{D}_G^\kappa. \tag{4.23}$$

Definition 4.1.6. *We call critical level the unique level κ_c on $\mathfrak{g}$ such that*

$$\kappa_c^* = \kappa_c.$$

We have the equality

$$\kappa_c = -\frac{1}{2}\kappa_{\mathfrak{g}}$$

where $\kappa_{\mathfrak{g}}$ is the Killing form of $\mathfrak{g}$. When $\mathfrak{g}$ is simple we have the equality

$$\kappa_c = -\check{h}\widetilde{\kappa_{\mathfrak{g}}}$$

where $\check{h}$ is the dual Coxeter number and

$$\widetilde{\kappa_{\mathfrak{g}}} = \frac{1}{2\check{h}}\kappa_{\mathfrak{g}}$$

is the normalized Killing form.

4.2 The conformal structure

We make a brief digression to discuss the conformal structure of affine vertex algebras and $\mathcal{D}_G^\kappa$. For the convenience of the reader we split the discussion between the abelian and the simple case. The reader familiar with the Segal–Sugawara construction in these contexts may go directly to the main statement (Theorem 4.2.2).

4.2.1 The case of an abelian Lie algebra

Assume that $G = T = \mathbb{G}_m^r$ is an algebraic torus of rank r with Lie algebra $\mathfrak{g} = \mathfrak{h} = \mathbb{C}^r$. Let κ be a symmetric bilinear form on $\mathfrak{h}$. Note that such a form is automatically $\mathfrak{h}$-invariant because $\mathfrak{h}$ is abelian. Recall that, because the Killing form is zero, we have

$$\kappa^* = -\kappa.$$

Assume that κ is nondegenerate. Let $(h^i)_{1\leqslant i\leqslant r}$ be an orthonormal basis of $\mathfrak{h}$ with respect to κ *i.e.*, a basis such that for all $1 \leqslant i, j \leqslant r$ we have the equality of complex numbers

$$\kappa(h^i, h^j) = \delta_{i,j}.$$

In that case, the vertex algebra $V^\kappa(\mathfrak{h})$ (resp. $V^{\kappa^*}(\mathfrak{h})$) is conformal (see Theorem 6.2.18 of [LL04]) and a choice of conformal vector is given by

$$T_L = \frac{1}{2}\sum_{i=1}^r h^i_{(-1)}h^i_{(-1)}|0\rangle \in V^\kappa(\mathfrak{h}),\ (\text{resp. } T_R = -\frac{1}{2}\sum_{i=1}^r h^i_{(-1)}h^i_{(-1)}|0\rangle \in V^{\kappa^*}(\mathfrak{h})).$$

Let

$$T = \pi_L(T_L) + \pi_R(T_R) \tag{4.24}$$

It is clear that T has conformal weight 2 and the goal is to show that it is a conformal vector of $\mathcal{D}^\kappa_G$.

Define $(\omega^i)_{1\leqslant i\leqslant r}$ to be the $\mathcal{O}_G$-basis of Ω^1_G dual to the $\mathcal{O}_G$-basis $(h^i_L)_{1\leqslant i\leqslant r}$ of $\mathrm{Der}(\mathcal{O}_G)$. Then from equation (4.17) we have for all $i \in \{1, \dots, r\}$ the equality

$$\pi_R(h^i) = -\pi_L(h^i) + \omega^i.$$

We have

$$\pi_R(T_R) = -\frac{1}{2}\sum_{i=1}^r(-\pi_L(h^i) + \omega^i)_{(-1)}(-\pi_L(h^i) + \omega^i)_{(-1)}|0\rangle.$$

So it is straightforward that

$$\pi_R(T_R) = -\frac{1}{2}\sum_{i=1}^r \pi_L(h^i)\pi_L(h^i) + \frac{1}{2}\sum_{i=1}^r(\pi_L(h^i)\omega^i + \omega^i\pi_L(h^i)) - \frac{1}{2}\sum_{i=1}^r \omega^i\omega^i.$$

For all $i \in \{1, \dots, r\}$ we have the equality $[\pi_L(h^i)_{(-1)}, \omega^i_{(-1)}] = 0$ in $\mathrm{End}(\mathcal{D}^\kappa_G)$. So finally, we have the equality

$$T = \sum_{i=1}^r \pi_L(h^i)\omega^i - \frac{1}{2}\sum_{i=1}^r \omega^i\omega^i. \tag{4.25}$$

Lemma 4.2.1. *The vector T is a conformal vector of central charge $2r$ of the vertex algebra $\mathcal{D}^\kappa_G$.*

Proof. We have to check that $T_{(0)}$ is the translation operator. Let $f \in \mathcal{O}_G$, then in view of equation (4.25) we have

$$T_{(0)}f = \sum_{i=1}^{r} (\pi_L(h^i)_{(-1)}\omega^i)_{(0)} f = \sum_{i=1}^{r} \pi_L(h^i)_{(0)}(f)\omega^i = \sum_{i=1}^{r} h^i_L(f)\omega^i = \partial(f).$$

The others properties are easily checked. $\square$

4.2.2 The case of a simple Lie algebra

Assume that G is a simple algebraic group, *i.e.*, a reductive algebraic group whose Lie algebra $\mathfrak{g}$ is simple. Because $\mathfrak{g}$ is simple there exists a unique complex number $k \in \mathbb{C}$ such that

$$\kappa = k\widetilde{\kappa_{\mathfrak{g}}}$$

and we refer interchangeably to κ and k as the level in that case. Here κ^* corresponds to the complex number

$$k^* = -k - 2\check{h}. \tag{4.26}$$

Let us assume that the level is noncritical *i.e.*, $k \neq -\check{h}$. In that case we clearly also have $k^* \neq -\check{h}$. Let $(x^i)_{i=1}^{\dim \mathfrak{g}}$ be an orthonormal basis of $\mathfrak{g}$ with respect to the normalized Killing form $\widetilde{\kappa_{\mathfrak{g}}}$ *i.e.*, a basis such that for all $i, j \in \{1, \dots, r\}$ we have the equality of complex numbers

$$\widetilde{\kappa_{\mathfrak{g}}}(x^i, x^j) = \delta_{i,j}.$$

In that case the vertex algebra $V^\kappa(\mathfrak{g})$ (resp. $V^{\kappa^*}(\mathfrak{g})$) is conformal (see *e.g.* Theorem 6.2.18 of [LL04]) and a choice of conformal vector is given by the Segal–Sugawara construction

$$T_L = \frac{1}{2(k+\check{h})} \sum_{i=1}^{\dim \mathfrak{g}} x^i_{(-1)}x^i_{(-1)}|0\rangle \in V^\kappa(\mathfrak{g})$$

$$T_R = -\frac{1}{2(k+\check{h})} \sum_{i=1}^{\dim \mathfrak{g}} x^i_{(-1)}x^i_{(-1)}|0\rangle \in V^{\kappa^*}(\mathfrak{g})$$

where the minus sign in the second equality comes from the equality of complex numbers

$$\frac{1}{k+\check{h}} = -\frac{1}{k^*+\check{h}}.$$

Let

$$T = \pi_L(T_L) + \pi_R(T_R).$$

It is clear that T is of conformal degree 2.

Define $(\omega^i)_{1\leqslant i\leqslant r}$ to be the $\mathcal{O}_G$-basis of Ω^1_G dual to the $\mathcal{O}_G$-basis $(x^i_L)_{1\leqslant i\leqslant r}$ of $\mathrm{Der}(\mathcal{O}_G)$. It follows from the Proposition 3.18 of [Zhu08] that the following equality holds in $\mathcal{D}^\kappa_G$

$$T = \sum_{i=1}^{\dim \mathfrak{g}} \pi_L(x^i)\omega^i + \frac{k^*}{2}\sum_{i=1}^{\dim \mathfrak{g}} \omega^i\omega^i. \tag{4.27}$$

and from there it is proven that T is a conformal vector of central charge $2\dim G$ of the vertex algebra $\mathcal{D}_G^\kappa$ (see Proposition 3.22 of *loc.cit.*).

4.2.3 The case of a reductive Lie algebra

Assume that G is a reductive algebraic group. Recall the decomposition (3.1) of $\mathfrak{g}$. Because κ is an invariant form it respects the previous decomposition, that is

$$\kappa = \kappa_{|\mathfrak{z}(\mathfrak{g})} \oplus \bigoplus_{i=1}^{r} \kappa_{|\mathfrak{g}_s}.$$

For each $s \in \{1, \dots, r\}$, because $\mathfrak{g}_s$ is simple and $\kappa_{|\mathfrak{g}_s}$ is invariant, there exists $k_s \in \mathbb{C}$ such that

$$\kappa_{|\mathfrak{g}_s} = k_s \widetilde{\kappa_{\mathfrak{g}_s}}.$$

Let us assume momentarily that κ is such that $\kappa - \kappa_c$ is nondegenerate where κ_c is the critical level (see Definition 4.1.6). This means that κ is such that $\kappa_{|\mathfrak{z}(\mathfrak{g})}$ is nondegenerate and $k_s \neq -\check{h}_s$ for all $1 \leqslant s \leqslant r$ where $\check{h}_s$ is the dual Coxeter number of the simple factor $\mathfrak{g}_s$.

We will apply the constructions of §4.2.1 to $\mathfrak{z}(\mathfrak{g})$ and of §4.2.2 to each simple factor of $\mathfrak{g}$ to construct a conformal vector in the general reductive case.

Fix an orthonormal basis $(h^i)_{i=1}^{\dim \mathfrak{z}(\mathfrak{g})}$ of the center $\mathfrak{z}(\mathfrak{g})$ with respect to $\kappa_{|\mathfrak{z}(\mathfrak{g})}$ and set

$$T_L^{ab} = \frac{1}{2}\sum_{i=1}^{r} h^i_{(-1)}h^i_{(-1)}|0\rangle \in V^{\kappa_{|\mathfrak{z}(\mathfrak{g})}}(\mathfrak{z}(\mathfrak{g})), T_R^{ab} = -\frac{1}{2}\sum_{i=1}^{r} h^i_{(-1)}h^i_{(-1)}|0\rangle \in V^{\kappa^*_{|\mathfrak{z}(\mathfrak{g})}}(\mathfrak{z}(\mathfrak{g})).$$

For each $s \in \{1, \dots, r\}$, fix an orthonormal basis $(x_s^{i_s})_{i_s=1}^{\dim \mathfrak{g}_s}$ of $\mathfrak{g}_s$ with respect to the normalized Killing form $\widetilde{\kappa}_{\mathfrak{g}_s}$ and set

$$T_L^s = \frac{1}{2(k+\check{h}_s)} \sum_{i_s=1}^{\dim \mathfrak{g}_s} x^{i_s}_{s\,(-1)} x^{i_s}_{s\,(-1)}|0\rangle \in V^{\kappa_{|\mathfrak{g}_s}}(\mathfrak{g}_s)_2,$$
$$T_R^s = -\frac{1}{2(k+\check{h}_s)} \sum_{i_s=1}^{\dim \mathfrak{g}_s} x^{i_s}_{s\,(-1)} x^{i_s}_{s\,(-1)}|0\rangle \in V^{\kappa^*_{|\mathfrak{g}_s}}(\mathfrak{g}_s)_2.$$

We have an isomorphism of vertex algebras

$$V^\kappa(\mathfrak{g}) \simeq V^{\kappa_{|\mathfrak{z}(\mathfrak{g})}}(\mathfrak{z}(\mathfrak{g})) \otimes \bigotimes_{s=1}^{r} V^{\kappa_{|\mathfrak{g}_s}}(\mathfrak{g}_s).$$

So it makes sense to define

$$T_L = T_L^{ab} + T_L^1 + \cdots + T_L^r \in V^\kappa(\mathfrak{g}),$$
$$T_R = T_R^{ab} + T_R^1 + \cdots + T_R^r \in V^{\kappa^*}(\mathfrak{g}).$$

Set

$$T = \pi_L(T_L) + \pi_R(T_R) \in \mathcal{D}_G^\kappa.$$

Clearly T is of conformal degree 2 and $(h^i)_{i=1}^{\dim \mathfrak{z}(\mathfrak{g})} \cup \bigcup_{s=1}^r (x_s^{i_s})_{i_s=1}^{\dim \mathfrak{g}_s}$ is a basis of $\mathfrak{g}$ and hence $(h_L^i)_{i=1}^{\dim \mathfrak{z}(\mathfrak{g})} \cup \bigcup_{s=1}^r ((x_s^{i_s})_L)_{i_s=1}^{\dim \mathfrak{g}_s}$ is a $\mathcal{O}_G$-basis of $\mathrm{Der}(\mathcal{O}_G)$. Denote by

$$(\omega^i)_{i=1}^{\dim \mathfrak{z}(\mathfrak{g})} \cup \bigcup_{s=1}^r (\omega_s^{i_s})_{i_s=1}^{\dim \mathfrak{g}_s}$$

its dual basis in Ω_G^1.

As a straightforward consequence of equalities (4.25) and (4.27) we have the equality:

$$T = \sum_{i=1}^r \pi_L(h^i)\omega^i + \sum_{s=1}^r \sum_{i_s=1}^{\dim \mathfrak{g}_s} \pi_L(x_s^{i_s})\omega_s^{i_s} - \frac{1}{2}\sum_{i=1}^{\dim \mathfrak{z}(\mathfrak{g})} \omega^i\omega^i + \sum_{s=1}^r \frac{k_s^*}{2} \sum_{i_s=1}^{\dim \mathfrak{g}_s} \omega_s^{i_s}\omega_s^{i_s}. \quad (4.28)$$

Theorem 4.2.2. *The vector T is a conformal vector of central charge $2\dim G$ of the vertex algebra $\mathcal{D}_G^\kappa$. Equivalently, the vertex algebra $\mathcal{D}_G^\kappa$ is a conformal extension of $V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$ with respect to the embedding*

$$\pi_L \otimes \pi_R : V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g}) \longrightarrow \mathcal{D}_G^\kappa.$$

Proof. Equality (4.28) shows that $T_{(0)}$ is the translation operator in a similar fashion as in the proof of Lemma 4.2.1. The rest is easily checked. □

Remark 4.2.3. *Note that here we assume a condition on the level κ such that the Segal–Sugawara constructions are available. It is remarkable that equation (4.28) does not involve the inverse of $(k + \check{h})$. The formula then makes sense at any level and one can check that $\mathcal{D}_G^\kappa$ is indeed always conformal of central charge $2\dim G$ (see Proposition 3.22 of [Zhu08]).*

4.3 Kazhdan–Lusztig category at generic levels

Let us first explain what we mean by a generic level.

Definition 4.3.1. *We say that κ is generic if $\kappa_{|\mathfrak{z}(\mathfrak{g})}$ is nondegenerate and $k_s \notin \mathbb{Q}$ for all $s \in \{1, \ldots, r\}$, where $k_s \in \mathbb{C}$ is the unique complex number such that $\kappa_{|\mathfrak{g}_s} = k_s\widetilde{\kappa_{\mathfrak{g}_s}}$. In what follows, we will sometimes identify the bilinear form $\kappa_{|\mathfrak{g}_s}$ with the complex number k_s defined above.*

Remark 4.3.2. *Note that if κ is generic then so is κ^* and the direct sum of two generic levels is again generic.*

From now on we assume that κ is a generic level.

The upshot is that for generic levels the Kazhdan–Lusztig category is easy to describe. The following statement can be found in the literature for the case of a simple group (see Lemma 30.1 of [KL94b]) or of an abelian Lie algebra (see §3 of [LW82]).

Theorem 4.3.3. *For all $\lambda \in P_+$, the Weyl module $\mathbb{V}^\kappa_\lambda$ is an irreducible $V^\kappa(\mathfrak{g})$-module. Moreover the functors*

$$M \in V^\kappa(\mathfrak{g})\text{−Mod}^{\mathcal{J}_\infty G} \longmapsto M^{t\mathfrak{g}[[t]]} \in G\text{−Mod}$$

and

$$V \in G\text{−Mod} \longmapsto \operatorname*{Ind}_{\hat{\mathfrak{g}}^\kappa \leftarrow \mathfrak{g}[[t]] \to \mathfrak{g}} V \in V^\kappa(\mathfrak{g})\text{−Mod}^{\mathcal{J}_\infty G}$$

are quasi-inverses of one another.

We are not aware of a reference stating it at the level of generality we need, so we devote the rest of this section to the proof.

We will use freely the notations introduced in §4.2. Recall equation (3.1)

$$\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus \bigoplus_{s=1}^{r} \mathfrak{g}_s$$

that expresses $\mathfrak{g}$ as a direct sum of its center and its simple factors. Accordingly, this decomposition induces an isomorphism of vertex algebras

$$V^\kappa(\mathfrak{g}) = V^{\kappa|_{\mathfrak{z}(\mathfrak{g})}}(\mathfrak{z}(\mathfrak{g})) \otimes \bigotimes_{s=1}^{r} V^{\kappa|_{\mathfrak{g}_s}}(\mathfrak{g}_s). \tag{4.29}$$

It is traditional to adopt the following notation for all $1 \leqslant s \leqslant r$:

$$L_0^{ab} = (T_L^{ab})_{(1)}, \qquad L_0^{s} = (T_L^{s})_{(1)}.$$

Lemma 4.3.4. *Let M be a $V^\kappa(\mathfrak{g})$-module, it is naturally a $\mathfrak{g}$-module by restriction. Then L_0^{ab} and $(L_0^s)_{1\leqslant s\leqslant r}$ define commuting $\mathfrak{g}$-endomorphisms of M.*

Proof. The fact that they pairwise commute is clear from the isomorphism (4.29). The fact that they commute with the $\mathfrak{g}$-action is easily checked. □

Definition 4.3.5. *A $V^\kappa(\mathfrak{g})$-module M is said to be graded by conformal degrees if L_0^{ab} and $(L_0^s)_{1\leqslant s\leqslant r}$ define locally finite endomorphisms of M. If M is such a module, we define for all $\mu^{ab}, \mu_1, \dots, \mu_r \in \mathbb{C}$ the $\mathfrak{g}$-submodule*

$$M[\mu^{ab}, \mu_1, \dots, \mu_r] = \{v \in M \mid (L_0^{ab})_{ss} v = \mu^{ab} v, (L_0^s)_{ss} v = \mu_s v \text{ for all } 1 \leqslant s \leqslant r\}$$

where the subscript $(*)_{ss}$ denotes the semisimple part of the Jordan–Chevalley decomposition of the corresponding locally finite endomorphism. We have the equality of $\mathfrak{g}$-modules:

$$M = \bigoplus_{\mu^{ab}, \mu_1, \dots, \mu_r \in \mathbb{C}} M[\mu^{ab}, \mu_1, \dots, \mu_r].$$

Lemma 4.3.6 (Theorem 6.2.21 of [LL04])**.** *Let $\lambda \in P_+$ be a dominant weight, then the Weyl module V^κ_λ is a $V^\kappa(\mathfrak{g})$-module on which $\mathfrak{g}$ acts locally finitely and $t\mathfrak{g}[[t]]$ acts locally nilpotently. As a $\mathfrak{g}$-module, its set of weights with respect to $\mathfrak{h}$ is equal to*

$$\lambda + Q \subset P.$$

Moreover, it is a $V^\kappa(\mathfrak{g})$-module such that L_0^{ab} and $(L_0^s)_{1\leqslant s\leqslant r}$ are semisimple endomorphisms, in particular, Weyl modules are graded by conformal degrees. Explicitly if one writes

$$\lambda = \lambda^{ab} + \sum_{s=1}^{r} \lambda_s \in \mathfrak{h}^* \simeq \mathfrak{z}(\mathfrak{g})^* \oplus \bigoplus_{s=1}^{r} \mathfrak{h}_s^*$$

then the conformal grading is given by

$$V^\kappa_\lambda = \bigoplus_{n^{ab}, n_1, \dots, n_r \in \mathbb{Z}_+} V^\kappa_\lambda\left[\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\lambda^{ab}, \lambda^{ab})}{2} + n_{ab}, \frac{C_{\mathfrak{g}_1}(\lambda_1)}{2(k_1 + \check{h}_1)} + n_1, \dots, \frac{C_{\mathfrak{g}_r}(\lambda_r)}{2(k_r + \check{h}_r)} + n_r\right]$$

where for each $1 \leqslant s \leqslant r$, if we let ρ_s be the half-sum of the positive roots in $\mathfrak{g}_s$, $C_{\mathfrak{g}_s}(\lambda_s)$ is the rational number given by

$$C_{\mathfrak{g}_s}(\lambda_s) = \kappa_{\mathfrak{g}_s}(\lambda_s + \rho_s, \lambda_s + \rho_s) - \kappa_{\mathfrak{g}_s}(\rho_s, \rho_s) = \kappa_{\mathfrak{g}_s}(\lambda_s + 2\rho, \lambda_s) \in \mathbb{Q}, \tag{4.30}$$

in which it is understood that we identify $\mathfrak{h}_s$ with $\mathfrak{h}_s^*$ via the Killing form $\kappa_{\mathfrak{g}_s}$. Moreover, for all $n^{ab}, n_1, \dots, n_r \in \mathbb{Z}_+$,

$$V^\kappa_\lambda\left[\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\lambda^{ab}, \lambda^{ab})}{2} + n_{ab}, \frac{C_{\mathfrak{g}_1}(\lambda_1)}{2(k_1 + \check{h}_1)} + n_1, \dots, \frac{C_{\mathfrak{g}_r}(\lambda_r)}{2(k_r + \check{h}_r)} + n_r\right]$$

is a finite dimensional $\mathfrak{g}$-submodule of V^κ_λ and the irreducible $\mathfrak{g}$-submodule V_λ of V^κ_λ coincides with the lowest conformal degree $\mathfrak{g}$-submodule, that is

$$V_\lambda = V^\kappa_\lambda\left[\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\lambda^{ab}, \lambda^{ab})}{2}, \frac{C_{\mathfrak{g}_1}(\lambda_1)}{2(k_1 + \check{h}_1)}, \dots, \frac{C_{\mathfrak{g}_r}(\lambda_r)}{2(k_r + \check{h}_r)}\right] \subset V^\kappa_\lambda.$$

Lemma 4.3.7. *Let $\mathfrak{g}$ be a reductive Lie algebra and κ a generic level. Then for all $\lambda \in P_+$, the Weyl module V^κ_λ is an irreducible $V^\kappa(\mathfrak{g})$-module. Moreover, they are pairwise nonisomorphic.*

Proof. Let M be a nonzero $V^\kappa(\mathfrak{g})$-submodule of V^κ_λ, it decomposes as a direct sum of the (possibly zero) $\mathfrak{g}$-submodules

$$M = \bigoplus_{n^{ab}, n_1, \dots, n_r \in \mathbb{Z}_+} M\left[\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\lambda^{ab}, \lambda^{ab})}{2} + n_{ab}, \frac{C_{\mathfrak{g}_1}(\lambda_1)}{2(k_1 + \check{h}_1)} + n_1, \dots, \frac{C_{\mathfrak{g}_r}(\lambda_r)}{2(k_r + \check{h}_r)} + n_r\right].$$

Let us pick $(n_{ab}, n_1, \dots, n_r) \in \mathbb{Z}_+$ such that

$$M\left[\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\lambda^{ab}, \lambda^{ab})}{2} + n_{ab}, \frac{C_{\mathfrak{g}_1}(\lambda_1)}{2(k_1 + \check{h}_1)} + n_1, \dots, \frac{C_{\mathfrak{g}_r}(\lambda_r)}{2(k_r + \check{h}_r)} + n_r\right] \neq 0$$

and such that the sum $n_{ab}+n_1+\cdots+n_r\in\mathbb{Z}_+$ is minimal. Note that it is not a priori unique.

It follows from Lemma 4.3.6 that

$$M\left[\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\lambda^{ab},\lambda^{ab})}{2}+n_{ab},\frac{C_{\mathfrak{g}_1}(\lambda_1)}{2(k_1+\check{h}_1)}+n_1,\ldots,\frac{C_{\mathfrak{g}_r}(\lambda_r)}{2(k_r+\check{h}_r)}+n_r\right]$$

is finite dimensional, hence it contains a highest weight vector v_μ for the Lie algebra $\mathfrak{g}$ with weight given by a certain $\mu=\mu_{ab}+\mu_1+\cdots+\mu_r\in P_+$ and it is easily seen that $\mu_{ab}=\lambda_{ab}$. From the minimality of $n_{ab}+n_1+\cdots+n_r$ we see that

$$t\mathfrak{g}[[t]]v_\mu=0$$

which proves that v_μ is a highest weight vector for the Lie algebra $\hat{\mathfrak{g}}^\kappa$. The universal property of Weyl modules (Proposition 3.2.13) yields a morphism of $V^\kappa(\mathfrak{g})$-modules $V_\mu^\kappa\to M$ mapping $|\mu\rangle$ to v_μ. This implies that the conformal weight of v_μ is equal on the one hand to

$$\left(\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\lambda^{ab},\lambda^{ab})}{2},\frac{C_{\mathfrak{g}_1}(\mu_1)}{2(k_1+\check{h}_1)},\ldots,\frac{C_{\mathfrak{g}_r}(\mu_r)}{2(k_r+\check{h}_r)}\right),$$

and on the other hand it is by construction equal to

$$\left(\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\lambda^{ab},\lambda^{ab})}{2}+n_{ab},\frac{C_{\mathfrak{g}_1}(\lambda_1)}{2(k_1+\check{h}_1)}+n_1,\ldots,\frac{C_{\mathfrak{g}_r}(\lambda_r)}{2(k_r+\check{h}_r)}+n_r\right).$$

It implies directly $n_{ab}=0$ and that for all $1\leqslant s\leqslant r$ we have the equality of complex numbers

$$\frac{C_{\mathfrak{g}_s}(\mu_s)}{2(k_s+\check{h}_s)}=\frac{C_{\mathfrak{g}_s}(\lambda_s)}{2(k_s+\check{h}_s)}+n_s,$$

better written as

$$n_s(2(k_s+\check{h}_s))+(C_{\mathfrak{g}_s}(\lambda_s)-C_{\mathfrak{g}_s}(\mu_s)))=0.$$

Now the fact that $k_s\notin\mathbb{Q}$ and $C_{\mathfrak{g}_s}(\lambda_s),C_{\mathfrak{g}_s}(\mu_s))\in\mathbb{Q}$ implies that $n_s=0$. We have $(n_{ab},n_1,\ldots,n_r)=(0,0,\ldots,0)$, now by Lemma 4.3.6, we see that M intersects non-trivially $V_\lambda\subset V_\lambda^\kappa$. But V_λ is a simple $\mathfrak{g}$-module so M contains it, and because V_λ generates V_λ^κ as $V^\kappa(\mathfrak{g})$-module, we see that $M=V_\lambda^\kappa$ which proves the simplicity.

It is easily seen that Weyl modules are pairwise nonisomorphic by looking at the lowest conformal degree spaces and using the corresponding statement at the level of finite dimensional Lie algebras. □

Proposition 4.3.8. *Let $\lambda\in P_+$ then $V_\lambda^\kappa\in V^\kappa(\mathfrak{g})\text{–Mod}^{\mathfrak{J}_\infty G}$ if and only if $\lambda\in X^*(T)_+$.*

Proof. According to Lemma 4.3.6, $\mathfrak{g}$ acts locally finitely on V_λ^κ and $t\mathfrak{g}[[t]]$ acts locally nilpotently, so we are left to show that the action of $\mathfrak{g}$ integrates to an action of the group G if and only if $\lambda\in X^*(T)_+$. In view of Corollary 3.1.2 it is enough to look at the $\mathfrak{h}$-weights, precisely, the action integrates if and only if they all lie in $X^*(T)$. According to Lemma 4.3.6 the $\mathfrak{h}$-weights of V_λ^κ are

$$\lambda + Q,$$

hence the action integrates if and only if $\lambda + Q \subset X^*(T)$ i.e. $\lambda \in X^*(T)$ since $Q \subset X^*(T)$. The result then follows from the equality $X^*(T)_+ = P_+ \cap X^*(T)$.

□

Let M be a $V^\kappa(\mathfrak{g})$-module, we define for all $i \geqslant 0$

$$\begin{aligned} M[0] &= 0 \\ M[1] &= \{m \in M \mid t\mathfrak{g}[[t]]m = 0\} = M^{t\mathfrak{g}[[t]]} \\ M[2] &= \{m \in M \mid t\mathfrak{g}[[t]]m \in M[1]\} \\ &\vdots \\ M[i] &= \{m \in M \mid t\mathfrak{g}[[t]]m \in M[i-1]\}. \end{aligned}$$

It is obvious that the following inclusions hold

$$0 = M[0] \subset M[1] \subset \cdots \subset M[i-1] \subset M[i] \subset \cdots \subset M$$

and that for all $i \geqslant 0$, $M[i]$ can be described as

$$M[i] = \{m \in M \mid \underbrace{t\mathfrak{g}[[t]] \ldots t\mathfrak{g}[[t]]}_{i \text{ factors}}\, m = 0\},$$

or more precisely

$$M[i] = \{m \in M \mid (x^1)_{(n_1)} \ldots (x^i)_{(n_i)} m = 0 \text{ for all } x^1, \ldots, x^i \in \mathfrak{g} \text{ and } n_1, \ldots, n_i > 0\ \}.$$

Lemma 4.3.9. *If $M \in V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$, then M is locally finite $\mathfrak{g}[[t]]$-module.*

Proof. Let $m \in M$ and $x^1, \ldots, x^n$ be an ordered basis of $\mathfrak{g}$. In view of the Poincaré–Birkhoff–Witt theorem, the $\mathfrak{g}[[t]]$-submodule generated by m is the vector space generated by elements of the form

$$(x^1_{(0)})^{\alpha^0_1} \ldots (x^n_{(0)})^{\alpha^0_n} (x^1_{(1)})^{\alpha^1_1} \ldots (x^n_{(1)})^{\alpha^1_n} \ldots (x^1_{(N)})^{\alpha^N_1} \ldots (x^n_{(N)})^{\alpha^N_n} m$$

where the α^i_j are nonnegative integers and N is a fixed integer such that for all $m > N$,

$$t^m \mathfrak{g}[[t]]m = 0.$$

Now using inductively the fact that $t\mathfrak{g}[[t]]$ acts locally nilpotently and that the $\mathfrak{g}$-action is locally finite concludes. □

Lemma 4.3.10. *Let M be a $V^\kappa(\mathfrak{g})$-module, then for all $i \geqslant 0$, $t\mathfrak{g}[[t]] \cdot M[i] \subset M[i-1]$ and $M[i]$ is a $\mathfrak{g}$-submodule of M. Moreover, if $M \in V^\kappa(\mathfrak{g})-\mathrm{Mod}^{\mathcal{J}_\infty G}$ then the filtration is exhaustive i.e.*

$$M = \bigcup_{i \geqslant 0} M[i].$$

Proof. It is clear from the definition that for all $i \geqslant 0$, $t\mathfrak{g}[[t]]M[i] \subset M[i-1]$. Fix $i \geqslant 0$, $x \in \mathfrak{g}$ and $m \in M[i]$ and let $n > 0$ and $y \in \mathfrak{g}$. The following equality

$$y_{(n)}x_{(0)}m = x_{(0)}y_{(n)}m + [y_{(n)}, x_{(0)}]m = x_{(0)}(y_{(n)}m) + [y, x]_{(n)}m$$

together with an induction on $i \geqslant 0$ shows that $M[i]$ is an $\mathfrak{g}$-module for all $i \geqslant 0$.

Assume that $M \in V^\kappa(\mathfrak{g})-\mathrm{Mod}^{\mathcal{J}_\infty G}$ and let $m \in M$, thanks to Lemma 4.3.9 and because M is smooth, there exists a finite dimensional $\mathfrak{g}[[t]]$-submodule W of M containing m and $N \geqslant 0$ such that $t^{N+1}\mathfrak{g}[[t]]W = 0$. We now see W as a $t\mathfrak{g}[[t]]$-module, by what we just said, it is naturally a $t\mathfrak{g}[[t]]/t^{N+1}\mathfrak{g}[[t]]$-module. Now $t\mathfrak{g}[[t]]/t^{N+1}\mathfrak{g}[[t]]$ is a finite dimensional nilpotent Lie algebra and W is a finite dimensional vector space. So Engel's theorem (see *e.g* Theorem 19.3.6 of [TY05]) applies and shows in particular that $W \subset M[\dim W]$ hence the $m \in M$ we started with belongs to some $M[i]$ and we are done.

□

Remark 4.3.11. *It is obvious from the definition of the $(M[i])_{i \geqslant 0}$ that if there exists $j \geqslant 0$ such that $M[j] = M[j+1]$ then for all $i \geqslant j$, we have*

$$M[j] = M[i].$$

Lemma 4.3.12. *Let $M \in V^\kappa(\mathfrak{g})-\mathrm{Mod}^{\mathcal{J}_\infty G}$. If $M[1] = 0$ then $M = 0$ so if M is nonzero then $M[1] \neq 0$ and M contains a Weyl module of highest weight $\lambda \in X^*(T)_+$. In particular, the only simple objects of the category $V^\kappa(\mathfrak{g})-\mathrm{Mod}^{\mathcal{J}_\infty G}$ are given by the V^κ_λ for $\lambda \in X^*(T)_+$.*

Proof. Notice that if the $\mathfrak{g}$-submodule $M[1]$ is equal to zero, then by Remark 4.3.11, all the $(M[i])_{i \geqslant 0}$ are equal to zero and then M would be too because $M = \bigcup_{i \geqslant 0} M[i]$ by Lemma 4.3.10. Now $M[1]$ is a $\mathfrak{g}$-submodule of M by Lemma 4.3.10 and because by assumption the $\mathfrak{g}$-action integrates to an action of G, then $M[1]$ contains by Theorem 3.1.1 a finite dimensional irreducible representation of $\mathfrak{g}$, say V_λ for a certain $\lambda \in X^*(T)_+$. That defines for us a nonzero $V^\kappa(\mathfrak{g})$-module map $V^\kappa_\lambda \to M$ by Proposition 3.2.13 which is injective since V^κ_λ is simple by Lemma 4.3.7. The second part of the statement is easily deduced from the first. □

The following lemma is clear:

Lemma 4.3.13. *The functor $M \in V^\kappa(\mathfrak{g})-\mathrm{Mod}^{\mathcal{J}_\infty G} \mapsto M[1] \in G-\mathrm{Mod}$ is left-exact and commutes with direct sums.*

Lemma 4.3.14. *If $\mathfrak{g}$ is a reductive Lie algebra and κ is a generic level then for all $\lambda \in P_+$ we have the equality of $\mathfrak{g}$-modules*

$$V_\lambda^\kappa[1] = V_\lambda.$$

Proof. It is clear that V_λ is contained in $V_\lambda^\kappa[1]$ by the very definition of V_λ^κ. Let us assume for a contradiction that V_λ is not all of $V_\lambda^\kappa[1]$. We know by Lemma 4.3.10 that $V_\lambda^\kappa[1]$ is a $\mathfrak{g}$-submodule of V_λ^κ. We saw in Lemma 4.3.6 that the action of $\mathfrak{g}$ is locally finite on V_λ^κ and hence on $V_\lambda^\kappa[1]$, as such, Theorem 3.1.1 applies and shows that there exists $v_\mu \in V_\lambda^\kappa[1]$ that is a highest weight vector of a certain weight $\mu \in P_+$ for $\mathfrak{g}$ that does not belong to V_λ. Proposition 3.2.13 applies and defines for us a nonzero morphism of $V^\kappa(\mathfrak{g})$-modules $V_\mu^\kappa \to V_\lambda^\kappa$. Both V_μ^κ and V_λ^κ are simple thanks to Lemma 4.3.7 so in particular this map is an isomorphism of $V^\kappa(\mathfrak{g})$-modules. Hence, $v_\mu \in V_\mu^\kappa$ being of lowest possible conformal degree is mapped to the lowest conformal degree part of V_λ^κ which is V_λ by Lemma 4.3.6, contradicting the fact that $v_\mu \notin V_\lambda$. □

If M is a $V^\kappa(\mathfrak{g})$-module and $S \subset M$ is a subset we denote by $\widehat{S} \subset M$ the $V^\kappa(\mathfrak{g})$-module generated by S, that is the smallest $V^\kappa(\mathfrak{g})$-submodule of M containing S. It is obvious that the following inclusions hold

$$0 = \widehat{M[0]} \subset \widehat{M[1]} \subset \cdots \subset \widehat{M[i-1]} \subset \widehat{M[i]} \subset \cdots \subset M$$

and it follows from Lemma 4.3.10 that if $M \in V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ then

$$M = \bigcup_{i \geqslant 0} \widehat{M[i]}.$$

Proposition 4.3.15. *Let $M \in V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ then M is isomorphic to a direct sum of Weyl modules if and only if $M = \widehat{M[1]}$.*

Proof. Assume there exists a set I and $(\lambda_i)_{i\in I} \in (X^*(T)_+)^I$ such that

$$M = \bigoplus_{i\in I} V_{\lambda_i}^\kappa.$$

For all $i \in I$ we have $V_{\lambda_i}^\kappa[1] = V_{\lambda_i}$ by Lemma 4.3.14. Thanks to Lemma 4.3.13 we have

$$M[1] = \Big(\bigoplus_{i\in I} V_{\lambda_i}^\kappa\Big)[1] = \bigoplus_{i\in I} V_{\lambda_i}^\kappa[1] = \bigoplus_{i\in I} V_{\lambda_i}.$$

It now follows that $M = \widehat{M[1]}$ because for all $i \in I$, $V_{\lambda_i}^\kappa = \widehat{V_{\lambda_i}}$.

Conversely, let us assume that $M = \widehat{M[1]}$. We know thanks to Lemma 4.3.10 that $M[1]$ is a $\mathfrak{g}$-submodule of M. As such it is locally finite as $\mathfrak{g}$-module and the action integrates to an action of the algebraic group G so by Theorem 3.1.1 there exists a set I and a family $(\lambda_i)_{i\in I} \in (X^*(T)_+)^I$ such that as $\mathfrak{g}$-module

$$M[1] = \bigoplus_{i\in I} V_{\lambda_i}.$$

Now Proposition 3.2.13 defines for us a morphism of $V^\kappa(\mathfrak{g})$-modules

$$\bigoplus_{i\in I} V^\kappa_{\lambda_i} \longrightarrow M$$

mapping isomorphically each $V_{\lambda_i} \subset V^\kappa_{\lambda_i}$ to the corresponding $V_{\lambda_i} \subset M[1]$. Since $M = \widehat{M[1]}$ this map is surjective. We now turn to the injectivity and denote by K the kernel of this map. The following sequence of $V^\kappa(\mathfrak{g})$-modules is obviously exact

$$0 \longrightarrow K \longrightarrow \bigoplus_{i\in I} V^\kappa_{\lambda_i} \longrightarrow M \longrightarrow 0$$

and it is clear that $K \in V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$. Applying the functor $M \mapsto M[1]$ yields, by Lemma 4.3.13, the following exact sequence

$$0 \longrightarrow K[1] \longrightarrow \left(\bigoplus_{i\in I} V^\kappa_{\lambda_i}\right)[1] \longrightarrow M[1].$$

Now it is true by Lemma 4.3.14 that $(\bigoplus_{i\in I} V^\kappa_{\lambda_i})[1] = \bigoplus_{i\in I} V_{\lambda_i}$ and so the last arrow is an isomorphism by the very definition of I. Hence, $K[1] = 0$ and so $K = 0$ by Lemma 4.3.12 and M is indeed a direct sum of Weyl modules. □

Lemma 4.3.16. *If $M \in V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$, then M is graded by conformal degrees.*

Proof. We fix $M \in V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ and write it as the nondecreasing union $M = \bigcup_{i\geqslant 0} \widehat{M[i]}$. In the rest of this proof, the symbol $*$ stands for either ab or an integer between 1 and r. To show that L_0^* is a locally finite endomorphism of M it is enough to show that each $\widehat{M[i]}$ is L_0^*-stable, which is straightforward as each $\widehat{M[i]}$ is a $V^\kappa(\mathfrak{g})$-submodule, and that L_0^* acts locally finitely on it as i ranges through nonnegative integers.

Let us show it by induction on $i \geqslant 0$, for $i = 0$, it is obvious since $\widehat{M[0]} = \widehat{\{0\}} = \{0\}$. Let $i \geqslant 0$ such that L_0^* acts locally finitely on $\widehat{M[i]}$, we have the following exact sequence in $V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$

$$0 \longrightarrow \widehat{M[i]} \longrightarrow \widehat{M[i+1]} \xrightarrow{\pi} \widehat{M[i+1]}/\widehat{M[i]} \longrightarrow 0.$$

Let $m \in M[i+1] \subset \widehat{M[i+1]} \subset M$ then $\pi(m) \in (\widehat{M[i+1]}/\widehat{M[i]})[1]$ indeed, let $x \in \mathfrak{g}$ and $n \geqslant 0$, we have

$$x_{(n)}\pi(m) = \pi(\underbrace{x_{(n)}m}_{\in M[i]}) = 0.$$

Now it is clear that $\pi(M[i+1])$ spans $\widehat{M[i+1]}/\widehat{M[i]}$ as a $V^\kappa(\mathfrak{g})$-module and so in particular it satisfies the conditions of Proposition 4.3.15 and as such is a direct sum of Weyl modules. It follows from Lemma 4.3.6 that L_0^* acts semisimply on Weyl modules, so L_0^* acts semisimply on $\widehat{M[i+1]}/\widehat{M[i]}$ and in particular locally finitely. Now, consider the following commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & \widehat{M[i]} \arrow[r] \arrow[d, "L_0^*"] & \widehat{M[i+1]} \arrow[r, "\pi"] \arrow[d, "L_0^*"] & \widehat{M[i+1]}/\widehat{M[i]} \arrow[r] \arrow[d, "L_0^*"] & 0 \\ 0 \arrow[r] & \widehat{M[i]} \arrow[r] & \widehat{M[i+1]} \arrow[r, "\pi"] & \widehat{M[i+1]}/\widehat{M[i]} \arrow[r] & 0, \end{tikzcd}$$

it implies together with the induction hypothesis that L_0^* acts locally finitely on $\widehat{M[i+1]}$ and we are done. □

To prove Theorem 4.3.3, it is enough to check that every object $M \in V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ is a direct sum of Weyl modules. Let $M \in V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$, in view of Proposition 4.3.15, to show that M is a direct sum of Weyl modules it is enough to show that $M = \widehat{M[1]}$. Denote by Q the $V^\kappa(\mathfrak{g})$-module $M/\widehat{M[1]}$ and consider the following exact sequence in $V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$

$$0 \longrightarrow \widehat{M[1]} \longrightarrow M \xrightarrow{\pi} N \longrightarrow 0.$$

Assume for a contradiction that $N \neq 0$. By Lemma 4.3.12 there exists $\mu \in X^*(T)_+$ such that the Weyl module V_μ^κ is contained in N, we let $v_\mu \in V_\mu = V_\mu^\kappa[1]$ denote a highest weight vector. As Lemma 4.3.6 shows, the conformal degree of v_μ is given by

$$\left(\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\mu^{ab}, \mu^{ab})}{2}, \frac{C_{\mathfrak{g}_1}(\mu_1)}{2(k_1 + \check{h}_1)}, \dots, \frac{C_{\mathfrak{g}_r}(\mu_r)}{2(k_r + \check{h}_r)}\right)$$

where we write $\mu = \mu^{ab} + \mu_1 + \cdots + \mu^r$ with respect to the decomposition $\mathfrak{h}^* = \mathfrak{z}(\mathfrak{g})^* \oplus \bigoplus_{i=1}^r (\mathfrak{h}^i)^*$. According to Lemma 4.3.16, M and N are graded by conformal degrees hence we can pick $\widetilde{v_\mu} \in M$ of conformal degree $\left(\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\mu^{ab}, \mu^{ab})}{2}, \frac{C_{\mathfrak{g}_1}(\mu_1)}{2(k_1 + \check{h}_1)}, \dots, \frac{C_{\mathfrak{g}_r}(\mu_r)}{2(k_r + \check{h}_r)}\right)$ such that $\pi(\widetilde{v_\mu}) = v_\mu$ we can moreover assume that $\widetilde{v_\mu}$ is a highest weight vector for $\mathfrak{g}$ thanks to Theorem 3.1.1. We have

$$t\mathfrak{g}[[t]]\widetilde{v_\mu} \in \widehat{M[1]},$$

indeed, if $x \in \mathfrak{g}$ and $n > 0$, then

$$\pi(x_{(n)}\widetilde{v_\mu}) = x_{(n)}\pi(\widetilde{v_\mu}) = x_{(n)}v_\mu = 0.$$

Say we can find $\gamma \in \widehat{M[1]}$ such that $\widetilde{v_\mu} - \gamma \in M[1]$. The existence of such a γ is a contradiction because then

$$\widetilde{v_\mu} = \underbrace{\widetilde{v_\mu} - \gamma}_{\in M[1]} + \underbrace{\gamma}_{\in \widehat{M[1]}},$$

which implies $\widetilde{v_\mu} \in \widehat{M[1]}$ and hence $0 = \pi(\widetilde{v_\mu}) = v_\mu$ contradicting the fact that v_μ is a highest weight, hence by definition nonzero, vector. We are looking for a $\gamma \in \widehat{M[1]}$ such that

$$t\mathfrak{g}[[t]](\widetilde{v_\mu} - \gamma) = 0$$

or equivalently such that for all $x \in \mathfrak{g}$ and $n > 0$

$$x_{(n)}\widetilde{v_\mu} = x_{(n)}\gamma.$$

It is clear that Proposition 4.3.15 applies to $\widehat{M[1]}$ thus there exists a set I and a family $(\lambda_i)_{i\in I} \in (X^*(T)_+)^I$ such that

$$\widehat{M[1]} = \bigoplus_{i\in I} V^\kappa_{\lambda_i}$$

as a $V^\kappa(\mathfrak{g})$-module. Let $1 \leqslant s \leqslant r$, $x \in \mathfrak{g}^s$ and $n > 0$. We want to prove that

$$x_{(n)}\widetilde{v_\mu} = 0. \tag{4.31}$$

Assume for a contradiction this is nonzero, because $\widetilde{v_\mu}$ is of conformal degree

$$\left(\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\mu^{ab},\mu^{ab})}{2}, \frac{C_{\mathfrak{g}_1}(\mu_1)}{2(k_1+\check{h}_1)}, \dots, \frac{C_{\mathfrak{g}^s}(\mu_s)}{2(k_s+\check{h}_s)}, \dots, \frac{C_{\mathfrak{g}_r}(\mu_r)}{2(k_r+\check{h}_r)}\right),$$

it follows that $xt^n\widetilde{v_\mu}$ is of conformal degree

$$\left(\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\mu^{ab},\mu^{ab})}{2}, \frac{C_{\mathfrak{g}_1}(\mu_1)}{2(k_1+\check{h}_1)}, \dots, \frac{C_{\mathfrak{g}^s}(\mu_s)}{2(k_s+\check{h}_s)} - n, \dots, \frac{C_{\mathfrak{g}_r}(\mu_r)}{2(k_r+\check{h}_r)}\right).$$

Now we know that $x_{(n)}\widetilde{v_\mu} \in \widehat{M[1]}$ and that the conformal degrees in $\widehat{M[1]} = \bigoplus_{i\in I} V^\kappa_{\lambda_i}$ are all of the form

$$\left(\frac{\kappa_{|\mathfrak{z}(\mathfrak{g})}(\lambda_i^{ab},\lambda_i^{ab})}{2} + n_{ab}, \frac{C_{\mathfrak{g}_1}(\lambda_i^1)}{2(k_1+\check{h}_1)} + n_1, \dots, \frac{C_{\mathfrak{g}^s}(\lambda_i^s)}{2(k_s+\check{h}_s)} + n_s, \dots, \frac{C_{\mathfrak{g}_r}(\lambda_i^r)}{2(k_r+\check{h}_r)} + n_r\right)$$

for some $i \in I$ and $n_{ab}, n_1, \dots, n_r \geq 0$ according to Lemma 4.3.6. In particular, since $x_{(n)}\widetilde{v_\mu} \neq 0$ then there exists $i \in I$ and $n_s \geq 0$ such that the following equality of complex numbers holds

$$\frac{C_{\mathfrak{g}^s}(\mu_s)}{2(k_s+\check{h}_s)} - n = \frac{C_{\mathfrak{g}^s}(\lambda_i^s)}{2(k_s+\check{h}_s)} + n_s$$

which is better written as

$$(2(k_s+\check{h}_s))(n+n_s) + (C_{\mathfrak{g}^s}(\lambda_i^s) - C_{\mathfrak{g}^s}(\mu_s)) = 0.$$

Since $n > 0$ and $n_s \geq 0$ we see that $n + n_s \neq 0$ and notice that $n + n_s$, $C_{\mathfrak{g}^s}(\lambda_i^s)$ and $C_{\mathfrak{g}^s}(\mu_s)$ are rational numbers. This contradicts the irrationality of k_s and proves equality (4.31). Let $x \in \mathfrak{z}(\mathfrak{g})$ and $n > 0$, because we proved that

$$t\mathfrak{g}^{ss}[[t]]\widetilde{v_\mu} = 0$$

and because $x \in \mathfrak{z}(\mathfrak{g})$ we have

$$t\mathfrak{g}^{ss}[[t]](x_{(n)}\widetilde{v_\mu}) = 0.$$

It follows from Lemma 4.3.14 that

$$x_{(n)}\widetilde{v_\mu} \in \bigoplus_{i\in I}\left(V^{\kappa_{|\mathfrak{z}(\mathfrak{g})}}_{\lambda_i^{ab}} \otimes \bigotimes_{s=1}^{r} V_{\lambda_i^s}\right).$$

Because we have chosen $\widetilde{v_\mu}$ to be a highest weight vector for $\mathfrak{g}$ and $x \in \mathfrak{z}(\mathfrak{g})$, it is easily checked that $x_{(n)}\widetilde{v_\mu}$ is again a highest weight vector for $\mathfrak{g}$ if it is nonzero. So in fact we have

$$x_{(n)}\widetilde{v_\mu} \in \bigoplus_{i\in I}\left(V^{\kappa_{|\mathfrak{z}(\mathfrak{g})}}_{\lambda_i^{ab}} \otimes \bigotimes_{s=1}^{r} \mathbb{C}|\lambda_i^s\rangle\right).$$

Choose $h^1,\dots,h^{\dim\mathfrak{z}(\mathfrak{g})}$ an orthonormal basis of $\mathfrak{z}(\mathfrak{g})$ with respect to $\kappa_{|\mathfrak{z}(\mathfrak{g})}$. For each $n>0$ and $1\leqslant j\leqslant \dim\mathfrak{z}(\mathfrak{g})$ we write

$$h^j t^n\widetilde{v_\mu} = \sum_{i\in I} s^i_{j,n}|\lambda_i^{ab}\rangle\otimes|\lambda_i^1\rangle\otimes\cdots\otimes|\lambda_i^r\rangle,$$

where $s^i_{j,n} \in \mathbb{C}\left[(h^\alpha t^{-\beta})_{\substack{1\leqslant\alpha\leqslant\dim\mathfrak{z}(\mathfrak{g})\\ 0<\beta}}\right]$ (recall Proposition 3.2.17). We start by studying some properties of the polynomials $s^i_{j,n}$. Fix $N\geqslant 0$ such that

$$t^{N+1}\mathfrak{g}[[t]]\widetilde{v_\mu} = 0,$$

then we see easily from the definitions that

$$s^i_{j,n} = 0 \text{ for all } n\geqslant N+1.$$

Let's fix $1\leqslant j\leqslant\dim\mathfrak{z}(\mathfrak{g})$ and $n>0$, then for all $1\leqslant\alpha\leqslant\dim\mathfrak{z}(\mathfrak{g})$ and $\beta\geqslant N+1$ the following equalities hold in M

$$h^\alpha t^\beta h^j t^n\widetilde{v_\mu} = h^j t^n(h^\alpha t^\beta\widetilde{v_\mu}) \overset{\beta\geqslant N+1}{=} 0,$$

that is to say

$$\sum_{i\in I}\beta\frac{\partial s^i_{j,n}}{\partial h^\alpha t^{-\beta}}|\lambda_i^{ab}\rangle\otimes|\lambda_i^1\rangle\otimes\cdots\otimes|\lambda_i^r\rangle = 0,$$

or equivalently

$$\beta\frac{\partial s^i_{j,n}}{\partial h^\alpha t^{-\beta}} = 0 \text{ for all } \beta\geqslant N+1 \text{ and all } 1\leqslant\alpha\leqslant\dim\mathfrak{z}(\mathfrak{g}).$$

We just proved that the polynomials we consider depend only of a finite number of variables depending on N, precisely

$$s^i_{j,n} \in \mathbb{C}[(h^\alpha t^{-\beta})_{\substack{1\leqslant\alpha\leqslant\dim\mathfrak{z}(\mathfrak{g})\\ 0<\beta\leqslant N}}].$$

Let $i,i'\in I$, $1\leqslant j,j'\leq\dim\mathfrak{z}(\mathfrak{g})$ and $0\leqslant n,n'\leq N$, the equality in M

$$h^{j'}t^{n'}h^j t^n\widetilde{v_\mu} = h^j t^n h^{j'}t^{n'}\widetilde{v_\mu}$$

implies that

$$n' \frac{\partial}{\partial h^{j'} t^{n'}} s^{i}_{j,n} = n \frac{\partial}{\partial h^{j} t^{n}} s^{i'}_{j',n'}. \tag{4.32}$$

It is clear that there exists a finite subset $J \subset I$ such that for all $i \notin J$, $s^{i}_{j,n} = 0$ for all j, n. This last remark legitimates what follows, say we have found for each $i \in J$ a certain $\gamma_i \in V^{\kappa|_{\mathfrak{z}(\mathfrak{g})}}_{\lambda^{ab}_i} \otimes \bigotimes_{s=1}^{r} \mathbb{C}|\lambda^{s}_{i}\rangle$ such that for all $1 \leqslant j \leqslant \dim \mathfrak{z}(\mathfrak{g})$

$$h^{i} t^{n} \gamma_i = s^{i}_{j,n} |\lambda^{ab}_{i}\rangle \otimes |\lambda^{1}_{i}\rangle \otimes \cdots \otimes |\lambda^{r}_{i}\rangle \text{ for all } 0 < n \leqslant N \tag{4.33}$$

$$h^{i} t^{n} \gamma_i = 0 \text{ for all } N < n. \tag{4.34}$$

It is then easily shown that $\gamma = \sum_{i \in J} \gamma_i$, which makes sense because J is finite, satisfies the desired properties. Hence, we can work with a fixed $i \in J$ which we do now. We are looking for γ_i in the form

$$\gamma_i = s^{i}(\gamma) |\lambda^{ab}_{i}\rangle \otimes |\lambda^{1}_{i}\rangle \otimes \cdots \otimes |\lambda^{r}_{i}\rangle$$

where the unknown polynomials are the $s^{i}(\gamma) \in \mathbb{C}[(h^{\alpha} t^{-\beta})_{\substack{1 \leqslant \alpha \leqslant \dim \mathfrak{z}(\mathfrak{g}) \\ 0 < \beta \leqslant N}}]$. The condition (4.34) is obviously satisfied because we require that β is less than or equal to N. The first condition is equivalent to the following system of differential equation

$$\frac{\partial s^{i}(\gamma)}{\partial h^{\alpha} t^{-\beta}} = \frac{1}{\beta} s^{i}_{\alpha,\beta}$$

for all $1 \leqslant \alpha \leqslant \dim \mathfrak{z}(\mathfrak{g})$ and $0 < \beta \leqslant N$ in the unknown polynomial $s^{i}(\gamma)$. The following algebraic version of Poincaré lemma, in addition to equation (4.32) shows the existence of a solution and finishes the proof.

Lemma 4.3.17. *Let Γ be a finite set and $\mathbb{C}[X_\alpha]_{\alpha \in \Gamma}$ be a polynomial ring. Let $(P_\gamma)_{\gamma \in \Gamma} \in \mathbb{C}[X_\alpha]_{\alpha \in \Gamma}$ such that for all $\alpha, \beta \in \Gamma$*

$$\frac{\partial P_\alpha}{\partial X_\beta} = \frac{\partial P_\beta}{\partial X_\alpha}.$$

Then there exists $P \in \mathbb{C}[X_\alpha]_{\alpha \in \Gamma}$ such that for all $\gamma \in \Gamma$

$$\frac{\partial P}{\partial X_\gamma} = P_\gamma.$$

Proof. Define a polynomial function by setting for all $x = (x_\gamma)_{\gamma \in \Gamma} \in \mathbb{C}^{\Gamma}$

$$P(x) = \sum_{\gamma \in \Gamma} x_\gamma \int_0^1 P_\gamma(tx) \mathrm{d}t.$$

Clearly this is again a polynomial and a straightforward computation concludes. □

The argument we just gave proves the following statement, which we record for later use.

Lemma 4.3.18. *Let $\mathfrak{h}$ be a finite dimensional abelian Lie algebra and κ be a non-degenerate level. Let A, B, C be $V^\kappa(\mathfrak{h})$-modules fitting in an exact sequence of $V^\kappa(\mathfrak{h})$-modules*

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0.$$

Assume that A is a direct sum of Fock spaces, then the following sequence of $\mathfrak{h}$-modules

$$0 \longrightarrow A^{t\mathfrak{h}[[t]]} \longrightarrow B^{t\mathfrak{h}[[t]]} \xrightarrow{p} C^{t\mathfrak{h}[[t]]} \longrightarrow 0$$

is exact.

4.4 Chiral Peter–Weyl decomposition

The map $\pi_L \otimes \pi_R$ makes $\mathcal{D}^\kappa_G$ into a $V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-module. When the level is generic we can give a very explicit description of this structure.

The following statement can be found in §4.4 of [AG02], we give a more detailed proof for the reader's convenience:

Theorem 4.4.1 (Chiral Peter–Weyl)**.** *For any level κ we have*

$$\mathcal{D}^\kappa_G \in V^{\kappa\oplus\kappa^*}(\mathfrak{g}\oplus\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}.$$

Moreover, if κ is generic, then $\mathcal{D}^\kappa_G$ decomposes as a $V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-module as the direct sum of Weyl modules

$$\mathcal{D}^\kappa_G = \bigoplus_{\lambda \in X^*(T)_+} V^\kappa_\lambda \otimes V^{\kappa^*}_{-\omega_0\lambda}. \tag{4.35}$$

Proof. We start by showing that

$$\mathcal{D}^\kappa_G \in V^{\kappa\oplus\kappa^*}(\mathfrak{g}\oplus\mathfrak{g})^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}.$$

Because the conformal degrees of $\mathcal{D}^\kappa_G$ are lower bounded, it is clear that the Lie algebra $t\mathfrak{g}[[t]] \oplus t\mathfrak{g}[[t]]$ acts locally nilpotently. It remains to show that the action of the Lie algebra $\mathfrak{g}\oplus\mathfrak{g}$ is locally finite and integrates to an action of the group G. Recall the generating subset (4.14) and choose an element $x \in \mathcal{D}^\kappa_G$ of this form

$$x = \pi_L(x^{i_1})_{(-m_1)} \dots \pi_L(x^{i_r})_{(-m_r)} f^{j_1}_{(-n_1)} \dots f^{j_s}_{(-n_s)}|0\rangle.$$

Because $\mathcal{O}_G$ is locally finite as a $\mathfrak{g}\oplus\mathfrak{g}$-module there exists $V \subset \mathcal{O}_G$ that is of finite dimension, contains $f^{j_1}, \dots, f^{j_s}$ and is $\mathfrak{g}\oplus\mathfrak{g}$-stable. It is easily checked that, in obvious notations, the vector space

$$\pi_L(\mathfrak{g})_{(-m_1)} \dots \pi_L(\mathfrak{g})_{(-m_r)} V_{(-n_1)} \dots V_{(-n_s)}|0\rangle$$

is finite dimensional, contains x and is $\mathfrak{g}\oplus\mathfrak{g}$-stable. Now to see that the action of $\mathfrak{g}\oplus\mathfrak{g}$ integrates to an action of the algebraic group $G \times G$, one checks that Corollary 3.1.2 applies. This proves the first part of the statement.

Let us now assume that κ is generic. Because the two actions of $\hat{\mathfrak{g}}^\kappa$ and $\hat{\mathfrak{g}}^{\kappa^*}$ commute, it is clear that

$$(\mathcal{D}_G^\kappa)^{t\mathfrak{g}[[t]]\oplus t\mathfrak{g}[[t]]} = ((\mathcal{D}_G^\kappa)^{0\oplus t\mathfrak{g}[[t]]})^{t\mathfrak{g}[[t]]\oplus 0}.$$

Using the description of $\mathcal{D}_G^\kappa$ as an induced module (recall equation (4.10)) and the equality of $\mathfrak{g}$-modules $\mathcal{O}_{\mathcal{J}_\infty G}^{t\mathfrak{g}[[t]]} = \mathcal{O}_G$ we see that

$$(\mathcal{D}_G^\kappa)^{0\oplus t\mathfrak{g}[[t]]} = V^\kappa(\mathfrak{g}) \otimes \mathcal{O}_G.$$

But since

$$(V^\kappa(\mathfrak{g}) \otimes \mathcal{O}_G)^{t\mathfrak{g}[[t]]\oplus 0} = \mathcal{O}_G$$

we have shown that

$$(\mathcal{D}_G^\kappa)^{t\mathfrak{g}[[t]]\oplus t\mathfrak{g}[[t]]} = \mathcal{O}_G$$

as a $\mathfrak{g} \oplus \mathfrak{g}$-module.

Finally, the description of the Kazhdan–Lusztig category of Theorem 4.3.3 together with the algebraic Peter–Weyl theorem (see *e.g.* §27.3.9 of [TY05]) proves equality (4.35). □

Remark 4.4.2. *In the case of a simple group, this was also proven by Minxian Zhu in [Zhu08] (see also [FS06] where $\mathcal{D}_{SL_2}^\kappa$ was introduced and studied under the name of modified regular representation and §5 of [Feĭ17]). A more general statement studying the situation for any level can be found in [Zhu11].*

For generic levels it is easy to deduce the simplicity of $\mathcal{D}_G^\kappa$ from the chiral Peter–Weyl theorem and the simplicity of $\mathcal{O}_G$ as a $\mathcal{D}_G$-module (but note that Proposition 4.1.3 holds for any level). The proof goes as follows, consider an ideal of $\mathcal{D}_G^\kappa$. In particular, it is a $V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-submodule of $\mathcal{D}_G^\kappa$. The chiral Peter–Weyl decomposition shows that it contains a nonzero function $f \in \mathcal{O}_G$. Now it is easily seen that it must contain all elements of the form $D(f)$ where $D \in \mathcal{D}_G$ is a differential operator on G. Since $\mathcal{O}_G$ is a simple $\mathcal{D}_G$-module, it must contain all of $\mathcal{O}_G$, and it is easy to conclude from there.

The following result expresses the rigidity of the chiral Peter–Weyl decomposition.

Theorem 4.4.3. *Let G be a reductive group and κ be a generic level. Then there exists on the $V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-module*

$$\bigoplus_{\lambda \in X^*(T)_+} V_\lambda^\kappa \otimes V_{-\omega_0\lambda}^{\kappa^*}$$

a unique structure of simple vertex algebra that is a conformal extension of $V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$ given by that of $\mathcal{D}_G^\kappa$.

We devote the rest of this paragraph to the proof of the theorem.

It follows from Proposition 4.1.3 and Theorem 4.4.1 that $\mathcal{D}^\kappa_G$ satisfies the conditions of the theorem. Conversely, let

$$\mathcal{A} = \bigoplus_{\lambda \in X^*(T)_+} V^\kappa_\lambda \otimes V^{\kappa^*}_{-\omega_0\lambda}$$

be a vertex algebra satisfying the hypothesis of the theorem. Because the conformal graded is enforced, one sees that $\mathcal{A}$ is $\mathbb{Z}_{\geq 0}$-graded and that

$$\mathcal{A}_0 = \bigoplus_{\lambda \in X^*(T)_+} V_\lambda \otimes V_{-\omega_0\lambda}. \tag{4.36}$$

Moreover, it follows from the Borcherds identities that the pair $(A, -_{(-1)}-)$ is an associative and commutative algebra with unit. For all $x \in \mathfrak{g} \subset V^\kappa(\mathfrak{g})$, $y \in \mathfrak{g} \subset V^{\kappa^*}(\mathfrak{g})$ and $f, g \in \mathcal{A}_0$ it follows from the Borcherds identities that

$$x_{(0)}(f_{(-1)}g) = (x_{(0)}f)_{(-1)}g + f_{(-1)}(x_{(0)}g), \tag{4.37}$$

$$y_{(0)}(f_{(-1)}g) = (y_{(0)}f)_{(-1)}g + f_{(-1)}(y_{(0)}g). \tag{4.38}$$

Assume that we can show that $\mathcal{A}_0 = \mathcal{O}_G$ as commutative $G \times G$-algebras. Then because of the generators of $\mathcal{D}^\kappa_G$ and the relations between them (recall equations (4.11), (4.12), and (4.13)) we have a nonzero morphism of vertex algebras

$$\mathcal{D}^\kappa_G \longrightarrow \mathcal{A}.$$

This morphism has to be injective by the simplicity of $\mathcal{D}^\kappa_G$ (Proposition 4.1.3) and hence is surjective because of the chiral Peter–Weyl decomposition of $\mathcal{D}^\kappa_G$ (equation (4.35)).

We are left to show that $(\mathcal{A}_0, -_{(-1)}-)$ is isomorphic to $\mathcal{O}_G$ as $G \times G$-commutative algebras. Introduce $\mathrm{Zhu}(\mathcal{A})$, the Zhu algebra of $\mathcal{A}$ (see §2 of [Zhu96]). Because $\mathcal{A}_0$ is the top space of the simple $\mathcal{A}$-module $\mathcal{A}$ then $\mathcal{A}_0$ is a simple $\mathrm{Zhu}(\mathcal{A})$-module (see Theorem 2.2.2 of *loc.cit.*). Given a homogeneous element $a \in \mathcal{A}$, its image in $\mathrm{Zhu}(\mathcal{A})$ acts by $a_{(\deg(a)-1)}$ on $\mathcal{A}_0$ (see Theorem 2.1.2 of *loc.cit.*).

Lemma 4.4.4. *The image $\mathcal{D}$ of the map*

$$\mathrm{Zhu}(\mathcal{A}) \longrightarrow \mathrm{End}(\mathcal{A}_0)$$

is generated as an associative algebra by elements of the form

$$x_{(0)}, y_{(0)}, f_{(-1)},$$

with $x \in \mathfrak{g} \subset V^\kappa(\mathfrak{g})$, $y \in \mathfrak{g} \subset V^{\kappa^*}(\mathfrak{g})$, $f \in \mathcal{A}_0$. In particular, $\mathcal{A}_0$ is a simple $\mathcal{D}$-module.

Proof. We proceed by induction on the conformal degree. Elements of conformal degree 0 are precisely elements of $\mathcal{A}_0$ so this step is clear. Now assume that the results holds for elements of $\mathcal{A}$ of degree at most d. A element of $\mathcal{A}$ of degree $d+1$ is a linear combination of elements of the form

$$(x_1)_{(-m_1-1)} \ldots (x_r)_{(-m_r-1)} (y_1)_{(-n_1-1)} \ldots (y_r)_{(-n_r-1)} f$$

where $r, s, m_1, \ldots, m_r, n_1, \ldots, n_s$ are nonnegative integers, $x_1, \ldots, x_r \in \mathfrak{g} \subset V^{\kappa}(\mathfrak{g})$, $y_1, \ldots, y_s \in \mathfrak{g} \subset V^{\kappa^*}(\mathfrak{g})$ and $f \in \mathcal{A}_0$. We fix an element of this form. Because it is of conformal degree $d+1 \geqslant 1$ we can always write it, by symmetry, in the form $x_{(-k-1)}v$ where v is of conformal degree $\leq d$. In fact,

$$\deg(v) = d - k.$$

Now applying Borcherds identity (3.5), we have the equality

$$(x_{(-k-1)}v)_{(d)} = \sum_{i \geqslant 0} (-1)^i \binom{-k-1}{i} \left(x_{(-k-1-i)} v_{(d+i)} - (-1)^{(-k-1)} v_{(-k-1+d-i)} x_{(i)} \right).$$

Let $i \geqslant 0$ then the degree of $v_{(d+i)}$ equals

$$\deg(v) - (d+i) - 1 = d - k - (d+i) - 1 = -k - i - 1 < 0.$$

Because x is of conformal degree 1, the operators $x_{(i)}$ for $i > 0$ act by zero on the top space. So all that remains in the previous sum when acting on the top space is up to sign

$$v_{(-k-1+d)} x_{(0)}$$

and we are done by applying the induction hypothesis to v since

$$\deg(v) - 1 = -k - 1 + d.$$

□

Equalities (4.37) and (4.38) mean that the map

$$-_{(-1)}- : f \otimes g \in \mathcal{A}_0 \otimes \mathcal{A}_0 \longmapsto f_{(-1)} g \in \mathcal{A}_0$$

is $\mathfrak{g} \oplus \mathfrak{g}$-equivariant. But the action of $\mathfrak{g} \oplus \mathfrak{g}$ on $\mathcal{A}_0$ integrates to an action of $G \times G$. As such, the map

$$-_{(-1)}- : f \otimes g \in \mathcal{A}_0 \otimes \mathcal{A}_0 \longmapsto f_{(-1)} g \in \mathcal{A}_0$$

is $G \times G$-equivariant. It is then clear from the definition of $\mathcal{D}$ that a $\mathcal{D}$-stable subset of $\mathcal{A}_0$ is the same as an ideal that is moreover a $\mathfrak{g} \oplus \mathfrak{g}$-submodule which is the same as a $G \times G$-submodule (recall that G is connected). This implies that $\mathcal{A}_0$ is reduced because the nilradical of $\mathcal{A}_0$ is obviously an ideal and is clearly stable by $G \times G$ by the $G \times G$-equivariance of the product. Hence, the nilradical of $\mathcal{A}_0$ is $\mathcal{D}$-stable and simplicity enforces it to be 0 because $1 \neq 0 \in \mathcal{A}_0$ is not nilpotent. We now show that $\mathcal{A}_0$ is an integral domain. Because we already know it is reduced, all we have to show is that $X = \mathrm{Spec}(\mathcal{A}_0)$ is irreducible. By what we explained earlier, $G \times G$ acts on the affine scheme X. Consider an irreducible component of X. It is closed by definition. Because $G \times G$ is connected, any irreducible component is $G \times G$-stable. This implies that the ideal defining the component is stable by $G \times G$. Hence, it is the zero ideal by simplicity. This proves that X is reduced and irreducible and, hence, $\mathcal{A}_0$ is an integral domain.

We now show that $\mathcal{A}_0$ is of finite type. Recall the decomposition of $\mathcal{A}_0$ as $\mathfrak{g} \oplus \mathfrak{g}$-module:

$$\mathcal{A}_0 = \bigoplus_{\lambda \in X^*(T)_+} V_\lambda \otimes V_{-\omega_0 \lambda}.$$

Because $\mathcal{A}_0$ is integral, the product of two highest weight vector is nonzero. In particular, the product of two highest weight vector of weight λ and μ is a highest weight vector of weight $\lambda + \mu$. Now pick elements of $X^*(T)_+$, say $\lambda_1, \ldots, \lambda_r \in X^*(T)_+$ such that any element of $X^*(T)_+$ can be written as a linear combination of these elements with nonnegative integral coefficients. Then the finite dimensional space

$$\bigoplus_{i=1}^{r} V_{\lambda_i} \otimes V_{-\omega_0 \lambda_i}$$

spans $\mathcal{A}_0$ as an algebra, hence $\mathcal{A}_0$ is of finite type.

Because X is a $G \times G$-scheme of finite type it contains a closed orbit. The ideal defining this orbit is $G \times G$-stable. Hence, by simplicity this ideal is zero. So X is a $G \times G$-homogeneous variety and therefore X is smooth, because the characteristic is zero.

We now use the theory of spherical varieties to conclude. The fact that $\mathcal{A}_0$ is spherical follows for instance from the Lemma 2.12 of [Bri10]. While there are many $G \times G$-commutative algebras having the decomposition

$$\bigoplus_{\lambda \in X^*(T)_+} V_\lambda \otimes V_{-\omega_0 \lambda}$$

only one may be smooth as a consequence of Knop's conjectures that were proven by Losev (Theorem 1.3 of [Los09]). This proves that $\mathcal{A}_0 = \mathcal{O}_G$ as a $G \times G$-commutative algebra and finishes the proof.

4.5 Degeneration of the geometric Satake equivalence

We start to investigate the representation theory of $\mathcal{D}_G^\kappa$. The conformal structure of $\mathcal{D}_G^\kappa$ is compatible under $\pi_L \otimes \pi_R$ with the conformal structure given by the Segal–Sugawara construction on $V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$ as we explained in §4.2. In particular any $\mathcal{D}_G^\kappa$-module is naturally a $V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-module by restriction. The following Lemma is an easy consequence of Lemmas 3.2.11 and 3.2.12.

Lemma 4.5.1. *Let M be a $\mathcal{D}_G^\kappa$-module. A vector $m \in M$ is vacuum-like with respect to the $\mathcal{D}_G^\kappa$-module structure if and only if it is vacuum-like with respect to the $V^{\kappa\oplus\kappa^*}(\mathfrak{g}\oplus\mathfrak{g})$-module structure. In particular, any morphism of $V^{\kappa\oplus\kappa^*}(\mathfrak{g}\oplus\mathfrak{g})$-module from $V^{\kappa\oplus\kappa^*}(\mathfrak{g}\oplus\mathfrak{g})$ to M extends uniquely to a morphism of $\mathcal{D}_G^\kappa$-modules from $\mathcal{D}_G^\kappa$ to M. Moreover, if M is a simple $\mathcal{D}_G^\kappa$-module, it is isomorphic as a $\mathcal{D}_G^\kappa$-module to the adjoint module $\mathcal{D}_G^\kappa$ if and only if it contains $V_0^\kappa \otimes V_0^{\kappa^*}$ as a $V^{\kappa\oplus\kappa^*}(\mathfrak{g}\oplus\mathfrak{g})$-submodule.*

We now define some of the categories that will be interesting for us

Definition 4.5.2. *Denote by*

$$\mathcal{D}_G^\kappa\text{−Mod}^{1\times\mathcal{J}_\infty G}$$

the full subcategory of $\mathcal{D}_G^\kappa$−Mod *consisting of modules belonging to* $V^\kappa(\mathfrak{g})$−Mod$^{\mathcal{J}_\infty G}$ *when seen as $V^{\kappa^*}(\mathfrak{g})$-modules. Let*

$$\mathcal{D}_G^\kappa\text{−Mod}^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$$

be the full subcategory of $\mathcal{D}_G^\kappa$−Mod *consisting of modules belonging to the Kazhdan–Lusztig category* $V^{\kappa\oplus\kappa^*}(\mathfrak{g}\oplus\mathfrak{g})$−Mod$^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$ *when seen as $V^{\kappa\oplus\kappa^*}(\mathfrak{g}\oplus\mathfrak{g})$-modules.*

Theorem 4.5.3 (Degeneration of the geometric Satake equivalence)**.** *Let G be a simple algebraic group and κ be a generic level. Then $\mathcal{D}_G^\kappa$ is the only simple object of the category* $\mathcal{D}_G^\kappa$−Mod$^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$*. Moreover, the category* $\mathcal{D}_G^\kappa$−Mod$^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$ *is equivalent to the category of $\mathbb{C}$-vector spaces.*

We devote the rest of this paragraph to the proof of the theorem.

Recall from Definition 4.3.1 that in this setting, the data of a generic level is that of an irrational complex number $k \in \mathbb{C}\setminus\mathbb{Q}$ such that $\kappa = k\widetilde{\kappa_\mathfrak{g}}$.

The fact that $\mathcal{D}_G^\kappa$ is a simple object of the category $\mathcal{D}_G^\kappa$−Mod$^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$ follows from Proposition 4.1.3 and Theorem 4.4.1.

Let M be an object of the category $\mathcal{D}_G^\kappa$−Mod$^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$. Thanks to Theorem 4.3.3 there exists a nonempty set I and two families $(\alpha)_{i\in I}, (\beta_i)_{i\in I} \in X^*(T)_+$ such that we have an isomorphism of $V^\kappa(\mathfrak{g})\otimes V^{\kappa^*}(\mathfrak{g})$-modules

$$M = \bigoplus_{i\in I} V_{\alpha_i}^\kappa \otimes V_{\beta_i}^{\kappa^*}. \tag{4.39}$$

Let $x, y \in \mathfrak{g}$ and $f, g \in \mathcal{O}_G$, then the following equalities hold in $\mathrm{End}(M)$

$$\begin{aligned}
&[\pi_L(x)_{(0)}, \pi_L(y)_{(0)}] = \pi_L([x,y])_{(0)},\\
&[\pi_L(x)_{(0)}, f_{(-1)}] = (x_L(f))_{(-1)},\\
&[f_{(-1)}, g_{(-1)}] = 0,
\end{aligned}$$

in view of equations (4.11), (4.12) and (4.13). This readily implies that M is a representation of the ring $\mathcal{D}_G$ of differential operators on the group G. Let $i \in I$ and consider the $\mathfrak{g}$-submodule $V_{\alpha_i}\otimes V_{\beta_i} \subset M$. Let M' be the $\mathcal{D}_G$-submodule generated by

$V_{\alpha_i} \otimes V_{\beta_i}$. Let $j \in I$, the conformal degrees appearing in the factor $V^{\kappa}_{\alpha_j} \otimes V^{\kappa^*}_{\beta_j}$ are of the form

$$\frac{c_j}{2(k+\check{h})} + n$$

where $c_j \in \mathbb{Q}$ and $n \in \mathbb{Z}_+$ in view of the conformal degrees appearing in Weyl modules (see *e.g.*, Proposition 2.7 of [KL93a] or Theorem 6.2.21 of [LL04]). Now since $k \notin \mathbb{Q}$ and hence $(k+\check{h}) \notin \mathbb{Q}$, for all $i, j \in I$ the equality

$$\frac{c_j}{2(k+\check{h})} + n = \frac{c_i}{2(k+\check{h})}$$

forces n to be zero. But it is clear that the action of $\mathcal{D}_G$ preserves conformal degree. These two facts show that there exists a subset $S \subset I$ such that

$$M' \subset \bigoplus_{s \in S} V_{\alpha_s} \otimes V_{\beta_s}.$$

Using Corollary 3.1.2 we see that M' is a $\mathcal{D}_G$-module such that the action of $\mathfrak{g}$ integrates to an action of the algebraic group G. Then using for instance Proposition 30.18 of [Eti24] we conclude that M' is a G-equivariant $\mathcal{D}_G$-module.

This is easily seen to imply that it is a direct sum of copies of $\mathcal{O}_G$ as a $\mathcal{D}_G$-module and so in particular as a $\mathfrak{g} \oplus \mathfrak{g}$-module. In particular, $0 \in S \subset I$. In other words, $V^{\kappa}_0 \otimes V^{\kappa^*}_0 = V^{\kappa}(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$ appears as a $V^{\kappa}(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-submodule of M. Lemma 4.5.1 then concludes that M contains a copy of $\mathcal{D}^{\kappa}_G$.

So it is now clear that $\mathcal{D}^{\kappa}_G$ is the only simple object of the category. Because any object contains a copy of $\mathcal{D}^{\kappa}_G$, we simply have to show that $\mathcal{D}^{\kappa}_G$ is projective to conclude. Let

$$0 \longrightarrow A \longrightarrow B \xrightarrow{\pi} \mathcal{D}^{\kappa}_G \longrightarrow 0$$

be an exact sequence in the category $\mathcal{D}^{k}_G\text{−Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}$. This is also an exact sequence in the category $V^{\kappa \oplus \kappa^*}(\mathfrak{g})\text{−Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}$. Since $V^{\kappa}(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g}) \subset \mathcal{D}^{\kappa}_G$ is projective in $V^{\kappa \oplus \kappa^*}(\mathfrak{g} \oplus \mathfrak{g})\text{−Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}$ (Theorem 4.3.3) we can lift it to B as a $V^{\kappa \oplus \kappa^*}(\mathfrak{g} \oplus \mathfrak{g})$-module via a map $s : V^{\kappa \oplus \kappa^*}(\mathfrak{g} \oplus \mathfrak{g}) \longrightarrow B$. Using Lemma 4.5.1, we can extend the map s to a map of $\mathcal{D}^{\kappa}_G$-modules

$$\tilde{s} : \mathcal{D}^{\kappa}_G \longrightarrow B.$$

Now it is clear that the map $\tilde{s}$ is a section because $\pi \circ \tilde{s} : \mathcal{D}^{\kappa}_G \longrightarrow \mathcal{D}^{\kappa}_G$ is a map of $\mathcal{D}^{\kappa}_G$-modules that coincides with the identity on $V^{\kappa}(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$. By the simplicity of $\mathcal{D}^{\kappa}_G$ (Proposition 4.1.3) and Schur's Lemma it must be the identity map.

Remark 4.5.4. *We note that the argument appearing at the end of this proof can be generalized as follows. Let V be a conformal vertex algebra and W be a simple conformal vertex algebra that is a conformal extension of V. Let $\mathcal{C}$ be a full subcategory of V*−Mod *such that $V, W \in \mathcal{C}$. Let $\mathcal{C}'$ be the full subcategory of W*−Mod *that consists of W-modules that belong to $\mathcal{C}$ when seen as V-modules. If V is projective in $\mathcal{C}$ then W is projective in $\mathcal{C}'$.*

Chapter 5

Spectral flow in the theory of vertex algebras

In view of Theorem 4.5.3, to build new interesting representations of $\mathcal{D}_G^\kappa$ we have to move away from the subcategory

$$\mathcal{D}_G^\kappa\text{–Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}.$$

This is a motivation to introduce the concept of spectral flow (see Remark 5.2.4). Let us introduce a general framework before applying it in our setting.

5.1 Generalities

Fix a finite dimensional abelian Lie algebra $\mathfrak{h}$ endowed with a nondegenerate symmetric bilinear form κ. Let $V^\kappa(\mathfrak{h})$ be the Heisenberg vertex algebra associated to this data. Let V be a vertex operator superalgebra with conformal vector ω endowed with a vertex superalgebra morphism

$$\varphi : V^\kappa(\mathfrak{h}) \longrightarrow V.$$

This morphism is necessarily injective since $V^\kappa(\mathfrak{h})$ is simple in view of Theorem 4.3.3, so we view $V^\kappa(\mathfrak{h})$ as a vertex subalgebra of V. We require the following two conditions

i) When viewing V as a $V^\kappa(\mathfrak{h})$-module then V is a direct sum of Fock spaces *i.e.*, there exists a set $i \in I$ and a map $\lambda : i \in I \mapsto \lambda_i \in \mathfrak{h}^*$ such that

$$V = \bigoplus_{i \in I} \mathcal{F}^\kappa_{\lambda_i}.$$

ii) All the elements $h \in \mathfrak{h} = V^\kappa(\mathfrak{h})_1$ are primary of weight one in V, that is, we have

$$[\omega_\lambda h] = (\partial + \lambda)h.$$

We associate to this data the following abelian group that depends on V, $\mathfrak{h}$, κ and φ. In the notation we only include the dependency on V as other should always be clear from the context.

Definition 5.1.1. *In the previous notations, we define*

$$\mathcal{SF}(V) = \{h \in V^{\kappa}(\mathfrak{h})_1 = \mathfrak{h} \mid h_{(0)} \text{ acts semisimply on } V \text{ with eigenvalues in } \mathbb{Z}\}.$$

We call $\mathcal{SF}(V)$ the spectral flow group of V.

Following [Li97] we define for any $h \in \mathcal{SF}(V)$ the associated Li's delta operator

$$\Delta(h, z) = z^{h_{(0)}} \exp\left(\sum_{n>0} \frac{h_{(n)}}{-n}(-z)^{-n}\right) \in \mathcal{F}(V). \tag{5.1}$$

We now digress a bit to discuss the meaning of this formula. First the term $z^{h_{(0)}}$ is to be understood in the following sense. Let V be a $\mathbb{C}$-vector space and $A \in \operatorname{End}(V)$ such that A acts semisimply with eigenvalues in $\mathbb{Z}$. Denote by $(\pi_n)_{n\in\mathbb{Z}}$ the family of projectors associated to the direct sum decomposition

$$V = \bigoplus_{n\in\mathbb{Z}} V_n$$

of V in eigenspaces for the semisimple endomorphism A. Now define

$$z^A = \sum_{n\in\mathbb{Z}} \pi_n z^n \in \mathcal{F}(V).$$

That is for each $n \in \mathbb{Z}$ and $v \in V_n$

$$z^A v = v z^n.$$

To understand the meaning of the exponential term appearing in formula (5.1), recall that the operators $(h_{(n)})_{n\in\mathbb{Z}}$ satisfy the Heisenberg commutation relations

$$[h_{(m)}, h'_{(n)}] = m\delta_{m+n}\kappa(h, h')\mathrm{id}$$

for all $h, h' \in \mathfrak{h}$ and $m, n \in \mathbb{Z}$. In particular, the annihilation operators *i.e.*, the non-negative modes $(h_{(n)})_{n\geqslant 0}$, pairwise commute. The exponential is then defined with the usual formula and it indeed defines a field on V. In fact, it is easily checked that more is true. Namely, for all $h \in \mathcal{SF}(V)$ and $v \in V$,

$$\Delta(h, z)v \in V[z, z^{-1}] \subset V((z)).$$

Note that for all $h, h' \in \mathfrak{h}$, we have the equalities

$$\Delta(h, z)\Delta(h', z) = \Delta(h + h', z),$$
$$\Delta(0, z) = \mathrm{id}_V,$$

so that we have

$$\Delta(h,z)^{-1} = \Delta(-h,z).$$

Now, given $h \in \mathcal{SF}(V)$ and a module M of the vertex algebra V, we can define a new module $h \cdot M$ as follows. The underlying vector space of $h \cdot M$ is the same, but the action gets twisted. If $Y_M : V \longrightarrow \mathcal{F}(M)$ is the map defining the module structure of M we define for $v \in V$

$$Y_{h\cdot M}(v,z) = Y_M(\Delta(h,z)^{-1}v,z) = Y_M(\Delta(-h,z)v,z),$$

where $Y_M : V \longrightarrow \mathcal{F}(M)$ is naturally extended to a $\mathbb{C}[z,z^{-1}]$-linear map denoted again $Y_M : V[z,z^{-1}] \longrightarrow \mathcal{F}(M)$. The following theorem summarizes the properties of spectral flow:

Theorem 5.1.2 ([Li97] and Proposition 5.4 of [Li96])**.** *Let V be an operator vertex superalgebra, $\mathfrak{h}$ be a finite dimensional abelian Lie algebra with a nondegenerate symmetric bilinear form κ. Let*

$$\varphi : V^{\kappa}(\mathfrak{h}) \longrightarrow V$$

be a vertex superalgebra morphism satisfying Conditions (i) and (ii).

Let $\mathcal{SF}(V)$ be the spectral flow group associated to this data as in Definition 5.1.1. Let M be a V-module and $h, h' \in \mathcal{SF}(V)$, then $h \cdot M$ is again a V-module and we have $h \cdot (h' \cdot M) \simeq (h + h') \cdot M$ and $0 \cdot M \simeq M$. In particular, $\mathcal{SF}(V)$ acts by invertible exact endofunctors on the category V-mod and sends simple objects to simple objects.

5.2 The spectral flow group of $V^{\kappa}(\mathfrak{g})$

We work out in detail the application of this theory to the Heisenberg vertex algebra and to the universal affine vertex algebra associated with a reductive Lie algebra. Of course, Heisenberg vertex algebras are particular cases of affine vertex algebras, so some results will be generalized later. Yet we believe that to understand what is going on, it is essential to keep this example in mind.

Let $\mathfrak{h}$ be a finite dimensional abelian Lie algebra and κ a nondegenerate symmetric bilinear form on $\mathfrak{h}$. Let $V^{\kappa}(\mathfrak{h})$ be the associated Heisenberg vertex algebra. In the previous notations, we choose φ to be the identity which is easily seen to satisfy conditions (i) and (ii). Recall that by definition

$$\mathcal{SF}(V^{\kappa}(\mathfrak{h})) = \{h \in V^{\kappa}(\mathfrak{h})_1 = \mathfrak{h} \,|\, h_{(0)} \text{ acts semisimply on } V \text{ with eigenvalues in } \mathbb{Z}\}.$$

Proposition 5.2.1. *The spectral flow group $\mathcal{SF}(V^{\kappa}(\mathfrak{h}))$ is equal to $\mathfrak{h}$.*

Proof. We have that for all $h \in \mathfrak{h}$, $h_{(0)} = 0$ in $\mathrm{End}(V^{\kappa}(\mathfrak{h}))$ because the Lie algebra $\mathfrak{h}$ is abelian and of the Heisenberg commutation relations. The result follows. $\square$

We can now compute what happens on any module now. If $x \in \mathcal{SF}(V^\kappa(\mathfrak{h})) = \mathfrak{h}$ and $h \in \mathfrak{h} = V^\kappa(\mathfrak{h})_1 \subset V^\kappa(\mathfrak{h})$ then

$$\Delta(x,z)h = z^{x_{(0)}} \exp\left(\sum_{n>0} \frac{x_{(n)}}{-n}(-z)^{-n}\right) h = h + \kappa(x,h)|0\rangle z^{-1}.$$

Because $V^\kappa(\mathfrak{h})$ is generated as a vertex algebra by elements in $\mathfrak{h} = V^\kappa(\mathfrak{h})_1$, this computation is all we need. Let M be a $V^\kappa(\mathfrak{h})$-module via the map $Y_M : V^\kappa(\mathfrak{h}) \longrightarrow \mathcal{F}(M)$. The action is twisted as follows. Set

$$Y_{x\cdot M}(h,z) := Y_M(\Delta(-x,z)h,z) = Y_M(h,z) - \kappa(x,h)\mathrm{id}z^{-1}.$$

That is to say $x \cdot M$ is the $\hat{\mathfrak{h}}^\kappa$-module such that for all $m \in M$ and $n \in \mathbb{Z}$

$$h_{(n)} \cdot_{new} m = h_{(n)} \cdot_{old} m - \delta_{n,0}\kappa(x,h)m \tag{5.2}$$

where the subscript *old* refers to the given module structure on M and *new* to the one obtained after spectral flow on $x \cdot M$ (recall that the underlying vector space is unchanged). The following is a fundamental example:

Proposition 5.2.2. *Let $\mathfrak{h}$ be an abelian Lie algebra and κ be a nondegenerate symmetric bilinear form on $\mathfrak{h}$. Let $V^\kappa(\mathfrak{h})$ be the associated Heisenberg vertex algebra. Let $x \in \mathcal{SF}(V^\kappa(\mathfrak{h})) = \mathfrak{h}$ and $\lambda \in \mathfrak{h}^*$ and $\mathcal{F}^\kappa_\lambda$ be the Fock space of weight λ then there exists an isomorphism of $V^\kappa(\mathfrak{h})$-modules*

$$x \cdot \mathcal{F}^\kappa_\lambda \simeq \mathcal{F}^\kappa_{\lambda-\kappa(x,\cdot)}.$$

Proof. It is obvious from formula (5.2) that after spectral flow the highest weight vector $|\lambda\rangle$ becomes a highest weight vector of weight $\lambda-\kappa(x,\cdot)$. So Proposition 3.2.13 defines for us a nonzero map of $V^\kappa(\mathfrak{h})$-modules

$$\mathcal{F}^\kappa_{\lambda-\kappa(x,\cdot)} \longrightarrow x \cdot \mathcal{F}^\kappa_\lambda,$$

and we know from Theorems 4.3.3 and 5.1.2 that both modules are simple, hence the map is an isomorphism. □

This result allows us to think of all Fock spaces as being an orbit under spectral flow of the vacuum (*i.e.*, weight zero) Fock space. In fact, since κ is nondegenerate, when h ranges through the spectral flow group $\mathcal{SF}(V^\kappa(\mathfrak{h})) = \mathfrak{h}$, $\kappa(h,\cdot)$ ranges through the whole dual space $\mathfrak{h}^*$, so starting from the vertex algebra $\mathcal{F}^\kappa_0 = V^\kappa(\mathfrak{h})$ itself we obtain every Fock space.

Let $\mathfrak{g}$ be a reductive Lie algebra and κ be a nondegenerate symmetric bilinear $\mathfrak{g}$-invariant form on $\mathfrak{g}$. If we fix a Cartan subalgebra $\mathfrak{h}$ of $\mathfrak{g}$ and denote again by κ the restriction of $\mathfrak{g}$ to $\mathfrak{h}$, it is still nondegenerate. We have a morphism of vertex algebras

$$\varphi : V^\kappa(\mathfrak{h}) \longrightarrow V^\kappa(\mathfrak{g})$$

satisfying conditions (i) and (ii).

Proposition 5.2.3 (See also §2 of [CR13]). *The spectral flow group* $\mathcal{SF}(V^\kappa(\mathfrak{g}))$ *is equal to the additive group of coweights* $\check{P}$.

Proof. Let $x \in \mathcal{SF}(V^\kappa(\mathfrak{g})) \subset \mathfrak{h} \subset \mathfrak{g}$. For every root $\alpha \in \Phi$ pick a nonzero root vector $e^\alpha \in \mathfrak{g}^\alpha$. The following equality holds in $V^\kappa(\mathfrak{g})$

$$x_{(0)}e^\alpha = \alpha(x)e^\alpha|0\rangle.$$

Because we assumed that $x \in \mathcal{SF}(V^\kappa(\mathfrak{g}))$, it forces that $\alpha(h) \in \mathbb{Z}$ for all $\alpha \in \Phi$, this is precisely asking that x belongs to $\check{P}$. Conversely, if $x \in \check{P}$ then one easily checks that $x_{(0)}$ acts semisimply with integral eigenvalues. □

Recall the root space decomposition (3.3) of $\mathfrak{g}$ and let $x \in \check{P}$ be a coweight, $\alpha \in \Phi$ be a root, e^α be a root vector of weight α and $h \in \mathfrak{h}$. The same computation as in the Heisenberg case shows that

$$\Delta(x, z)h = h + \kappa(x, h)|0\rangle z^{-1}.$$

A simple computation shows that

$$\Delta(x, z)e^\alpha = z^{\alpha(x)}e^\alpha.$$

Because $V^\kappa(\mathfrak{g})$ is generated as a vertex algebra by elements in $\mathfrak{g} = V^\kappa(\mathfrak{h})_1$ this computation is all we need. Let M be a $V^\kappa(\mathfrak{g})$-module via $Y_M : V^\kappa(\mathfrak{g}) \longrightarrow \mathcal{F}(M)$. The twist is as follows, for all $h \in \mathfrak{h} \subset V^\kappa(\mathfrak{g})_1$

$$Y_{x \cdot M}(h, z) := Y_M(\Delta(-x, z)h, z) = Y_M(h, z) - \kappa(x, h)\mathrm{id}z^{-1}.$$

For all $\alpha \in \Phi$ and root vector $e^\alpha \in \mathfrak{g}^\alpha \subset V^\kappa(\mathfrak{g})$,

$$Y_{x \cdot M}(e^\alpha, z) := Y_M(\Delta(-x, z)e^\alpha, z) = z^{-\alpha(x)}Y_M(e^\alpha, z).$$

That is to say, $x \cdot M$ is the smooth $\mathfrak{g}^\kappa$-module such that for all $m \in M$ and $n \in \mathbb{Z}$ we have

$$e^\alpha{}_{(n)} \cdot_{new} m = e^\alpha{}_{(n-\alpha(x))} \cdot_{old} m \tag{5.3}$$

$$h_{(n)} \cdot_{new} m = h_{(n)} \cdot_{old} m - \delta_{n,0}\kappa(x, h)m \tag{5.4}$$

where the subscripts *old* refers to the given module structure on M and *new* to the new one obtained after spectral flow on $x \cdot M$ (recall that the underlying vector space is unchanged).

Remark 5.2.4. *It is clear that, for a simple Lie algebra, the Kazhdan–Lusztig category is not stable by spectral flow. Namely, for any nonzero* $x \in \check{P}$ *and* $M \in V^\kappa(\mathfrak{g})\text{–}\mathrm{Mod}^{\mathcal{J}_\infty G}$, *equation* (5.3) *shows that*

$$x \cdot M \notin V^\kappa(\mathfrak{g})\text{–}\mathrm{Mod}^{\mathcal{J}_\infty G}.$$

This highly contrasts with the case of an abelian Lie algebra (compare with Proposition 5.2.2).

5.3 The spectral flow group of $\mathcal{D}_G^\kappa$

Denote again by

$$\pi_L \otimes \pi_R : V^{\kappa \oplus \kappa^*}(\mathfrak{h} \oplus \mathfrak{h}) \longrightarrow \mathcal{D}_G^\kappa$$

the restriction to the Cartan subalgebra $\mathfrak{h} \oplus \mathfrak{h}$ of $\mathfrak{g} \oplus \mathfrak{g}$ of $\pi_L \otimes \pi_R$ (recall (4.23)).

This morphism satisfies condition (i) in view of Theorem 4.4.1, and condition (ii) is clearly satisfied. That allows us to define the spectral flow group:

$$\begin{aligned}\mathcal{SF}(\mathcal{D}_G^\kappa) =& \{(h, h') \in V^{\kappa \oplus \kappa^*}(\mathfrak{h} \oplus \mathfrak{h})_1 \simeq \mathfrak{h} \oplus \mathfrak{h} \mid \\ & \pi_L(h)_{(0)} + \pi_R(h')_{(0)} \text{ acts semisimply with integral eigenvalues on } \mathcal{D}_G^\kappa\}.\end{aligned}$$

Proposition 5.3.1. *The group $\mathcal{SF}(\mathcal{D}_G^\kappa)$ is isomorphic to $X_*(T) \times \check{P}$, an isomorphism is given by*

$$(\gamma, x) \in X_*(T) \times \check{P} \mapsto (\gamma + x, \omega_0^T(x)) \in \mathcal{SF}(\mathcal{D}_G^\kappa).$$

Proof. As explained before, we have the inclusions of vertex algebras

$$V^{\kappa \oplus \kappa^*}(\mathfrak{h} \oplus \mathfrak{h}) \subset V^{\kappa \oplus \kappa^*}(\mathfrak{g} \oplus \mathfrak{g}) \subset \mathcal{D}_G^\kappa.$$

This readily implies the inclusion

$$\mathcal{SF}(\mathcal{D}_G^\kappa) \subset \mathcal{SF}(V^{\kappa \oplus \kappa^*}(\mathfrak{g} \oplus \mathfrak{g})).$$

Using Proposition 5.2.3 to compute the right-hand side, we see that

$$\mathcal{SF}(\mathcal{D}_G^\kappa) \subset \check{P} \times \check{P}.$$

Let $(h, h') \in \mathcal{SF}(\mathcal{D}_G^\kappa)$. In particular, $\pi_L(h)_{(0)} + \pi_R(h)_{(0)}$ should act on $\mathcal{O}_G \subset \mathcal{D}_G^\kappa$ with integral eigenvalues. For all $\lambda \in X^*(T)$ there exists a nonzero function in $\mathcal{O}_G$ of weight $(\lambda, -\omega_0(\lambda))$ with respect to $\mathfrak{h} \oplus \mathfrak{h}$. So, by assumption, we must have

$$\lambda(h) - \omega_0\lambda(h') = \lambda(h - \omega_0^T(h')) \in \mathbb{Z}.$$

Now recall that (3.2) defines a perfect pairing between the free $\mathbb{Z}$-modules $X^*(T)$ and $X_*(T)$ inside of $\mathfrak{h}^*$ and $\mathfrak{h}$. So there exists $\gamma \in X_*(T)$ such that

$$h - \omega_0^T(h') = \gamma.$$

Now, if we write $h' = {\omega_0}^T(x) \in \check{P}$ we get using $(\omega_0^T)^2 = \mathrm{id}$ the equality

$$(h, h') = (x + \gamma, \omega_0^T x).$$

Conversely if we pick $(\gamma, x) \in X_*(T) \times \check{P}$, one can check that $\pi_L(x+\gamma)_{(0)} + \pi_R(\omega_0^T x)_{(0)}$ acts semisimply (this is clear) with integral eigenvalues on $\mathcal{D}_G^\kappa$ using for instance the generators (4.14). □

Remark 5.3.2. *The isomorphism given in the previous proposition is noncanonical. We could have for instance chosen the following one*

$$(\gamma, x) \in X_*(T) \times \check{P} \longmapsto (x, \omega_0^T(x) + \gamma) \in \mathcal{SF}(\mathcal{D}_G^\kappa).$$

The reason why we make this specific choice should become clear later. From now on, we identify $\mathcal{SF}(\mathcal{D}_G^\kappa)$ with $X_*(T) \times \check{P}$ using the isomorphism of Proposition 5.3.1. Note that this results holds for any level.

Chapter 6

Quantum Hamiltonian reduction and spectral flow

6.1 Reminders on quantum Hamiltonian reduction

A good reference is Chapter 15 of [FB04]. Let $\mathfrak{g}$ be a simple Lie algebra with a given Cartan subalgebra $\mathfrak{h}$ and a choice of a Borel $\mathfrak{b}$ containing it. Let $r = \dim \mathfrak{h}$ be the rank of $\mathfrak{g}$. Let $\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$ be the associated triangular decomposition. That is $\mathfrak{n}_\pm = \sum_{\alpha \in \Phi_\pm} \mathfrak{g}^\alpha$ in terms of the notations introduced in §3.1. For each $\alpha \in \Phi_+$, denote by e^α a nonzero root vector in $\mathfrak{g}^\alpha$ and by f^α a nonzero root vector in $\mathfrak{g}^{-\alpha}$ such that

$$\widetilde{\kappa_{\mathfrak{g}}}(e^\alpha, f^\alpha) = 1.$$

Introduce the structure constants $(c_\gamma^{\alpha,\beta})_{\alpha,\beta,\gamma \in \Phi_+} \in \mathbb{C}$ to be such that

$$[e^\alpha, e^\beta] = \sum_{\gamma \in \Phi_+} c_\gamma^{\alpha,\beta} e^\gamma$$

for all $\alpha, \beta \in \Phi_+$. Let $\alpha_1, \dots, \alpha_r \in \Phi_+$ be a basis of simple roots, and let

$$f = \sum_{i=1}^r f^{\alpha_i}.$$

The following formula

$$\chi : x \in \mathfrak{n}_+ \longmapsto \widetilde{\kappa_{\mathfrak{g}}}(f, x) \in \mathbb{C}$$

defines a character of the Lie algebra $\mathfrak{n}_+$ that takes on e^α the value 1 if α is simple and 0 else. Let us denote by $\bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+$ the fermionic ghost vertex algebra associated to $\mathfrak{n}_+$. As a vector space, it is equal to

$$\Big(\bigwedge (\varphi_{\alpha,n})_{\alpha \in \Phi_+, n<0} \otimes \bigwedge (\varphi^*_{\alpha,n})_{\alpha \in \Phi_+, n \leqslant 0}\Big) |0\rangle,$$

where $\bigwedge(a_i)_{i\in I}$ denotes the exterior algebra on the elements $(a_i)_{i\in I}$. The following (super)commutation relations,

$$[\varphi_{\alpha,m}, \varphi_{\beta,n}]_+ = [\varphi^*_{\alpha,m}, \varphi^*_{\beta,n}]_+ = 0,$$
$$[\varphi_{\alpha,m}, \varphi^*_{\beta,n}]_+ = \delta_{\alpha,\beta}\delta_{m,-n},$$

for all $\alpha, \beta \in \Phi_+$ and $m, n \in \mathbb{Z}$, together with the conditions

$$\varphi_{\alpha,m}|0\rangle = 0 \text{ for } m \geqslant 0,$$
$$\varphi^*_{\alpha,n}|0\rangle = 0 \text{ for } n > 0,$$

determine the vertex superalgebra structure. Namely, if we set

$$\varphi_\alpha = \varphi_{\alpha,-1}|0\rangle,$$
$$\varphi^*_\alpha = \varphi^*_{\alpha,0}|0\rangle,$$

and define

$$Y(\varphi_\alpha, z) = \sum_{n\in\mathbb{Z}} \varphi_{\alpha,n} z^{-n-1},\ Y(\varphi^*_\alpha, z) = \sum_{n\in\mathbb{Z}} \varphi^*_{\alpha,n} z^{-n}$$

so that we have for all $n \in \mathbb{Z}$ the equalities

$$(\varphi_\alpha)_{(n)} = \varphi_{\alpha,n},\ (\varphi^*_\alpha)_{(n)} = \varphi_{\alpha,n+1}.$$

It comes equipped with two gradations, to define them fix an element of the form

$$\varphi_{\alpha_1,m_1} \cdots \varphi_{\alpha_r,m_r} \varphi^*_{\beta_1,n_1} \cdots \varphi^*_{\beta_s,n_s}|0\rangle$$

where $r, s \geqslant 0$, $\alpha_1, \ldots, \alpha_r, \beta_1, \ldots, \beta_s \in \Phi_+$, $m_1, \ldots, m_s < 0$ and $n_1, \ldots, n_s \leqslant 0$. Then define the degree of this element to be the integer

$$-(m_1 + \cdots + m_s + n_1 + \cdots + n_s)$$

and its charge to be the integer

$$s - r.$$

Finally, define the parity of such an element to be its charge modulo 2 to obtain a structure of a super vector space on $\bigwedge^{\frac{\infty}{2}+\bullet}(\mathfrak{n}_+)$. For a given $c \in \mathbb{Z}$, denote the vector subspace of charge c of $\bigwedge^{\frac{\infty}{2}+\bullet}(\mathfrak{n}_+)$ to be $\bigwedge^{\frac{\infty}{2}+c}(\mathfrak{n}_+)$.

We now introduce the functor of quantum Drinfeld–Sokolov reduction. Let V be a vertex algebra equipped with a morphism

$$V^\kappa(\mathfrak{g}) \longrightarrow V.$$

Let M be a V-module, define

$$C(M) = M \otimes \bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+.$$

This is naturally a $C(V) = V \otimes \bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+$-module which is $\mathbb{Z}$-graded by the charge gradation inherited from the second factor. We define two odd elements of degree 1 of $C(V)$ as follows

$$Q^{st} = \left(\sum_{\alpha\in\Phi_+} e^{\alpha}_{(-1)}(\varphi^*_\alpha)_{(-1)} - \frac{1}{2}\sum_{\alpha,\beta,\gamma} c^{\alpha,\beta}_{\gamma}(\varphi^*_\alpha)_{(-1)}(\varphi^*_\beta)_{(-1)}(\varphi_\gamma)_{(-1)} \right)|0\rangle,$$

$$Q^{\chi} = \sum_{\alpha\in\Phi_+} \chi(e^\alpha)\varphi^*_\alpha = \sum_{i=1}^{r}(\varphi^*_{\alpha_i})_{(-1)}|0\rangle,$$

and set

$$Q = Q^{st} + Q^{\chi}.$$

Let

$$\mathrm{d}^{st} = Q^{st}_{(0)}, \quad \mathrm{d}^{\chi} = Q^{\chi}_{(0)}, \quad \mathrm{d} = \mathrm{d}^{st} + \mathrm{d}^{\chi}.$$

Because d is a mode of the vertex algebra $C(V)$ it acts on $C(M)$, and it is clear from its definition that it is of degree 1, *i.e.*, for all $c \in \mathbb{Z}$ we have

$$\mathrm{d} : M \otimes \bigwedge^{\frac{\infty}{2}+c}(\mathfrak{n}_+) \longrightarrow \otimes \bigwedge^{\frac{\infty}{2}+c+1}(\mathfrak{n}_+).$$

Moreover $\mathrm{d}\circ\mathrm{d} = 0$ so we can define $H^*_{DS}(M)$ to be the total cohomology of the complex $(C(M), \mathrm{d})$.

Definition 6.1.1. *The vertex algebra*

$$W^\kappa(\mathfrak{g}) = H^*_{DS}(V^\kappa(\mathfrak{g}))$$

is called the principal universal affine $\mathcal{W}$-algebra of level κ for the Lie algebra $\mathfrak{g}$.

Assume we are given a vertex algebra V together with a morphism

$$V^\kappa(\mathfrak{g}) \longrightarrow V$$

and fix a V-module M. Then $H^*_{DS}(V)$ is naturally a vertex algebra such that we have a morphism of vertex algebras

$$W^\kappa(\mathfrak{g}) \longrightarrow H^*_{DS}(V)$$

and moreover, $H^*_{DS}(M)$ is naturally a $H^*_{DS}(V)$-module. So in fact this procedure defines a functor

$$H^*_{DS} : V\text{–Mod} \longrightarrow H^*_{DS}(V)\text{–Mod}.$$

Theorem 6.1.2 (Theorem 15.1.9 of [FB04], Proposition 2 of [FG10] and Theorem 4.15 of [Ara15])**.** *Let $\mathfrak{g}$ be a simple Lie algebra, κ be any level and $V^\kappa(\mathfrak{g})$ the universal affine vertex algebra, then*

$$H^i_{DS}(V^\kappa(\mathfrak{g})) = 0 \text{ if } i \neq 0,$$
$$H^0_{DS}(V^\kappa(\mathfrak{g})) = W^\kappa(\mathfrak{g}).$$

Moreover, if G is a connected reductive algebraic group with Lie algebra $\mathfrak{g}$ and $M \in V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ then for all $i \neq 0$, we have

$$H^i_{DS}(M) = 0.$$

In particular, the functor

$$H^0_{DS} : V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G} \longrightarrow W^\kappa(\mathfrak{g})\text{–Mod}$$

is exact.

6.2 Twisted quantum Hamiltonian reduction

The quantum Hamiltonian reduction functor H^*_{DS} introduced in the previous paragraph can be twisted by a coweight $\check{\mu} \in \check{P}$ to obtain a new functor

$$H^*_{DS,\check{\mu}} : V^\kappa(\mathfrak{g})\text{–Mod} \longrightarrow W^\kappa(\mathfrak{g})\text{–Mod}.$$

It was studied for instance in [AF19] and [ACF22]. We now recall the construction while collecting some facts that will be useful later.

Let

$$\mathfrak{a} = \bigoplus_{\alpha\in\Phi_+} \mathbb{C} b_\alpha$$

be an abelian Lie algebra with basis $(b_\alpha)_{\alpha\in\Phi_+}$ and $\kappa_\mathfrak{a}$ be the unique nondegenerate bilinear form on $\mathfrak{a}$ such that

$$\kappa_\mathfrak{a}(b_\alpha, b_\beta) = \delta_{\alpha,\beta}$$

for all positive root α,β. Let $(\lambda_\alpha)_{\alpha\in\Phi_+}$ be the basis of $\mathfrak{a}^*$ dual to the basis $(b_\alpha)_{\alpha\in\Phi_+}$. Let $V^{\kappa_\mathfrak{a}}(\mathfrak{a})$ be the associated Heisenberg vertex algebra. The following is easy to prove.

Proposition 6.2.1. *There exists an injective morphism of vertex superalgebras*

$$V^{\kappa_\mathfrak{a}}(\mathfrak{a}) \longrightarrow \bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+$$

uniquely determined by the condition that it maps b_α to

$$\varphi^*_{\alpha(-1)}\varphi_{\alpha(-1)}|0\rangle$$

for all $\alpha \in \Phi_+$. Moreover, the charge gradation on $\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+$ coincides with the eigenvalues of the endomorphism

$$\sum_{\alpha\in\Phi_+} b_{\alpha(0)}.$$

This endows $\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+$ with the structure of a $V^{\kappa_\mathfrak{a}}(\mathfrak{a})$-module. The following statement is an instance of a boson-fermion correspondence.

Theorem 6.2.2 (Theorem 5.3.2 of [FB04])**.** *The vertex superalgebra $\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+$ decomposes as a $V^{\kappa_\mathfrak{a}}(\mathfrak{a})$-module as*

$$\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+ = \bigoplus_{(n_\alpha)\in\mathbb{Z}^{\Phi_+}} \mathcal{F}^{\kappa_\mathfrak{a}}_{\sum_{\alpha\in\Phi_+} n_\alpha\lambda_\alpha}. \tag{6.1}$$

Remark 6.2.3. *The right-hand side of equation (6.1) is naturally equipped with the structure of a vertex superalgebra and is called a lattice vertex superalgebra (see §10.2 for the definition). In fact, this is an equality of vertex superalgebras. We will not need it in this chapter.*

We can define, with respect to the morphism of Proposition 6.2.1, the spectral flow group $\mathcal{SF}\left(\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+\right)$.

Lemma 6.2.4. *We have an isomorphism of abelian groups*

$$\mathcal{SF}\left(\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+\right) \simeq \mathbb{Z}^{\Phi_+}$$

obtained by mapping each $(n_\alpha)\in\mathbb{Z}^{\Phi_+}$ to

$$\sum_{\alpha\in\Phi_+} n_\alpha b_\alpha \in \mathfrak{a} \subset V^{\kappa_\mathfrak{a}}(\mathfrak{a}) \subset \bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+.$$

Proof. This is a straightforward consequence of Theorem 6.2.2. □

The morphism of vertex algebras

$$V^\kappa(\mathfrak{h}) \otimes V^{\kappa_\mathfrak{a}}(\mathfrak{a}) \longrightarrow C(V^\kappa(\mathfrak{g}))$$

allows us to consider the spectral flow group $\mathcal{SF}(C(V^\kappa(\mathfrak{g}))$.

Lemma 6.2.5. *The map*

$$(\check{\mu}, (n_\alpha)) \in \check{P}\times\mathbb{Z}^{\Phi_+} \longmapsto \check{\mu}\otimes 1 + 1\otimes \sum_{\alpha\in\Phi_+} n_\alpha b_\alpha \in V^\kappa(\mathfrak{h})\otimes V^{\kappa_\mathfrak{a}}(\mathfrak{a}) \subset C(V^\kappa(\mathfrak{g}))$$

induces an isomorphism of abelian groups

$$\mathcal{SF}(C(V^\kappa(\mathfrak{g}))) \simeq \check{P}\times\mathbb{Z}^{\Phi_+}.$$

Proof. This follows from Proposition 5.2.3, Lemma 6.2.4 and the fact that

$$\mathcal{SF}\left(V^\kappa(\mathfrak{g}) \otimes \bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+\right) \simeq \mathcal{SF}(V^\kappa(\mathfrak{g})) \times \mathcal{SF}\left(\bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+\right).$$

□

The canonical pairing between $\mathfrak{h}$ and its dual $\mathfrak{h}^*$ and the very definition of the group of coweights implies that the following map

$$\iota : \check{\mu} \in \check{P} \longmapsto (\alpha(\check{\mu}))_{\alpha\in\Phi_+} \in \mathbb{Z}^{\Phi_+}$$

is a well-defined group morphism. In particular, to each coweight $\check{\mu} \in \check{P}$ we can associate the element

$$(\check{\mu}, -\iota(\check{\mu})) \in \check{P} \times \mathbb{Z}^{\Phi_+}$$

which, under the isomorphism of Lemma 6.2.5, corresponds to the element

$$C(\check{\mu}) := \check{\mu} - \sum_{\alpha\in\Phi_+} \alpha(\check{\mu}) b_\alpha \in \mathcal{SF}(C(V^\kappa(\mathfrak{g}))). \tag{6.2}$$

Let $\check{\mu} \in \check{P}$ be a coweight and M a $V^\kappa(\mathfrak{g})$-module, then

$$C(M) = M \otimes \bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+$$

is naturally a $C(V^\kappa(\mathfrak{g}))$-module. Applying a spectral flow twist of parameter $C(\check{\mu})$, we see that

$$C(\check{\mu}) \cdot C(M) = (\check{\mu} \cdot M) \otimes (-\iota(\check{\mu}) \cdot \bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+)$$

is again a $C(V^\kappa(\mathfrak{g}))$-module. Recall that, as vector spaces,

$$C(\check{\mu}) \cdot C(M) = C(M)$$

so we can equip it with the charge gradation and with respect to this gradation, the operator d acts with degree 1 and squares to zero. Define $H^*_{DS,\check{\mu}}(M)$ to be the cohomology of the complex $(C(\check{\mu}) \cdot C(M), \mathrm{d})$. It is naturally a $W^\kappa(\mathfrak{g})$-module and we thus get a well-defined functor

$$H^*_{DS,\check{\mu}} : V^\kappa(\mathfrak{g})\text{–Mod} \longrightarrow W^\kappa(\mathfrak{g})\text{–Mod}.$$

6.3 Compatibility with spectral flow

The twisted quantum Hamiltonian reduction functors and the untwisted one are related by the following result:

Theorem 6.3.1. *Let M be a $V^\kappa(\mathfrak{g})$-module and $\check\mu \in \check P$. Then there exists an isomorphism of $W^\kappa(\mathfrak{g})$-modules*

$$H^*_{DS}(\check\mu \cdot M) \simeq H^{*-2\rho(\check\mu)}_{DS,\check\mu}(M)$$

where $\rho = \frac{1}{2}\sum_{\alpha\in\Phi_+}\alpha$.

The rest of this section is devoted to the proof of this theorem. The only difference between the two functors

$$H^*_{DS}(\check\mu \cdot -) \text{ and } H^*_{DS,\check\mu}(-)$$

is a spectral flow twist on the fermionic part but, in view of Theorem 6.2.2, this only amounts to shifting the charge gradation and hence the cohomological degree (see §9.1.1 of [CG20]). We now give the details.

Lemma 6.3.2. *Let $(m_\alpha) \in \mathcal{SF}\left(\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+\right)$. Then we have an isomorphism*

$$S_{(m_\alpha)} : \bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+ \longrightarrow (m_\alpha)\cdot\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+ \tag{6.3}$$

of $\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+$-modules. As vector spaces we have the equality

$$(m_\alpha)\cdot\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+ = \bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+$$

and, endowing $(m_\alpha)\cdot\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+$ with the charge gradation of the right-hand side, the isomorphism $S_{(m_\alpha)}$ is of charge $\sum_{\alpha\in\Phi_+} m_\alpha$.

Proof. Theorem 6.2.2 shows that as $V^{\kappa_\mathfrak{a}}(\mathfrak{a})$-modules we have the equality

$$\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+ = \bigoplus_{(n_\alpha)\in\mathbb{Z}^{\Phi_+}} \mathcal{F}^{\kappa_\mathfrak{a}}_{\sum_{\alpha\in\Phi_+} n_\alpha\lambda_\alpha}.$$

By the computation of the spectral flow group in Lemma 6.2.4, the element $(n_\alpha) \in \mathbb{Z}^{\Phi_+}$ is associated to the element

$$\sum_{\alpha\in\Phi_+} n_\alpha b_\alpha \in V^{\kappa_\mathfrak{a}}(\mathfrak{a})$$

and so, by Proposition 5.2.2, the $\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+$-module

$$(m_\alpha)\cdot\bigwedge^{\frac{\infty}{2}+\bullet}\mathfrak{n}_+$$

decomposes as a $V^{\kappa_\mathfrak{a}}(\mathfrak{a})$-module as

$$\begin{aligned}(m_\alpha)\cdot\bigoplus_{(n_\alpha)\in\mathbb{Z}^{\Phi_+}} \mathcal{F}^{\kappa_\mathfrak{a}}_{\sum_{\alpha\in\Phi_+} n_\alpha\lambda_\alpha} &= \bigoplus_{(n_\alpha)\in\mathbb{Z}^{\Phi_+}} (m_\alpha)\cdot\mathcal{F}^{\kappa_\mathfrak{a}}_{\sum_{\alpha\in\Phi_+} n_\alpha\lambda_\alpha} \\ &= \bigoplus_{(n_\alpha)\in\mathbb{Z}^{\Phi_+}} \mathcal{F}^{\kappa_\mathfrak{a}}_{\sum_{\alpha\in\Phi_+} (n_\alpha-m_\alpha)\lambda_\alpha}.\end{aligned}$$

In particular, $(m_\alpha) \cdot \bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+$ contains the vacuum Fock space $\mathcal{F}_0^{\kappa_\mathfrak{a}} = V^{\kappa_\mathfrak{a}}(\mathfrak{a})$. It is generated by the vector $|(m_\alpha)\rangle$. By Lemma 3.2.11 we see that there exists a unique morphism of $\bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+$-modules

$$S_{(m_\alpha)} : \bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+ \longrightarrow (m_\alpha) \cdot \bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+$$

mapping $|0\rangle$ to $|\sum_{\alpha\in\Phi_+} m_\alpha \lambda_\alpha\rangle$. In particular, this morphism is nonzero. So by the simplicity of $\bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+$ it is an isomorphism. The assertion on the charge of $S_{(m_\alpha)}$ follows from the observation that the charge of the Fock space

$$\mathcal{F}^{\kappa_\mathfrak{a}}_{\sum_{\alpha\in\Phi_+} n_\alpha \lambda_\alpha} \subset \bigwedge^{\frac{\infty}{2}+\bullet} \mathfrak{n}_+$$

is equal to $\sum_{\alpha\in\Phi_+} n_\alpha$ and the fact that $S_{(m_\alpha)}$ is in particular a morphism of $V^{\kappa_\mathfrak{a}}(\mathfrak{a})$-modules. □

Let M be a $V^\kappa(\mathfrak{g})$-module and $\check{\mu} \in \check{P}$, we have an isomorphism of $C(V^\kappa(\mathfrak{g}))$-modules

$$(\mathrm{id} \otimes S_{(-\alpha(\check{\mu}))}) : (\check{\mu}, 0) \cdot C(M) \longrightarrow (\check{\mu}, -\iota(\check{\mu})) \cdot C(M).$$

This isomorphism is an isomorphism of complexes that shifts the gradation by

$$\sum_{\alpha\in\Phi_+} -\alpha(\check{\mu}) = -\left(\sum_{\alpha\in\Phi_+} \alpha\right)(\check{\mu}) = -2\rho(\check{\mu})$$

as explained in Lemma 6.3.2. It induces in cohomology the isomorphism

$$H^*_{DS}(\check{\mu} \cdot M) \simeq H^{*-2\rho(\check{\mu})}_{DS,\check{\mu}}(M)$$

of $W^\kappa(\mathfrak{g})$-modules, and that finishes the proof of Theorem 6.3.1.

Chapter 7

Building modules on $\mathcal{W}_G^\kappa$

7.1 Construction and basic properties of $\mathcal{W}_G^\kappa$

We now introduce the equivariant affine $\mathcal{W}$-algebras following [Ara18]. Assume that G is a simple algebraic group. Recall that we have morphisms of vertex algebras

$$\begin{tikzcd} & V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g}) \arrow[d, "\pi_L \otimes \pi_R"] & \\ V^\kappa(\mathfrak{g}) \arrow[ur] \arrow[r, "\pi_L"] & \mathcal{D}_G^\kappa & V^{\kappa^*}(\mathfrak{g}) \arrow[ul] \arrow[l, "\pi_R"'] \end{tikzcd}$$

so we can perform the Hamiltonian reduction with respect to $V^\kappa(\mathfrak{g})$. We obtain a vertex algebra

$$\mathcal{W}_G^\kappa = H_{DS}^*(\mathcal{D}_G^\kappa),$$

called the equivariant affine $\mathcal{W}$-algebra at level κ of the group G. It comes equipped with morphisms of vertex algebras

$$\begin{tikzcd} & W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g}) \arrow[d, "\pi_L \otimes \pi_R"] & \\ W^\kappa(\mathfrak{g}) \arrow[ur] \arrow[r, "\pi_L"] & \mathcal{W}_G^\kappa & V^{\kappa^*}(\mathfrak{g}). \arrow[ul] \arrow[l, "\pi_R"'] \end{tikzcd}$$

In particular, any $\mathcal{W}_G^\kappa$-module is naturally $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-module by restriction *i.e.*, it carries two commuting actions of $W^\kappa(\mathfrak{g})$ and $V^{\kappa^*}(\mathfrak{g})$.

For all $i \neq 0$ we have

$$H_{DS}^i(\mathcal{D}_G^\kappa) = 0$$

and hence

$$H_{DS}^0(\mathcal{D}_G^\kappa) = \mathcal{W}_G^\kappa$$

as a direct consequence of Theorem 6.1.2 because $\mathcal{D}_G^\kappa \in V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$.

It was proven by Arakawa that the associated Poisson scheme of $\mathcal{W}_G^\kappa$ is the principal equivariant Slodowy slice $\mathcal{S}_G$ which is smooth, reduced and symplectic (see the discussion following Corollary 6.2 of [Ara18]). Using Proposition 4.1.2 we have the following result:

Lemma 7.1.1. *The vertex algebra $\mathcal{W}_G^\kappa$ is simple.*

To understand the structure of $\mathcal{W}_G^\kappa$ as a $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-module one needs to understand the behaviour of Hamiltonian reduction on the Kazhdan–Lusztig category:

Proposition 7.1.2 (Corollary 9.1.6. of [Ara07])**.** *Assume that $\kappa \in \mathbb{C} \setminus \mathbb{Q}$ and let $\lambda \in P_+$ then*

$$T^\kappa_{\lambda,0} := H^0_{DS}(V^\kappa_\lambda)$$

is a simple $W^\kappa(\mathfrak{g})$-module.

Remark 7.1.3. *In fact, $T^\kappa_{\lambda,0}$ is a highest weight module of $W^\kappa(\mathfrak{g})$. See §5 of [Ara07] for what it means to be of highest weight in this context.*

Lemma 7.1.4. *We have a decomposition*

$$\mathcal{W}_G^\kappa = \bigoplus_{\lambda \in X^*(T)_+} T^\kappa_{\lambda,0} \otimes V^{\kappa^*}_{-\omega_0\lambda}$$

as $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-modules.

Proof. Applying the quantum Hamiltonian reduction functor with respect to the left action to the decomposition of Theorem 4.4.1 we obtain that, as $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-modules,

$$\mathcal{W}_G^\kappa = \bigoplus_{\lambda \in X^*(T)_+} H^0_{DS}(V^\kappa_\lambda) \otimes V^{\kappa^*}_{-\omega_0\lambda}$$

and the results now follows from Proposition 7.1.2. □

As explained before we have a morphism of vertex algebras

$$V^{\kappa^*}(\mathfrak{g}) \longrightarrow \mathcal{W}_G^\kappa.$$

It allows us to define the spectral flow group $\mathcal{SF}(\mathcal{W}_G^\kappa)$ of $\mathcal{W}_G^\kappa$. The following is an easy consequence of Lemma 7.1.4:

Corollary 7.1.5. *We have the equality of abelian groups*

$$\mathcal{SF}(\mathcal{W}_G^\kappa) = X_*(T).$$

The following property is a very useful trick we will apply later to prove the simplicity of certain modules (compare with Lemma 8.1.3).

Proposition 7.1.6. *Let M be a $\mathcal{W}^{\kappa}_{G}$-module and assume that the module M contains $V^{\kappa^*}(\mathfrak{g})$ as a $V^{\kappa^*}(\mathfrak{g})$-submodule. Then for all $\lambda \in X^*(T)_+$, M contains $V^{\kappa^*}_{-\omega_0\lambda}$ as a $V^{\kappa^*}(\mathfrak{g})$-submodule.*

Proof. Let $|-,0\rangle \in M$ denote the highest weight vector of weight 0 that spans the given copy of $V^{\kappa^*}(\mathfrak{g})$. In view of the decomposition of Lemma 7.1.4, we can choose for each $\lambda \in X^*(T)_+$ an element $|-,-\omega_0\lambda\rangle =\in T^{\kappa}_{\lambda,0} \otimes V^{\kappa^*}_{-\omega_0\lambda}$ that is a highest weight vector of weight $-\omega_0\lambda$. Because $\mathcal{W}^{\kappa}_{G}$ is simple by Lemma 7.1.1 it follows from Proposition 3.2.10 that there exists a unique $m \in \mathbb{Z}$ such that

$$\begin{aligned}
&|-,-\omega_0\lambda\rangle_{(m)}|-,0\rangle \neq 0,\\
&|-,-\omega_0\lambda\rangle_{(k)}|-,0\rangle = 0 \text{ for all } k > m.
\end{aligned}$$

Let $x \in \mathfrak{g}$ and $n \geqslant 0$,

$$x_{(n)}|-,-\omega_0\lambda\rangle_{(m)}|-,0\rangle = |-,-\omega_0\lambda\rangle_{(m)}x_{(n)}|-,0\rangle + [x_{(n)}, |-,-\omega_0\lambda\rangle_{(m)}]|-,0\rangle.$$

It is clear that the first term vanishes, then we have by Borcherds identity (3.4)

$$x_{(n)}|-\omega_0\lambda\rangle_{(m)}|-,0\rangle = \sum_{i\geqslant 0}\binom{n}{i}(x_{(i)}|-,-\omega_0\lambda\rangle)_{(m+n-i)}|-,0\rangle = (x_{(0)}|-,-\omega_0\lambda\rangle)_{(m+n)}|-,0\rangle \tag{7.1}$$

where the last equality follows from the fact that $|-,-\omega_0\lambda\rangle$ is a highest weight vector and hence is annihilated by $t\mathfrak{g}[[t]]$.

Assume that $n = 0$ and x is a positive root vector, then equation (7.1) shows that

$$x_{(0)}|-\omega_0\lambda\rangle_{(m)}|-,0\rangle = 0.$$

If $x \in \mathfrak{h}$ we see from the same equation that

$$x_{(0)}|-\omega_0\lambda\rangle_{(m)}|-,0\rangle = -\omega_0(\lambda)(x)|-\omega_0\lambda\rangle_{(m)}|-,0\rangle.$$

Finally, we see that the $\mathfrak{g}$-submodule generated by $|-\omega_0\lambda\rangle_{(m)}|-,0\rangle$ is clearly of finite dimension since the one generated by $|-\omega_0\lambda\rangle$ is.

Now if $n > 0$, using Borcherds identity (3.5) in equation (7.1) we see that

$$x_{(n)}|-\omega_0\lambda\rangle_{(m)}|-,0\rangle = x_{(0)}|-\omega_0\lambda\rangle_{(m+n)}|-,0\rangle - |-\omega_0\lambda\rangle_{(m+n)}x_{(0)}|-,0\rangle. \tag{7.2}$$

The choice of m shows that

$$x_{(0)}|-\omega_0\lambda\rangle_{(m+n)}|-,0\rangle = 0.$$

Hence, equality (7.2) together with the obvious fact that

$$x_{(0)}|-,0\rangle = 0$$

shows that $|-\omega_0\lambda\rangle_{(m)}|-,0\rangle$ is annihilated by $t\mathfrak{g}[[t]]$. We therefore see that Proposition 3.2.13 applies and we are done. $\square$

7.2 Building simple modules

Recall from §6.1 that we have a well-defined functor

$$H^*_{DS} : \mathcal{D}^\kappa_G\text{–Mod} \longrightarrow \mathcal{W}^\kappa_G\text{–Mod}.$$

The techniques developed in §5.3 show that for each $(\gamma, x) \in \mathcal{SF}(\mathcal{D}^\kappa_G)$ we have a simple $\mathcal{D}^\kappa_G$-module $((\gamma, x) \cdot \mathcal{D}^\kappa_G)_{\gamma\in X_*(T), x\in\check{P}}$ as a direct application of Theorem 5.1.2 together with Proposition 4.1.3.

The goal now is, given $\gamma \in X_*(T)$ and $x \in \check{P}$, to understand the module

$$H^*_{DS}((\gamma, x) \cdot \mathcal{D}^\kappa_G) \in \mathcal{W}^\kappa_G\text{–Mod}.$$

As $V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-modules we have the equality

$$(\gamma, x) \cdot \mathcal{D}^\kappa_G = \bigoplus_{\lambda\in X^*(T)_+} (\gamma + x) \cdot V^\kappa_\lambda \otimes (\omega_0^T x) \cdot V^{\kappa^*}_{-\omega_0\lambda}$$

so that, as $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-modules, we have the equality

$$H^*_{DS}((\gamma, x) \cdot \mathcal{D}^\kappa_G) = \bigoplus_{\lambda\in X^*(T)_+} H^*_{DS}((\gamma + x) \cdot V^\kappa_\lambda) \otimes ({}^t\omega_0 x) \cdot V^{\kappa^*}_{-\omega_0\lambda}. \tag{7.3}$$

To understand the structure of these modules as $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-module one needs to understand the behaviour of twisted Hamiltonian reduction on the Kazhdan–Lusztig category. The following was conjectured by Creutzig and Gaiotto (see Conjecture 1.5 of [CG20]) and is now a theorem of Arakawa and Frenkel ([AF19]).

Lemma 7.2.1. *Let $\lambda \in P_+$ and $\check{\mu} \in \check{P} = \mathcal{SF}(V^\kappa(\mathfrak{g}))$. Then if $\check{\mu} \notin \check{P}_+$ we have*

$$H^*_{DS}(\check{\mu} \cdot V^\kappa_\lambda) = 0.$$

If $\check{\mu} \in \check{P}_+$, then

$$H^i_{DS}(\check{\mu} \cdot V^\kappa_\lambda) = 0 \textit{ if } i \neq 2\rho(\check{\mu}),$$

and we have that

$$T^\kappa_{\lambda,\check{\mu}} := H^{2\rho(\check{\mu})}_{DS}(\check{\mu} \cdot V^\kappa_\lambda)$$

is a simple $W^\kappa(\mathfrak{g})$-module.

Proof. In view of Theorem 6.3.1 we have

$$H^*_{DS}(\check{\mu} \cdot V^\kappa_\lambda) = H^{*-2\rho(\check{\mu})}_{DS,\check{\mu}}(V^\kappa_\lambda)$$

and the result then follows from Theorems 2.1 and 4.3 of [AF19]. □

Remark 7.2.2. *Again, $T^\kappa_{\lambda,\check{\mu}}$ is a highest weight module of $W^\kappa(\mathfrak{g})$. This family of $W^\kappa(\mathfrak{g})$-modules enjoys remarkable duality properties with respect to the Feigin–Frenkel duality of $\mathcal{W}$-algebras that were independently conjectured by Creutzig–Gaiotto and Gaitsgory and later proven by Arakawa and Frenkel (see Theorem 2.2 of [AF19]).*

Proposition 7.2.3. *Let $\gamma \in X_*(T)$. We have the equalities as $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-modules*

$$\begin{aligned} &H^*_{DS}(\gamma \cdot \mathcal{D}^\kappa_G) = 0, && \text{if } \gamma \notin X_*(T)_+,\\ &H^*_{DS}(\gamma \cdot \mathcal{D}^\kappa_G) = H^{2\rho(\gamma)}_{DS}(\gamma \cdot \mathcal{D}^\kappa_G) = \bigoplus_{\lambda \in X^*(T)_+} T^\kappa_{\lambda,\gamma} \otimes V^{\kappa^*}_{-\omega_0\lambda}, && \text{if } \gamma \in X_*(T)_+. \end{aligned}$$

Let $x \in \check{P}$, more generally we have the equalities as $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-modules

$$\begin{aligned} &H^*_{DS}((\gamma, x) \cdot \mathcal{D}^\kappa_G) = 0 && \text{if } \gamma + x \notin \check{P}_+,\\ &H^n_{DS}((\gamma, x) \cdot \mathcal{D}^\kappa_G) = 0 && \text{if } n \neq 2\rho(\gamma + x).\\ &H^{2\rho(\gamma+x)}_{DS}((\gamma, x) \cdot \mathcal{D}^\kappa_G) = \bigoplus_{\lambda \in X^*(T)_+} T^\kappa_{\lambda,\gamma+x} \otimes (\omega_0^T x) \cdot V^{\kappa^*}_{-\omega_0\lambda} && \text{if } \gamma + x \in \check{P}_+. \end{aligned}$$

Proof. This follows directly from equation (7.3) and Lemma 7.2.1. □

Define

$$\mathcal{W}^\kappa_G\text{–Mod}^{\mathcal{J}_\infty G}$$

to be the full subcategory of $\mathcal{W}^\kappa_G$-modules that belong to the Kazhdan–Lusztig category $V^{\kappa^*}(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ when seen as $V^{\kappa^*}(\mathfrak{g})$-modules by restriction.

We can now state a conjecture which is a version of the fundamental local equivalence of the quantum geometric Langlands program in our language:

Conjecture 7.2.4. *For generic κ, the category $\mathcal{W}^\kappa_G\text{–Mod}^{\mathcal{J}_\infty G}$ is semisimple with simple objects given by the $H^{2\rho(\gamma)}_{DS}(\gamma \cdot \mathcal{D}^\kappa_G)$ for $\gamma \in X_*(T)_+$. That is, we have an equivalence of abelian categories*

$$\mathcal{W}^\kappa_G\text{–Mod}^{\mathcal{J}_\infty G} \simeq \check{G}\text{–Mod}$$

that sends for all $\gamma \in X_*(T)_+$ the simple $\mathcal{W}^\kappa_G$-module $H^{2\rho(\gamma)}_{DS}(\gamma \cdot \mathcal{D}^k_G)$ to the simple $\check{G}$-module V_γ.

Theorem 7.2.5. *The family of $\mathcal{W}^\kappa_G$-modules $(H^{2\rho(\gamma)}_{DS}(\gamma \cdot \mathcal{D}^\kappa_G))_{\gamma \in X_*(T)_+}$ consists of pairwise nonisomorphic simple objects of $\mathcal{W}^\kappa_G\text{–Mod}^{\mathcal{J}_\infty G}$.*

Proof. It is clear from Proposition 7.2.3 that these modules belong to $\mathcal{W}^\kappa_G\text{–Mod}^{\mathcal{J}_\infty G}$ and are pairwise nonisomorphic. Let M be a $\mathcal{W}^\kappa_G$-submodule of $H^{2\rho(\gamma)}_{DS}(\gamma \cdot \mathcal{D}^\kappa_G)$ and define

$$M' = H^{2\rho(\gamma)}_{DS}(\gamma \cdot \mathcal{D}^\kappa_G)/M.$$

We want to show that one of M or M' is the zero module. For all $\lambda \in X^*(T)_+$ the $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-module $T^\kappa_{\lambda,\gamma} \otimes V^{\kappa^*}_{-\omega_0\lambda}$ is simple by Theorem 4.3.3 and Lemma 7.2.1.

Proposition 7.2.3 expresses $H^{2\rho(\gamma)}_{DS}(\gamma \cdot \mathcal{D}^\kappa_G)$ as a direct sum of simple $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-modules each appearing with multiplicity one. It follows that there exists a subset $S \subset X^*(T)_+$ such that

$$M = \bigoplus_{\lambda \in S} T^\kappa_{\lambda,\gamma} \otimes V^{\kappa^*}_{-\omega_0\lambda} \tag{7.4}$$

and

$$M' = \bigoplus_{\lambda \in X^*(T)_+ \setminus S} T^\kappa_{\lambda,\gamma} \otimes V^{\kappa^*}_{-\omega_0\lambda} \tag{7.5}$$

as $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-modules. It is obvious that $0 \in S$ or $0 \in X^*(T)_+ \setminus S$. Let us focus on the second case, this means that

$$T^\kappa_{0,\gamma} \otimes V^{\kappa^*}_0 = T^\kappa_{0,\gamma} \otimes V^{\kappa^*}(\mathfrak{g}) \subset M'.$$

Proposition 7.1.6 applies and shows that for all $\lambda \in X^*(T)_+$ there exists a highest weight vector for $\hat{\mathfrak{g}}^{\kappa^*}$ of weight $-\omega_0\lambda$. Using equation (7.5) it means that $-\omega_0\lambda \in X^*(T)_+ \setminus S$ for all $\lambda \in X^*(T)_+$. This proves that $S = \emptyset$ and hence M is the zero module. The other case is treated similarly. □

In fact, we can say a bit more, the situation is especially nice when we assume the group G to be simple and of adjoint type, that is, the center of G is trivial, or equivalently, $X_*(T) = \check{P}$. A typical example is $G = \mathrm{PSL}_2$.

Proposition 7.2.6. *Let G be simple of adjoint type. For all $(\gamma, x) \in X_*(T) \times \check{P} \in \mathcal{SF}(\mathcal{D}^\kappa_G)$ such that $\gamma + x \in X^*(T)_+$, the $\mathcal{W}^\kappa_G$-module $H^{2\rho(\gamma+x)}_{DS}((\gamma, x) \cdot \mathcal{D}^\kappa_G)$ is simple.*

Proof. By Corollary 7.1.5 we have that $\mathcal{SF}(\mathcal{W}^\kappa_G) = X_*(T)$. But since G is of adjoint type we have $X_*(T) = \check{P}$ so that we can consider the $\mathcal{W}^\kappa_G$-module

$$(-\omega_0^T x) \cdot H^{2\rho(\gamma+x)}_{DS}((\gamma, x) \cdot \mathcal{D}^\kappa_G).$$

Clearly it decomposes as a $W^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})$-module as

$$\bigoplus_{\lambda \in X^*(T)_+} T^\kappa_{\lambda,\gamma+x} \otimes V^{\kappa^*}_{-\omega_0\lambda}.$$

Now the proof of Theorem 7.2.5 applies to prove that $(-\omega_0^T x) \cdot H^{2\rho(\gamma+x)}_{DS}((\gamma, x) \cdot \mathcal{D}^\kappa_G)$ is simple and hence so is $H^{2\rho(\gamma+x)}_{DS}((\gamma, x) \cdot \mathcal{D}^\kappa_G)$. □

Remark 7.2.7. *We made the choice when considering quantum Hamiltonian reduction to restrict our attention to the case where $\mathfrak{g}$ (and hence the underlying group G) is simple. The generalization of those concepts and of the proof of Theorem 7.2.5 is straightforward to the full reductive case. Namely, on the abelian part this reduction procedure is trivial and on the semisimple part of $\mathfrak{g}$ it works separately simple factor by simple factor. Conjecture 7.2.4 then makes sense for any reductive group G. In the same spirit, the further perspectives discussed in Chapter 11 can be generalized to the reductive case.*

Chapter 8

The case of an algebraic torus

Let $T = \mathbb{G}_m^r$ be an algebraic torus. Its Lie algebra $\mathfrak{h}$ is a finite dimensional abelian Lie algebra of dimension r. Let κ be a nondegenerate level. Recall that in this case $\kappa^* = -\kappa$. For all $\alpha \in X^*(T) = X^*(T)_+$, we let $e^\alpha \in \mathcal{O}_T$ be a function of highest weight $(\alpha, -\alpha)$. In this setting, Theorem 4.4.1 says that we have a decomposition

$$\mathcal{D}_T^\kappa = \bigoplus_{\alpha \in X^*(T)} \mathcal{F}_\alpha^\kappa \otimes \mathcal{F}_{-\alpha}^{-\kappa}$$

as $V^\kappa(\mathfrak{h}) \otimes V^{\kappa^*}(\mathfrak{h})$-modules.

It can be seen from Theorem 4.4.3 that $\mathcal{D}_T^\kappa$ is a half-lattice vertex algebra (in the sense of [BDT02]). We learned the theory of lattice vertex algebras from [DLM97], [BDT02] and §6.4 and §6.5 of [LL04]. It is more convenient for our purposes to reproduce fully the results of the previously cited references. Our main input is the reformulation of certain well-known results using the spectral flow group introduced in Chapter 5, while systematically keeping track of the level κ.

Lemma 8.1.1. *Let M be a $V^\kappa(\mathfrak{h})$-module. Then for all $h \in \mathfrak{h}$ the formula*

$$E^-(h,z) = \exp\left(\sum_{n<0} \frac{h_{(n)}}{-n} z^{-n}\right) = \sum_{r \leqslant 0} \left(\sum_{k \geqslant 0} \frac{1}{k!} \sum_{\substack{i_1+\dots+i_k=r \\ i_1,\dots,i_k<0}} \frac{h_{(i_1)}}{-i_1} \cdots \frac{h_{(i_k)}}{-i_k}\right) z^{-r}$$

defines an element of $\mathrm{End}(M)[[z]]$*. Moreover, for all $x \in \mathfrak{h}$ and $m \in \mathbb{Z}$, the following commutation relations hold*

$$[x_{(m)}, E^-(h,z)] = 0, \qquad \text{if } m \leqslant 0 \tag{8.1}$$

$$[x_{(m)}, E^-(h,z)] = \kappa(x,h) z^m E^-(h,z) \qquad \text{if } m \geqslant 1. \tag{8.2}$$

Proof. The fact that this is well-defined is easy. Let

$$H^-(z) = \sum_{n<0} \frac{h_{(n)}}{-n} z^{-n} \in \mathrm{End}(M)[[z]].$$

By definition, we have

$$E^-(h,z) = \sum_{k\geqslant 0} \frac{1}{k!} H^-(z)^k.$$

The case $m \leqslant 0$ is straightforward, so assume that $m \geqslant 1$. Then

$$[x_{(m)}, H^-(z)] = \sum_{n<0} [x_{(m)}, h_{(n)}] \frac{z^{-n}}{-n} = \sum_{n<0} m\delta_{m,-n}\kappa(x,h)\frac{z^{-n}}{-n} = \kappa(x,h)z^m,$$

hence

$$\begin{aligned}
[x_{(m)}, E^-(h,z)] = \sum_{k\geqslant 0} \frac{1}{k!}[x_{(m)}, H^-(z)^k] &= \sum_{k\geqslant 0} \frac{1}{k!} kH^-(z)^{k-1}[x_{(m)}, H^-(z)] \\
&= \kappa(x,h)z^m E^-(h,z).
\end{aligned}$$

□

Lemma 8.1.2. *Let M be a $\mathcal{D}_T^\kappa$-module. In particular, this is a $V^{\kappa\oplus\kappa^*}(\mathfrak{h}\oplus\mathfrak{h})$-module and for all $\alpha \in X^*(T)$, set*

$$A(\alpha)(z) = E^-((-\alpha,-\alpha),z)e^\alpha(z)$$

and write

$$A(\alpha)(z) = \sum_{n\in\mathbb{Z}} A(\alpha)_n z^{-n-1}.$$

Then $A(\alpha)(z)$ is a well-defined element of $\mathrm{End}(M)[[z,z^{-1}]]$ and for all $x \in \mathfrak{h}$, and $m,n \in \mathbb{Z}$, we have:

$$[\pi_L(x)_{(m)}, A(\alpha)_n] = \kappa(x,\alpha)A(\alpha)_{m+n}, \qquad \text{if } m \leqslant 0 \tag{8.3}$$

$$[\pi_R(x)_{(m)}, A(\alpha)_n] = -\kappa(x,\alpha)A(\alpha)_{m+n}, \qquad \text{if } m \leqslant 0 \tag{8.4}$$

$$[\pi_L(x)_{(m)}, A(\alpha)_n] = 0, \qquad \text{if } m \geqslant 1 \tag{8.5}$$

$$[\pi_R(x)_{(m)}, A(\alpha)_n] = 0, \qquad \text{if } m \geqslant 1 \tag{8.6}$$

$$(-n-1)A(\alpha)_n = \sum_{i\geqslant 0} A(\alpha)_{n-i}(\pi_L(\alpha)_{(i)} + \pi_R(\alpha)_{(i)}). \tag{8.7}$$

Proof. It is easy to see that $A(\alpha)(z)$ is well-defined because $E^-((-\alpha,-\alpha),z) \in \mathrm{End}(M)[[z]]$ and $e^\alpha(z)$ is a field. Recall that for all $m \in \mathbb{Z}$ the following equalities hold

$$[\pi_L(x)_{(m)}, e^\alpha(z)] = \kappa(x,\alpha)z^m e^\alpha(z), \tag{8.8}$$

$$[\pi_R(x)_{(m)}, e^\alpha(z)] = -\kappa(x,\alpha)z^m e^\alpha(z). \tag{8.9}$$

Assume for instance that $m \leqslant 0$. Then

$$\begin{aligned}
[\pi_R(x)_{(m)}, A(\alpha)(z)] &= [\pi_R(x)_{(m)}, E^-((-\alpha,-\alpha),z)]e^\alpha(z) + E^-((-\alpha,-\alpha),z)[\pi_R(x)_{(m)}, e^\alpha(z)] \\
&= 0 - \kappa(x,\alpha)z^m E^-((-\alpha,-\alpha),z)e^\alpha(z) \\
&= -\kappa(x,\alpha)z^m A(\alpha)(z),
\end{aligned}$$

which prove the announced equality by looking at the coefficients. The one for $\pi_L(x)$ is proven similarly. Now assume that $m \geqslant 1$. Then

$$\begin{aligned}[\pi_R(x)_{(m)}, A(\alpha)(z)] &= -\kappa(x,-\alpha)z^m E^-((-\alpha,-\alpha),z)e^\alpha(z) - \kappa(x,\alpha)z^m E^-((-\alpha,-\alpha),z)e^\alpha(z)\\ &= 0,\end{aligned}$$

which proves the equality. Again, the one for $\pi_L(x)$ is proven similarly. We now compute

$$\begin{aligned}\frac{\partial}{\partial z}A(\alpha)(z) &= \frac{\partial}{\partial z}(E^-((-\alpha,-\alpha),z)e^\alpha(z))\\ &= \frac{\partial}{\partial z}(E^-((-\alpha,-\alpha),z))e^\alpha(z) + E^-((-\alpha,-\alpha),z)\frac{\partial}{\partial z}(e^\alpha(z)).\end{aligned}$$

But we have

$$\begin{aligned}\frac{\partial}{\partial z}E^-((-\alpha,-\alpha),z) &= \left(-\sum_{n<0}(\pi_L(\alpha)_{(n)} + \pi_R(\alpha)_{(n)}\right)z^{-n-1})E^-((-\alpha,-\alpha),z)\\ &= E^-((-\alpha,-\alpha),z)\left(-\sum_{n<0}(\pi_L(\alpha)_{(n)} + \pi_R(\alpha)_{(n)})z^{-n-1}\right),\end{aligned}$$

and

$$\frac{\partial}{\partial z}e^\alpha(z) = (\pi_L(\alpha)(z) + \pi_R(\alpha)(z))e^\alpha(z).$$

Hence, we have

$$\begin{aligned}\frac{\partial}{\partial z}A(\alpha)(z) &= E^-((-\alpha,-\alpha),z)e^\alpha(z)\left(-\sum_{n<0}(\pi_L(\alpha)_{(n)} + \pi_R(\alpha)_{(n)})z^{-n-1}\right)\\ &\quad + E^-((-\alpha,-\alpha),z)e^\alpha(z)(\pi_L(\alpha)(z) + \pi_R(\alpha)(z))\\ &= A(\alpha)(z)\left(\sum_{n\geqslant 0}(\pi_L(\alpha)_{(n)} + \pi_R(\alpha)_{(n)})z^{-n-1}\right),\end{aligned}$$

where the first equality follows from the fact that for all $n \in \mathbb{Z}$, $h \in \mathfrak{h}$ and $\alpha \in X^*(T)$ we have

$$[\pi_L(h)_{(n)} + \pi_R(h)_{(n)}, e^\alpha(z)] = 0$$

by summing equations (8.8) and (8.9). Hence, we have proven that

$$\frac{\partial}{\partial z}A(\alpha)(z) = A(\alpha)(z)\left(\sum_{n\geqslant 0}(\pi_L(\alpha)_{(n)} + \pi_R(\alpha)_{(n)})z^{-n-1}\right)$$

which, by looking at the coefficients, is the announced equality. □

Lemma 8.1.3. *Let M be a $\mathcal{D}^\kappa_T$-module. Let $\lambda,\mu \in \mathfrak{h}^*$ and assume that M contains a Fock space $\mathcal{F}^\kappa_\lambda \otimes \mathcal{F}^{\kappa^*}_\mu$ with respect to its $V^\kappa(\mathfrak{h}) \otimes V^{\kappa^*}(\mathfrak{h})$-module structure. Then for all $\alpha \in X^*(T)$, M contains a Fock space of weight $(\lambda+\alpha, \mu-\alpha)$ and*

$$\lambda + \mu \in \kappa(X_*(T)),$$

where we think of κ as an isomorphism $\kappa : \mathfrak{h} \longrightarrow \mathfrak{h}^*$.

Proof. Let $|\lambda, \mu\rangle$ be a highest weight vector that spans the given Fock space in M. Because $\mathcal{D}^\kappa_T$ is a simple vertex algebra by Proposition 4.1.3, Proposition 3.2.10 applies and shows that there exists a unique $m \in \mathbb{Z}$ such that

$$\begin{aligned} e^\alpha_{(m)}|\lambda, \mu\rangle &\neq 0, \\ e^\alpha_{(k)}|\lambda, \mu\rangle &= 0 \text{ for all } k > m. \end{aligned}$$

It is easy to check that $e^\alpha_{(m)}|\lambda, \mu\rangle$ is a highest weight vector of weight $(\lambda + \alpha, \mu - \alpha)$, we denote it by

$$|\lambda + \alpha, \mu - \alpha\rangle.$$

It is clear by its definition that $|\lambda + \alpha, \mu - \alpha\rangle$ lies in the $\mathcal{D}^\kappa_T$-submodule generated by $|\lambda, \mu\rangle$. Applying Lemma 3.2.9, we see that for all $\alpha \in X^*(T)$, the difference of the conformal weights of $|\lambda, \mu\rangle$ and that of $|\lambda + \alpha, \mu - \alpha\rangle$ should be an integer. That is

$$\frac{\kappa(\lambda + \alpha, \lambda + \alpha)}{2} - \frac{\kappa(\mu - \alpha, \mu - \alpha)}{2} - \left(\frac{\kappa(\lambda, \lambda)}{2} - \frac{\kappa(\mu, \mu)}{2}\right) = \kappa(\lambda + \mu, \alpha) \in \mathbb{Z}$$

i.e.,

$$\alpha(\kappa^{-1}(\lambda + \mu)) \in \mathbb{Z}.$$

As this holds for an arbitrary $\alpha \in X^*(T)$, we get the result. □

Proposition 8.1.4. *Let M be an object of $\mathcal{D}^\kappa_T$–*$\mathrm{Mod}^{\{1\} \times \mathcal{J}_\infty T}$*. If $M \neq 0$ then*

$$M^{t\mathfrak{h}[[t]] \oplus t\mathfrak{h}[[t]]} \neq 0.$$

Moreover, $M^{t\mathfrak{h}[[t]] \oplus t\mathfrak{h}[[t]]}$ contains a weight vector for $\mathfrak{h} \oplus \mathfrak{h}$ of weight $(\lambda + \kappa(\gamma, \cdot), -\lambda)$, where $\lambda \in X^*(T)$ and $\gamma \in X_*(T)$. In particular, M contains the Fock space $\mathcal{F}^\kappa_{\lambda + \kappa(\gamma, \cdot)} \otimes \mathcal{F}^{\kappa^*}_{-\lambda}$ as a $V^\kappa(\mathfrak{h}) \otimes V^{\kappa^*}(\mathfrak{h})$-submodule.

Proof. Because M is nonzero and belongs to the Kazhdan–Lusztig category with respect to the right action, it follows from Theorem 4.3.3 that $M^{0 \oplus t\mathfrak{h}[[t]]} \neq 0$. For all $m \in M$, since M is a smooth $\hat{\mathfrak{h}}^\kappa \oplus \hat{\mathfrak{h}}^{\kappa^*}$-module and $\mathfrak{h}$ is finite dimensional, it is clear that the vector space $\{(\pi_L(x)_{(n)} + \pi_R(x)_{(n)})m \mid x \in \mathfrak{h}, n > 0\}$ is finite dimensional. We denote by $d(m)$ its dimension. Let $m \in M^{0 \oplus t\mathfrak{h}[[t]]}$ be a nonzero vector such that $d(m)$ is minimal in $M^{0 \oplus t\mathfrak{h}[[t]]}$. If $d(m)$ is zero we are done. Otherwise there exists $k \geqslant 1$ such that there exists $\alpha \in X^*(T)$ satisfying $(\pi_L(\alpha)_{(k)} + \pi_R(\alpha)_{(k)})m \neq 0$ and $(\pi_L(h')_{(n)} + \pi_R(h')_{(n)})m = 0$ for all $h' \in \mathfrak{h}$ and $n > k$ (recall that $X^*(T)$ spawns $\mathfrak{h}^*$ as a $\mathbb{C}$-vector space).

Now because $\mathcal{D}^\kappa_T$ is a simple vertex algebra, it follows from Proposition 3.2.10 that $e^\alpha(z)m \neq 0$, and hence

$$A(\alpha)(z)m = E^-((-\alpha, -\alpha), z)e^\alpha(z)m \neq 0.$$

But since $e^\alpha(z)m \in M((z))$ and $E^-((-\alpha,-\alpha),z) \in \mathrm{End}(M)[[z]]$ we see that $A(\alpha)(z) \in M((z))$. So there exists an integer $k' \in \mathbb{Z}$ such that $A(\alpha)_{k'}m \neq 0$ and for all $n > k'$, $A(\alpha)_{k'}m = 0$. Let us now apply equation (8.7) with $n = k + k'$. We get

$$\begin{aligned}
0 &= (-k-k'-1)A(\alpha)_{k+k'}m \\
&= \sum_{i=0}^{k} A(\alpha)_{k+k'-i}(\pi_L(\alpha)_{(i)} + \pi_R(\alpha)_{(i)})m \\
&= \sum_{i=0}^{k} (\pi_L(\alpha)_{(i)} + \pi_R(\alpha)_{(i)})A(\alpha)_{k+k'-i}m \\
&= (\pi_L(\alpha)_{(k)} + \pi_R(\alpha)_{(k)})A(\alpha)_{k'}m.
\end{aligned}$$

Set $m' = A(\alpha)_{k'}m$. Clearly it is again in $M^{0\oplus t\mathfrak{h}[[t]]}$ but $d(m') < d(m)$ since the previous computation shows that the inclusion of vector spaces

$$\{(\pi_L(x)_{(n)} + \pi_R(x)_{(n)})m \mid x \in \mathfrak{h}, n > 0\} \subset \{(\pi_L(x)_{(n)} + \pi_R(x)_{(n)})m' \mid x \in \mathfrak{h}, n > 0\}$$

is strict. By infinite descent, this proves that there exists a nonzero $m \in M^{0\oplus t\mathfrak{h}[[t]]}$ such that $d(m) = 0$ *i.e.*, such that for all $x \in \mathfrak{h}$ and $n > 0$

$$(\pi_L(x)_{(n)} + \pi_R(x)_{(n)})m = 0,$$

but by assumption $\pi_R(x)_{(n)}m = 0$ so that implies $\pi_L(x)_{(n)}m = 0$. This proves the claim.

Because $M^{t\mathfrak{h}[[t]]\oplus t\mathfrak{h}[[t]]}$ is a $\mathfrak{h}\oplus\mathfrak{h}$-submodule and M belongs to the Kazhdan–Lusztig category with respect to the right action, we can choose a nonzero vector $m \in M^{t\mathfrak{h}[[t]]\oplus t\mathfrak{h}[[t]]}$ of weight $\lambda \in X^*(T)$ with respect to the right action. Let $\alpha \in X^*(T)$, as before we can find $k \in \mathbb{Z}$ such that $A(\alpha)_k(w) \neq 0$, now applying equation (8.7) with $n = k$ we find that

$$\begin{aligned}
(-k-1)A(\alpha)_k m &= A(\alpha)_k(\pi_L(\alpha)_{(0)} + \pi_R(\alpha)_{(0)})m \\
&= A(\alpha)_k\pi_L(\alpha)_{(0)}m + A(\alpha)_k\pi_R(\alpha)_{(0)}m \\
&= A(\alpha)_k\pi_L(\alpha)_{(0)}m + \lambda(\alpha)A(\alpha)_k m.
\end{aligned}$$

Recall from equation (8.3) that $[\pi_L(\alpha)_{(0)}, A(\alpha)_k] = \kappa(\alpha,\alpha)A(\alpha)_k$ so that the previous equality becomes

$$\pi_L(\alpha)_{(0)}A(\alpha)_k m = (-k-1-\lambda(\alpha)+\kappa(\alpha,\alpha))A(\alpha)_k m.$$

Notice that $A(\alpha)_k m$ is still a nonzero element of $M^{t\mathfrak{h}[[t]]\oplus t\mathfrak{h}[[t]]}$ (recall equations (8.5) and (8.6)) but is now an eigenvector of $\pi_L(\alpha)_{(0)}$. It is also clear from equation (8.4) that $A(\alpha)_k m$ is still a weight vector for the right action and that if m is an eigenvector for a certain operator $x_{(0)}$ where $x \in \mathfrak{h}$, so is $A(\alpha)_k m$ by equation (8.3). Pick a $\mathbb{Z}$-basis $\alpha_1,\dots,\alpha_r$ of $X^*(T)$. It is a basis of $\mathfrak{h}^*$ as $\mathbb{C}$-vector space. Iterating the previous construction starting from m and having α range through $\{\alpha_1,\dots,\alpha_r\}$ yields a common eigenvector for the action of $\mathfrak{h} \oplus \mathfrak{h}$ *i.e.*, the desired weight vector. The fact that its weight is of the form $(\lambda + \kappa(\gamma, \cdot), -\lambda)$ where $\lambda \in X^*(T)$ and $\gamma \in X_*(T)$ follows from Lemma 8.1.3. □

Corollary 8.1.5. *The only simple objects of the category $\mathcal{D}^\kappa_T\text{–Mod}^{1\times\mathcal{J}_\infty T}$, up to isomorphism, are the $\gamma \cdot \mathcal{D}^\kappa_T$ where γ ranges through $X_*(T) \subset \mathcal{SF}(\mathcal{D}^\kappa_T)$.*

Proof. Let M be a simple object of $\mathcal{D}^\kappa_T\text{–Mod}^{1\times\mathcal{J}_\infty T}$. Thanks to Proposition 8.1.4 we know that there exists $\lambda \in X^*(T)$ and $\gamma \in X_*(T)$ such that the Fock space $\mathcal{F}^\kappa_{\lambda+\kappa(\gamma,\cdot)}\otimes\mathcal{F}^{\kappa^*}_{-\lambda}$ is contained in M. It follows from Proposition 5.2.2 that the module $\gamma\cdot M$ contains the Fock space $\mathcal{F}^\kappa_\lambda\otimes\mathcal{F}^{\kappa^*}_{-\lambda}$. Lemma 8.1.3 shows that $\gamma\cdot M$ contains the vacuum Fock space $\mathcal{F}^\kappa_0\otimes\mathcal{F}^{\kappa^*}_0$. Since $\gamma\cdot M$ is again a simple $\mathcal{D}^\kappa_T$-module, Lemma 4.5.1 applies to show that $\gamma \cdot M$ is isomorphic to $\mathcal{D}^\kappa_T$ as $\mathcal{D}^\kappa_T$-modules so that M is isomorphic to $-\gamma \cdot \mathcal{D}^\kappa_T$. □

Theorem 8.1.6. *The category $\mathcal{D}^\kappa_T\text{–Mod}^{1\times\mathcal{J}_\infty T}$ is semisimple with simple objects given by the $\gamma \cdot \mathcal{D}^\kappa_T$ where $\gamma \in X_*(T) \subset \mathcal{SF}(\mathcal{D}^\kappa_T)$.*

Proof. The statement concerning the simple objects is Corollary 8.1.5. Let M be an object of $\mathcal{D}^\kappa_T\text{–Mod}^{1\times\mathcal{J}_\infty T}$ and denote by M' the sum of all simple modules contained in M. By definition we have an exact sequence of $\mathcal{D}^\kappa_T$-modules so, in particular, of $V^\kappa(\mathfrak{h}) \otimes V^{\kappa^*}(\mathfrak{h})$-modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M/M' \xrightarrow{\pi} 0.$$

We want to show that M/M' is zero. Let us assume for a contradiction it is not. It follows from Corollary 8.1.5 that M' is a direct sum of Fock spaces, so by Lemma 4.3.18 we have an exact sequence of $\mathfrak{h} \oplus \mathfrak{h}$-modules

$$0 \longrightarrow (M')^{t\mathfrak{h}[[t]]\oplus t\mathfrak{h}[[t]]} \longrightarrow M^{t\mathfrak{h}[[t]]\oplus t\mathfrak{h}[[t]]} \longrightarrow (M/M')^{t\mathfrak{h}[[t]]\oplus t\mathfrak{h}[[t]]} \longrightarrow 0.$$

Let $x \in (M/M')^{t\mathfrak{h}[[t]]\oplus t\mathfrak{h}[[t]]}$ be a weight vector with respect to $\mathfrak{h}\oplus\mathfrak{h}$ (it exists by Proposition 8.1.4). By assumption, the right factor $0 \oplus \mathfrak{h} \subset \mathfrak{h} \oplus \mathfrak{h}$ acts semisimply on every object so we can find a lift $\widetilde{x} \in M^{t\mathfrak{h}[[t]]\oplus t\mathfrak{h}[[t]]}$ which is of weight $\lambda \in \mathfrak{h}^*$ with respect to the right factor $0 \oplus \mathfrak{h} \subset \mathfrak{h} \oplus \mathfrak{h}$.

Let $\alpha \in X^*(T)$, as in the proof of Proposition 8.1.4. There exists an integer k such that $A(\alpha)_k x \neq 0$. Applying equation (8.7) with $n = k$ we have (recall that $\widetilde{x} \in M^{t\mathfrak{h}[[t]]\oplus t\mathfrak{h}[[t]]}$)

$$(-k-1)A(\alpha)_k\widetilde{x} = A(\alpha)_k(\pi_L(\alpha)_{(0)} + \pi_R(\alpha)_{(0)})\widetilde{x} = A(\alpha)_k\pi_L(\alpha)_{(0)}\widetilde{x} + \lambda(\alpha)A(\alpha)_k\widetilde{x}.$$

So by equation (8.3) we have

$$\pi_L(\alpha)_{(0)}A(\alpha)_k\widetilde{x} = (-k - 1 - \lambda(\alpha) + \kappa(\alpha,\alpha))A(\alpha)_k\widetilde{x}.$$

Now, $A(\alpha)_k\widetilde{x}$ does not belong to M' because $\pi(A(\alpha)_k\widetilde{x}) = A(\alpha)_k\widetilde{x} \neq 0$ by our choice of k. It is still in $M^{t\mathfrak{h}[[t]]\oplus t\mathfrak{h}[[t]]}$ by equations (8.5) and (8.6). Thanks to equations (8.3) and (8.4), $A(\alpha)_k\widetilde{x}$ is again a common eigenvector for the right action of $\mathfrak{h}$ and of the left action of $\alpha \in \mathfrak{h}$. Iterating the previous construction starting from $\widetilde{x}$ as α runs through a $\mathbb{Z}$-basis of $X^*(T)$ yields a vector $m \in M^{t\mathfrak{h}[[t]]\oplus t\mathfrak{h}[[t]]}$ such that $m \notin M'$ and that is a weight vector for the action of $\mathfrak{h}\oplus\mathfrak{h}$. As such the $\hat{\mathfrak{h}}^\kappa\oplus\hat{\mathfrak{h}}^{\kappa^*}$-module generated by m in M is a Fock space of the form $\mathcal{F}^\kappa_{\lambda+\kappa(\gamma,\cdot)}\otimes\mathcal{F}^{\kappa^*}_{-\lambda}$ where $\lambda \in X^*(T)$ and $\gamma \in X_*(T)$ by Lemma 8.1.3. As in the proof of Corollary 8.1.5, we see that there exists a morphism of $\mathcal{D}^\kappa_T$-module from $-\gamma\cdot\mathcal{D}^\kappa_T$ to M such that $m \neq 0$ belongs to the image. But because $\gamma\cdot\mathcal{D}^\kappa_T$ is simple, it means in particular that the $\mathcal{D}^\kappa_T$-submodule generated by m inside of M is simple and is not contained in M', because m is not. This contradicts the definition of M' and finishes the proof. $\square$

Remark 8.1.7. *A striking feature of this result is that our hypothesis enforces that the module we start with is a direct sum of Fock spaces with respect to the right action but requires nothing with respect to the left action. But a posteriori the module is indeed a direct sum of Fock spaces with respect to the left action. This is even more remarkable in the case of a simple Lie algebra (see Theorem 9.4.1).*

Recall that, inside $\mathfrak{h}$, lies the lattice $X_*(T)$ of cocharacters, and inside $\mathfrak{h}^*$, lies the lattice $X^*(T)$ of characters. Let us identify $\mathfrak{h}$ with $\mathfrak{h}^*$ using κ. Denote by $\kappa : \mathfrak{h} \longrightarrow \mathfrak{h}^*$ the isomorphism and by κ^{-1} its inverse. It is clear that $\kappa^{-1}(X^*(T)) \subset \mathfrak{h}$ is a full rank lattice of $\mathfrak{h}$. We define

$$Y = X_*(T) \cap \kappa^{-1}(X^*(T)).$$

It is a free abelian group of rank possibly lower than r.

Corollary 8.1.8. *The simple objects of the category $\mathcal{D}^\kappa_T\text{–Mod}^{\mathcal{J}_\infty T\times\mathcal{J}_\infty T}$ are in bijection with the subgroup Y of $X_*(T)$ where for each $\gamma \in Y \subset X_*(T) \subset \mathcal{SF}(\mathcal{D}^\kappa_T)$, the corresponding simple object is given by $\gamma\cdot\mathcal{D}^\kappa_T$. Moreover, the category $\mathcal{D}^\kappa_T\text{–Mod}^{\mathcal{J}_\infty T\times\mathcal{J}_\infty T}$ is semisimple, that is, every object of $\mathcal{D}^\kappa_T\text{–Mod}^{\mathcal{J}_\infty T\times\mathcal{J}_\infty T}$ is a direct sum of simple objects.*

Proof. This is a direct consequence of Theorem 8.1.6 and Corollary 3.1.2. $\square$

Remark 8.1.9. *For almost every nondegenerate κ, we have $Y = 0$ and in this case, the category $\mathcal{D}^\kappa_T\text{–Mod}^{\mathcal{J}_\infty T\times\mathcal{J}_\infty T}$ is equivalent to the category of $\mathbb{C}$-vector spaces (compare with Theorem 4.5.3). For general values of κ, we believe this result should be compared with the twisted geometric Satake equivalence of [FL10] and to the metaplectic geometric Langlands theory ([Rei12], [GL18]).*

Chapter 9

The case of a simple adjoint group of classical simply laced type

We now assume that G is a simple adjoint group of type A or D. That is, G equals PSL_n or the quotient of SO_{2n} by its center. In particular, we have the equalities of abelian groups $X^*(T) = Q$ and $X_*(T) = \check{P}$.

9.1 Shifted chiral differential operators on a simple group

The (chiral) Peter–Weyl decomposition is at the technical heart of our arguments so far. We can even go further, in a sense, all that mattered was the existence a simple vertex algebra structure on

$$\bigoplus_{\lambda \in X^*(T)_+} V^k_\lambda \otimes V^{k^*}_{-\omega_0 \lambda}$$

which is a conformal extension of $V^k(\mathfrak{g}) \otimes V^{k^*}(\mathfrak{g})$. Even the relation between k and k^* is irrelevant to some extent. This motivates the following question, given $k, l \in \mathbb{C} \setminus \mathbb{Q}$, does there exist a simple vertex algebra structure on

$$\bigoplus_{\lambda \in X^*(T)_+} V^k_\lambda \otimes V^{l}_{-\omega_0 \lambda}$$

which is a conformal extension of $V^k(\mathfrak{g}) \otimes V^l(\mathfrak{g})$?

It is easily seen that k and l may not be arbitrary if we require our vertex algebras to be $\mathbb{Z}$-graded. We can rewrite the relation between k and k^* as

$$\frac{1}{k + \check{h}} + \frac{1}{k^* + \check{h}} = 0. \tag{9.1}$$

Given $n \in \mathbb{Z}$ and $k \in \mathbb{C} \setminus \mathbb{Q}$ it is natural to allow l to be defined by the relation

$$\frac{1}{k + \check{h}} + \frac{1}{l + \check{h}} = n. \tag{9.2}$$

We call l the n-shifted level of k and denote it by $k[n]$. It is easy to see that if k is irrational then so is $k[n]$.

The following was conjectured by Creutzig and Gaiotto (see Conjecture 1.1 of [CG20]):

Theorem 9.1.1 (Main Theorem B of [Mor22], see also Theorem 2.2 of [CLNS25])**.** *Let $k \in \mathbb{C} \setminus \mathbb{Q}$ and $n \in \mathbb{Z}$ then there exists a simple vertex algebra structure on*

$$\bigoplus_{\lambda \in X^*(T)_+} V_\lambda^k \otimes V_{-\omega_0\lambda}^{k[n]} \tag{9.3}$$

that is a conformal extension of $V^k(\mathfrak{g}) \otimes V^{k[n]}(\mathfrak{g})$.

Remark 9.1.2. *If the previous theorem can be generalized to the exceptional type E, our techniques apply as well to prove Conjecture 7.2.4 in this case.*

We will prove there is a unique such vertex algebra structure (see Theorem 9.3.9). So it makes sense to denote this vertex algebra by $\mathcal{D}_G^k[n]$ and call it the n-shifted vertex algebra of chiral differential operators on the group G at level k. We have morphisms of vertex algebras

$$\begin{tikzcd} & V^k(\mathfrak{g}) \otimes V^{k[n]}(\mathfrak{g}) \arrow[d] & \\ V^k(\mathfrak{g}) \arrow[ur] \arrow[r] & \mathcal{D}_G^k[n] & V^{k[n]}(\mathfrak{g}) \arrow[ul] \arrow[l] \end{tikzcd}$$

so we can perform the Hamiltonian reduction with respect to $V^k(\mathfrak{g})$. We obtain a vertex algebra

$$\mathcal{W}_G^k[n] = H_{DS}^*(\mathcal{D}_G^k[n]),$$

that we call the n-shifted affine equivariant $\mathcal{W}$-algebra on the group G at level k. It comes equipped with morphisms of vertex algebras

$$\begin{tikzcd} & W^k(\mathfrak{g}) \otimes V^{k[n]}(\mathfrak{g}) \arrow[d] & \\ W^k(\mathfrak{g}) \arrow[ur] \arrow[r] & \mathcal{W}_G^k[n] & V^{k[n]}(\mathfrak{g}). \arrow[ul] \arrow[l] \end{tikzcd}$$

In particular, any $\mathcal{W}_G^k[n]$-module is naturally a $W^k(\mathfrak{g}) \otimes V^{k[n]}(\mathfrak{g})$-module by restriction *i.e.*, carries two commuting actions of $W^k(\mathfrak{g})$ and $V^{k[n]}(\mathfrak{g})$. We define, as in the unshifted case (see Definition 4.5.2), the categories

$$\mathcal{D}_G^k[n]\text{–Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}, \qquad \mathcal{D}_G^k[n]\text{–Mod}^{1 \times \mathcal{J}_\infty G}, \qquad \mathcal{W}_G^k[n]\text{–Mod}^{\mathcal{J}_\infty G}.$$

Remark 9.1.3. *It is an open problem whether the vertex algebras $\mathcal{D}_G^k[n]$ and $\mathcal{W}_G^k[n]$ come from a deformable family of vertex algebras where the structure constants are continuous functions of k. This question is settled in the case of* PSL_2 *for $n = 1$ in [CG20] and for $n = 2$ in [CGL20].*

9.2 Vertex convolution operation

Let V and W be two vertex algebras such that there exists two morphisms of vertex algebra

$$V \longleftarrow V^{\kappa^*}(\mathfrak{g}),\ V^{\kappa}(\mathfrak{g}) \longrightarrow W.$$

By definition, we have the equality

$$\kappa + \kappa^* = -\kappa_{\mathfrak{g}}$$

so that, with respect to the diagonal action, $V \otimes W$ is a vertex algebra that comes equipped with a morphism from $V^{-\kappa_{\mathfrak{g}}}(\mathfrak{g})$. This is exactly the level for which the relative semi-infinite cohomology

$$V \circ W = H^{\frac{\infty}{2}+\bullet}(\mathfrak{g}^{-\kappa_{\mathfrak{g}}}, \mathfrak{g}, V \otimes W)$$

is defined. By construction, $V \circ W$ is again a vertex algebra. If we are given $M \in V$−Mod and $N \in W$−Mod then by viewing $M \otimes N$ as a $V^{-\kappa_{\mathfrak{g}}}(\mathfrak{g})$-module, we can consider the relative semi-infinite cohomology

$$M \circ N = H^{\frac{\infty}{2}+\bullet}(\mathfrak{g}^{-\kappa_{\mathfrak{g}}}, \mathfrak{g}, M \otimes N),$$

and it is naturally a $V \circ W$-module. This convolution operation, known as the quantized Moore–Tachikawa operation, is especially well-behaved on the Kazhdan–Lusztig category.

Theorem 9.2.1 ([FGZ86] and Theorem 3.3 of [CLNS25]). *Assume that V (resp. W) belongs to $V^{\kappa^*}(\mathfrak{g})$−Mod$^{\mathcal{J}_\infty G}$ (resp. $V^{\kappa}(\mathfrak{g})$−Mod$^{\mathcal{J}_\infty G}$) then*

$$H^{\frac{\infty}{2}+k}(\mathfrak{g}^{-\kappa_{\mathfrak{g}}}, \mathfrak{g}, V \otimes W) = 0 \text{ if } k \neq 0. \tag{9.4}$$

More precisely, we have the equality of vertex algebras

$$V^{\kappa^*}(\mathfrak{g}) \circ V^{\kappa}(\mathfrak{g}) = \mathbb{C}$$

and if $\lambda, \mu \in P_+$, then we have the equality as $\mathbb{C}$-module

$$V_\lambda^{\kappa^*} \circ V_\mu^{\kappa} = \delta_{\lambda,-\omega_0\mu}\mathbb{C}. \tag{9.5}$$

It makes sense to consider the functors

$$\begin{aligned} \mathcal{D}_G^\kappa \circ - &: V^\kappa(\mathfrak{g})\text{−Mod} \longrightarrow (\mathcal{D}_G^\kappa \circ V^\kappa(\mathfrak{g}))\text{−Mod}, \\ - \circ \mathcal{D}_G^\kappa &: V^{\kappa^*}(\mathfrak{g})\text{−Mod} \longrightarrow (V^{\kappa^*}(\mathfrak{g}) \circ \mathcal{D}_G^\kappa)\text{−Mod}. \end{aligned}$$

The following property is fundamental and expresses that $\mathcal{D}_G^\kappa$ is the unit for the convolution operation:

Proposition 9.2.2 (Theorem 6.2 of [AG02], discussion following Proposition 5.6 of [Ara18] and Theorem 9.2.8 of [AM25])**.** *We have the equalities of vertex algebras*

$$\mathcal{D}_G^\kappa \circ V^\kappa(\mathfrak{g}) = V^\kappa(\mathfrak{g}), \quad V^{\kappa^*}(\mathfrak{g}) \circ \mathcal{D}_G^\kappa = V^{\kappa^*}(\mathfrak{g}). \tag{9.6}$$

More generally, if V (resp. W) is a vertex algebra that belongs to $V^{\kappa^*}(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ (resp. $V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$) then, as vertex algebras,

$$V \circ \mathcal{D}_G^\kappa = V, \quad \mathcal{D}_G^\kappa \circ W = W. \tag{9.7}$$

Moreover, if M (resp. N) is a V-module (resp. W-module) in $V^{\kappa^*}(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ (resp. $V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$) then, as a V-module (resp. W-module),

$$M \circ \mathcal{D}_G^\kappa = M, \quad \mathcal{D}_G^\kappa \circ N = N. \tag{9.8}$$

Remark 9.2.3. *The definitions and results given so far in this paragraph extend to the case of a reductive algebraic group for generic levels (this is the setting of [CLNS25]). In particular, when G is an algebraic torus, equation (9.5) holds for Fock spaces (see §6.1 of [Cre+22]). We also note that the semisimplicity of the Kazhdan–Lusztig category $V^\kappa(\mathfrak{g}) \otimes V^{\kappa^*}(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}$ (recall Theorem 4.3.3) together with equalities (9.5) and (9.8) give another proof of the chiral Peter–Weyl Theorem 4.4.1.*

The vertex algebra $\mathcal{W}_G^\kappa$ comes equipped with a morphism

$$\mathcal{W}_G^\kappa \longleftarrow V^{\kappa^*}(\mathfrak{g}).$$

As such, it defines a functor

$$\mathcal{W}_G^\kappa \circ - : V^\kappa(\mathfrak{g})\text{–Mod} \longrightarrow (\mathcal{W}_G^\kappa \circ V^\kappa(\mathfrak{g}))\text{–Mod}$$

The following result expresses the fact that taking the convolution with $\mathcal{W}_G^\kappa$ "represents" the quantum Hamiltonian reduction functor on the Kazhdan–Lusztig category.

Proposition 9.2.4 (Theorem 6.7 of [Ara18])**.** *We have the equality of vertex algebras*

$$\mathcal{W}_G^\kappa \circ V^\kappa(\mathfrak{g}) = W^\kappa(\mathfrak{g}). \tag{9.9}$$

More generally, if V is a vertex algebra that belongs to $V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ then we have the equality of vertex algebras

$$\mathcal{W}_G^\kappa \circ V = H^0_{DS}(V). \tag{9.10}$$

Moreover, if M is a V-module in $V^\kappa(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ then we have the equality as $H^*_{DS}(V)$-modules

$$\mathcal{W}_G^\kappa \circ M = H^0_{DS}(M). \tag{9.11}$$

9.3 The strategy

Conjecture 7.2.4 gives a description of the category

$$\mathcal{W}^k_G\text{–Mod}^{\mathcal{J}_\infty G}$$

in terms of the Langlands dual group of G. The study of this category feels a bit beyond the scope of usual vertex algebraic techniques. For instance $\mathcal{W}^k_G$ is a $\mathbb{Z}$-graded vertex algebra that is not lower bounded. So it is unclear at face value how the Zhu algebra may be useful. In the case of a torus (see Chapter 8), some explicit computations in Fock spaces solve the problem. Here, the lack of knowledge of the relations between strong generators of the principal $\mathcal{W}$-algebra makes this approach seemingly impossible: one needs a new idea to proceed.

Let us forget momentarily about Hamiltonian reduction and consider $\mathcal{D}^k_G$ again. This is a somewhat complicated vertex algebra, for instance its 0-degree space is $\mathbb{C}[G]$ which is infinite dimensional. In view of Theorem 9.1.1 we see that $\mathcal{D}^k_G = \mathcal{D}^k_G[0]$ is a member of a $\mathbb{Z}$-family of simple vertex algebras. Members of this family have very different behaviours as vertex algebras, for instance $\mathcal{D}^k_G[1]$ is positively graded and $(\mathcal{D}^k_G)_0 = \mathbb{C}$ while $\mathcal{D}^k_G[-1]$ is a $\mathbb{Z}$-graded, not lower bounded vertex algebra with every conformal degree of infinite dimension. Yet, as we will see shortly, the categories $\mathcal{D}^k_G[0]\text{–Mod}^{\mathcal{J}_\infty G}$ and $\mathcal{D}^k_G[1]\text{–Mod}^{\mathcal{J}_\infty G}$ are naturally equivalent under the convolution operation. The equivalence is well behaved enough to induce an equivalence between $\mathcal{W}^k_G[0]\text{–Mod}^{\mathcal{J}_\infty G}$ and $\mathcal{W}^k_G[1]\text{–Mod}^{\mathcal{J}_\infty G}$. The hope being that the latter category is easier to relate to the representation of the Langlands dual group.

Let us be more precise, let $m, n \in \mathbb{Z}$ and $k, l \in \mathbb{C} \setminus \mathbb{Q}$, the vertex algebras $\mathcal{D}^k_G[m]$ and $\mathcal{D}^l_G[n]$ each come equipped with two morphisms

$$\begin{tikzcd}
 & \mathcal{D}^k_G[m] & & & \mathcal{D}^l_G[n] & \\
V^k(\mathfrak{g}) \arrow[ur] & & V^{k[m]}(\mathfrak{g}) \arrow[ul] & V^l(\mathfrak{g}) \arrow[ur] & & V^{l[n]}(\mathfrak{g}) \arrow[ul]
\end{tikzcd} .$$

In particular, the convolution $\mathcal{D}^k_G[m] \circ \mathcal{D}^l_G[n]$ makes sense if and only if $l = k[m][0]$. An easy computation using relation (9.2) shows that

$$k[m][0][n] = k[m+n].$$

Hence, we have two morphisms of vertex algebras

$$\begin{tikzcd}
 & \mathcal{D}^k_G[m] \circ \mathcal{D}^{k[m][0]}_G[n] & \\
V^k(\mathfrak{g}) \arrow[ur] & & V^{k[m+n]}(\mathfrak{g}) \arrow[ul]
\end{tikzcd}.$$

More precisely, in view of equations (9.3) and (9.5), as a $V^k(\mathfrak{g}) \otimes V^{k[m+n]}(\mathfrak{g})$-module, we have the decomposition

$$\mathcal{D}^k_G[m] \circ \mathcal{D}^{k[m][0]}_G[n] = \bigoplus_{\lambda \in X^*(T)_+} V^k_\lambda \otimes V^{k[m+n]}_{-\omega_0\lambda}. \tag{9.12}$$

This strongly suggests the following:

Proposition 9.3.1. *We have the following equality of vertex algebras*

$$\mathcal{D}^k_G[m] \circ \mathcal{D}^{k[m][0]}_G[n] = \mathcal{D}^k_G[m+n].$$

We begin by collecting some consequences. The following technical Lemma is essential:

Lemma 9.3.2 (Lemma 4.8 of [BN22] and Theorem 10.11 of [Ara18])**.** *Let A, B and C be vertex algebras. Assume that there exists morphisms of vertex algebra*

$$\begin{array}{rl} & A \longleftarrow V^{k^*}(\mathfrak{g}) \\ V^k(\mathfrak{g}) \longrightarrow & B \longleftarrow V^{k^*}(\mathfrak{g}) \\ V^k(\mathfrak{g}) \longrightarrow & C \end{array}$$

such that A, B and C belong to the Kazhdan–Lusztig category. Then, as vertex algebras,

$$(A \circ B) \circ C \simeq A \circ (B \circ C). \tag{9.13}$$

Moreover, if we are given, in obvious notations,

$$U \in A\text{−Mod}^{\mathcal{J}_\infty G}, \quad V \in B\text{−Mod}^{\mathcal{J}_\infty G \times \mathcal{J}_\infty G}, \quad W \in C\text{−Mod}^{\mathcal{J}_\infty G},$$

then, as modules over $A \circ B \circ C$ (which is unambiguous by (9.13)), we have

$$(U \circ V) \circ W \simeq U \circ (V \circ W).$$

Remark 9.3.3. *The proof of the associativity property (9.13) given in [BN22] is a spectral sequence argument that relies on the fact that in the Kazhdan–Lusztig category, the cohomology is concentrated in degree 0. Because the vanishing also holds in the case of an algebraic torus (recall equation (9.5) and Remark 9.2.3), their proof also works for an abelian Lie algebra.*

Let $n \in \mathbb{Z}$ and V be any vertex algebra that comes equipped with a vertex algebra morphism

$$V \longleftarrow V^{k^*}(\mathfrak{g})$$

such that V is in the Kazhdan–Lusztig category $V^{k^*}(\mathfrak{g})\text{−Mod}^{\mathcal{J}_\infty G}$. Set

$$V[n] = V \circ \mathcal{D}^k_G[n].$$

Remark 9.3.4. *Note that this is indeed compatible with the previously introduced notations as we have for all $n \in \mathbb{Z}$ the equality of vertex algebras*

$$\mathcal{W}^k_G[n] = H^*_{DS}(\mathcal{D}^k_G[n])$$

thanks to Proposition 9.2.4.

Convolution by $\mathcal{D}^{k[n][0]}_G[1]$ defines a functor

$$- \circ \mathcal{D}^{k[n][0]}_G[1] : V[n]\text{–Mod}^{\mathcal{J}_\infty G} \longmapsto V[n+1]\text{–Mod}^{\mathcal{J}_\infty G}.$$

Convolution by $\mathcal{D}^{k[n+1][0]}_G[-1]$ defines a functor in the opposite direction in view of Proposition 9.3.1 and Lemma 9.3.2. As a direct consequence of Proposition 9.2.2 together with Lemma 9.3.2 we have:

Proposition 9.3.5. *These two functors are quasi-inverse to one another.*

This readily implies that the following diagrams consists of equivalences of categories

$$\begin{tikzcd}
\mathcal{D}^k_G[-1]\text{–Mod}^{1\times\mathcal{J}_\infty G} \arrow[r, bend left, "-\circ\mathcal{D}^{k[-1][0]}_G[1]"] & \mathcal{D}^k_G\text{–Mod}^{1\times\mathcal{J}_\infty G} \arrow[l, bend left, "-\circ\mathcal{D}^k_G[-1]"] \arrow[r, bend left, "-\circ\mathcal{D}^k_G[1]"] & \mathcal{D}^k_G[1]\text{–Mod}^{1\times\mathcal{J}_\infty G}, \arrow[l, bend left, "-\circ\mathcal{D}^{k[1][0]}_G[-1]"]
\end{tikzcd}$$

$$\begin{tikzcd}
\mathcal{W}^k_G[-1]\text{–Mod}^{\mathcal{J}_\infty G} \arrow[r, bend left, "-\circ\mathcal{D}^{k[-1][0]}_G[1]"] & \mathcal{W}^k_G\text{–Mod}^{\mathcal{J}_\infty G} \arrow[l, bend left, "-\circ\mathcal{D}^k_G[-1]"] \arrow[r, bend left, "-\circ\mathcal{D}^k_G[1]"] & \mathcal{W}^k_G[1]\text{–Mod}^{\mathcal{J}_\infty G}. \arrow[l, bend left, "-\circ\mathcal{D}^{k[1][0]}_G[-1]"]
\end{tikzcd}$$

Remark 9.3.6. *Clearly, the same technique shows that for all $n \in \mathbb{Z}$ the categories $\mathcal{D}^k_G[n]\text{–Mod}^{\mathcal{J}_\infty G\times\mathcal{J}_\infty G}$ are equivalent. So Theorem 4.5.3 describes them all.*

We now turn to the proof of Proposition 9.3.1.

Lemma 9.3.7 (Corollary 3.6 of [CLNS25])**.** *Let*

$$V = \bigoplus_{\lambda\in X^*(T)_+} V^k_\lambda \otimes V^{k[m]}_{-\omega_0\lambda}$$

and

$$W = \bigoplus_{\lambda\in X^*(T)_+} V^{k[m][0]}_\lambda \otimes V^{k[m+n]}_{-\omega_0\lambda}$$

be simple vertex algebras. Assume that V (resp. W) is a conformal extension of $V^k(\mathfrak{g}) \otimes V^{k[m]}(\mathfrak{g})$ (resp. $V^{k[m][0]} \otimes V^k(\mathfrak{g})^{k[m+n]}$). Then $V \circ W$ is a simple vertex algebra that is a conformal extension of $V^k(\mathfrak{g}) \otimes V^{k[m+n]}(\mathfrak{g})$.

Remark 9.3.8. *The proof given in [CLNS25] carries over without changes to the case of an algebraic torus.*

It is clear from this lemma that Proposition 9.3.1 follows from equation (9.12) together with the following statement (see §3.1 of [CG20]).

Theorem 9.3.9. *There exists on the $V^k(\mathfrak{g}) \otimes V^{k[n]}(\mathfrak{g})$-module*

$$\bigoplus_{\lambda \in X^*(T)_+} V^k_\lambda \otimes V^{k[n]}_{-\omega_0\lambda}$$

a unique structure of simple vertex algebra that is a conformal extension of $V^k(\mathfrak{g}) \otimes V^{k[n]}(\mathfrak{g})$ given by that of $\mathcal{D}^k[n]$.

Proof. The existence part of the statement is Theorem 9.1.1. For the unicity, assume we are given two families $\mathcal{A}^k[n]$ and $\mathcal{B}^k[n]$ depending on $k \in \mathbb{C} \setminus \mathbb{Q}$ and $n \in \mathbb{Z}$ with obvious notations. Let $k \in \mathbb{C} \setminus \mathbb{Q}$ and $n \in \mathbb{Z}$. It follows from Theorem 4.4.3, Lemma 9.3.7 and equation (9.12) that we have the equalities of vertex algebras:

$$\begin{aligned} \mathcal{D}^{k[n][0]}_G &= \mathcal{A}^{k[n][0]}[-n] \circ \mathcal{B}^k[n], \\ \mathcal{D}^k_G &= \mathcal{A}^k[n] \circ \mathcal{A}^{k[n][0]}[-n]. \end{aligned}$$

By successive applications of Lemma 9.3.2, Proposition 9.2.2 and the previous two equalities we have the equalities of vertex algebras:

$$\begin{aligned} \mathcal{A}^k[n] = \mathcal{A}^k[n] \circ \mathcal{D}^{k[n][0]}_G &= \mathcal{A}^k[n] \circ (\mathcal{A}^{k[n][0]}[-n] \circ \mathcal{B}^k[n]) \\ &= (\mathcal{A}^k[n] \circ \mathcal{A}^{k[n][0]}[-n]) \circ \mathcal{B}^k[n] = \mathcal{D}^k_G \circ \mathcal{B}^k[n] = \mathcal{B}^k[n]. \end{aligned}$$

□

9.4 Proof of the fundamental local equivalence

The goal of this paragraph is to prove the following result.

Theorem 9.4.1. *Conjecture 7.2.4 holds for a simple adjoint group of type A or D. That is, for $k \in \mathbb{C} \setminus \mathbb{Q}$ there is an equivalence of abelian categories*

$$\mathcal{W}^k_G\text{–Mod}^{\mathcal{J}_\infty G} \simeq \check{G}\text{–Mod}$$

that sends for all $\gamma \in X_*(T)_+$ the simple object $H^{2\rho(\gamma)}_{DS}(\gamma \cdot \mathcal{D}^k_G)$ to V_γ.

We devote the rest of this paragraph to the proof of this theorem. By assumption we have $\mathrm{Lie}(\check{G}) = \mathfrak{g}$ and $\check{G}$ is simply connected so the simple objects of $\check{G}$–Mod are in bijection with P_+.

We apply the strategy explained in §9.3. A central ingredient is the following relation between the equivariant affine $\mathcal{W}$-algebra and the Goddard–Kent–Olive coset realization of $\mathcal{W}$-algebra:

Theorem 9.4.2. *For $k \in \mathbb{C} \setminus \mathbb{Q}$ there is an isomorphism of vertex algebras*

$$\mathcal{W}^k_G[1] \simeq V^{k[1]-1}(\mathfrak{g}) \otimes L_1(\mathfrak{g})$$

where $L_1(\mathfrak{g})$ is the simple quotient of $V^1(\mathfrak{g})$.

Proof. By Lemma 7.1.1, $\mathcal{W}^k_G$ is simple, so it follows from Proposition 9.3.5 that $\mathcal{W}^k_G[1]$ is simple. Now we notice that (the proof of) Theorem 1.1 of [CN23] (see §4 of *loc.cit.*) applies and shows the desired result. □

According to Proposition 9.3.5 the following functors are equivalences of categories

$$\begin{tikzcd} \mathcal{W}^k_G\text{–Mod}^{\mathcal{J}_\infty G} \arrow[r, bend left, "-\circ\mathcal{D}^k_G[1]"] & \mathcal{W}^k_G[1]\text{–Mod}^{\mathcal{J}_\infty G}. \arrow[l, bend left, "-\circ\mathcal{D}^{k[1][0]}_G[-1]"] \end{tikzcd}$$

Under the isomorphism of Theorem 9.4.2 one checks that

$$\mathcal{W}^k_G[1]\text{–Mod}^{\mathcal{J}_\infty G} \simeq V^{k[1]-1}(\mathfrak{g}) \otimes L_1(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$$

where the right-hand side consists of the $V^{k[1]-1}(\mathfrak{g}) \otimes L_1(\mathfrak{g})$-modules that when seen as $V^{k[1]}(\mathfrak{g})$-modules via the diagonal embedding

$$V^{k[1]}(\mathfrak{g}) \longrightarrow V^{k[1]-1}(\mathfrak{g}) \otimes L_1(\mathfrak{g})$$

belong to $V^{k[1]}(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$. The representation theory of $L_1(\mathfrak{g})$ is well-understood (see *e.g.*, §6.6 of [LL04]), its simple modules are parametrized by the set of

$$P^1_+ = \{\lambda \in P_+ \mid \lambda(\theta) \leqslant 1\}$$

where θ is the longest root of $\mathfrak{g}$. Given $\lambda \in P^1_+$ we denote the corresponding simple module by $L_1(\lambda)$, it is the unique simple quotient of the Weyl module V^1_λ.

Because $L_1(\mathfrak{g})$ is a regular vertex algebra (see *e.g.*, Theorem 3.7 of [DLM97]), any $V^{k[1]-1} \otimes L_1(\mathfrak{g})$-module can be written

$$\bigoplus_{\lambda \in P^1_+} M_\lambda \otimes L_1(\lambda) \tag{9.14}$$

with $M_\lambda \in V^{k[1]-1}(\mathfrak{g})\text{–Mod}$. If we assume moreover that $M \in V^{k[1]-1}(\mathfrak{g}) \otimes L_1(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ then it is easy to see that for all $\lambda \in P^1_+$ we must have $M_\lambda \in V^{k[1]}(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty \check{G}}$.

Now if we fix $\mu \in P_+$ and $\lambda \in P^1_+$ we can consider the simple $V^{k[1]-1}(\mathfrak{g}) \otimes L_1(\mathfrak{g})$-module

$$V^{k[1]-1}_\mu \otimes L_1(\lambda).$$

It belongs to $V^{k[1]-1}(\mathfrak{g}) \otimes L_1(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$ if and only if

$$\mu + \lambda \in Q,$$

i.e.,

$$\lambda \in -\mu + Q.$$

The set P_+^1 consists of 0 and the minuscule weights (see Chapter VI, §1, Exercice 24 of [Bou02]). One can show that minuscule weights are lifts of nonzero classes of the quotient P/Q (Chapter VI, §2, Exercice 5 of *loc.cit.*). This means that given $\mu \in P_+$ there exists a unique $\lambda = \lambda(\mu) \in P_+^1$ such that

$$\mu + \lambda(\mu) \in Q.$$

This observation paired with equation (9.14) and Theorem 4.3.3 shows that the category

$$V^{k[1]-1}(\mathfrak{g}) \otimes L_1(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G}$$

is semisimple with simple objects given by the

$$V_\lambda^{k[1]-1} \otimes L_1(\lambda(\mu))$$

with $\lambda \in P_+$. Hence, as abelian categories we indeed have

$$V^{k[1]-1}(\mathfrak{g}) \otimes L_1(\mathfrak{g})\text{–Mod}^{\mathcal{J}_\infty G} \simeq \check{G}\text{–Mod}.$$

Now to see that the simple objects are mapped to one another under this correspondence, it suffices to combine the branching rules (Theorem 11.1 of [ACL19]) with Arakawa–Frenkel duality (Theorem 2.2 of [AF19]).

Remark 9.4.3. *The semisimplicity of the category $\mathcal{W}_G^k\text{–Mod}^{\mathcal{J}_\infty G}$ is very surprising to us. For instance, it does not seem to be known (beyond the $\mathfrak{sl}_2$ case) that the full subcategory of $W^k(\mathfrak{g})\text{–Mod}$ generated by the $T_{\lambda,0}^k$ with $\lambda \in P_+$ (or the $T_{0,\check{\mu}}^k$ with $\check{\mu} \in \check{P}_+$) is semisimple.*

Remark 9.4.4. *Remarkably, Theorem 9.4.2 also shows that $\mathcal{W}_G^k[1]$ comes from a 1-parameter family of vertex algebras (compare with Remark 9.1.3). The key fact we use from [CN23] is that there exists exactly one structure of a simple vertex algebra on the $W^k(\mathfrak{g}) \otimes V^{k[1]}(\mathfrak{g})$-module*

$$\bigoplus_{\lambda \in X^*(T)_+} T_{\lambda,0}^k \otimes V_{-\omega_0\lambda}^{k[1]}$$

that is a conformal extension of $W^k(\mathfrak{g}) \otimes V^{k[1]}(\mathfrak{g})$ given by that of $V^{k[1]-1} \otimes L_1(\mathfrak{g})$. By the proof of Theorem 9.3.9, we see that for all $n \in \mathbb{Z}$, there exists a unique structure of a simple vertex algebra on the $W^k(\mathfrak{g}) \otimes V^{k[n]}(\mathfrak{g})$-module

$$\bigoplus_{\lambda \in X^*(T)_+} T_{\lambda,0}^k \otimes V_{-\omega_0\lambda}^{k[n]}$$

that is a conformal extension of $W^k(\mathfrak{g}) \otimes V^{k[n]}(\mathfrak{g})$ *given by that of* $\mathcal{W}^k_G[n]$. *In particular, for* $n = 0$*, it shows that the vertex algebra structure on* $\mathcal{W}^k_G$ *is determined by the fact that it is a simple conformal extension of* $W^k(\mathfrak{g}) \otimes V^{k^*}(\mathfrak{g})$ *together with its Peter–Weyl decomposition (Lemma 7.1.4). It seems reasonable to expect that the same kind of unicity statements hold beyond the simply laced and principal cases, as well as for the chiral universal centralizer (see §11.2).*

Chapter 10

Shifted chiral differential operators on an algebraic torus

For simplicity, we assume that $G = T = \mathbb{G}_m$ is a one-dimensional torus. Let $\mathfrak{h}$ be its Lie algebra and κ a nondegenerate level. Let $\gamma \in X_*(T)$ be a $\mathbb{Z}$-basis of $X_*(T)$. This fixes an identification between $\mathfrak{h} = X_*(T) \otimes_{\mathbb{Z}} \mathbb{C}$ and $\mathbb{C}$ and between $\mathfrak{h}^* = X^*(T) \otimes_{\mathbb{Z}} \mathbb{C}$ and $\mathbb{C}$ via $\lambda \in X^*(T) \mapsto \lambda(\gamma) \in \mathbb{C}$. We use these identifications from now on. The datum of the level κ is then nothing but that of the complex number $k = \kappa(\gamma, \gamma)$. The fact that κ is nondegenerate means that $k \neq 0$.

The remarkable family of vertex algebras we discuss in this chapter has already been introduced and studied in [Cre+22].

10.1 Existence

The Heisenberg vertex algebra $V^k(\mathfrak{h})$ is strongly generated by γ under the following relation:

$$[\gamma_\lambda \gamma] = k\lambda. \tag{10.1}$$

A conformal vector is given by

$$\omega = \frac{1}{2}\frac{\gamma}{\sqrt{k}}\frac{\gamma}{\sqrt{k}} = \frac{1}{2k}\gamma_{(-1)}\gamma_{(-1)}|0\rangle.$$

Let $\lambda \in \mathfrak{h}^* = \mathbb{C}$ and $\mathcal{F}^\kappa_\lambda$ the corresponding Fock space. More precisely, it is generated by the highest weight vector $|\lambda\rangle$ such that

$$\gamma_{(0)}|\lambda\rangle = \lambda|\lambda\rangle.$$

Recall from Lemma 4.3.6 that the lowest conformal degree of this Fock space equals $\frac{\lambda^2}{2k}$.

Let $l \in \mathbb{C} \setminus \{0\}$ and think of it as another level for $\mathfrak{h}$. Consider the $V^k(\mathfrak{h}) \otimes V^l(\mathfrak{h})$-module

$$\bigoplus_{\lambda \in \mathbb{Z}} \mathcal{F}^k_\lambda \otimes \mathcal{F}^l_{-\lambda}.$$

Then for each $\lambda \in \mathbb{Z}$, the conformal degree of $|\lambda, -\lambda\rangle = |\lambda\rangle \otimes |-\lambda\rangle$ equals

$$\frac{1}{2k}\lambda^2 + \frac{1}{2l}(-\lambda)^2 = \frac{\lambda^2}{2}\left(\frac{1}{k} + \frac{1}{l}\right).$$

In particular, if we let $n = \frac{1}{k} + \frac{1}{l}$, the previous quantity is a half-integer (resp. integer) for all $\lambda \in \mathbb{Z}$ if and only if $n \in \mathbb{Z}$ (resp. $n \in 2\mathbb{Z}$). Conversely, fix $n \in \mathbb{Z}$ and $k \in \mathbb{C} \setminus \{0\}$ and assume that

$$\frac{1}{k} \notin \mathbb{Z}.$$

Define $l = k[n]$ by the relation

$$\frac{1}{k} + \frac{1}{k[n]} = n. \tag{10.2}$$

A natural question in view of §9.1 is to decide whether or not there exists a unique structure of a simple vertex superalgebra on the $V^k(\mathfrak{h}) \otimes V^{k[n]}(\mathfrak{h})$-module

$$\bigoplus_{\lambda \in \mathbb{Z}} \mathcal{F}^k_\lambda \otimes \mathcal{F}^{-k}_\lambda$$

that is a conformal extension of $V^k(\mathfrak{h}) \otimes V^{k[n]}(\mathfrak{h})$.

Theorem 10.1.1. *Let $k \in \mathbb{C} \setminus \{0\}$ and assume that*

$$\frac{1}{k} \notin \mathbb{Z},$$

then for all $n \in \mathbb{Z}$ there exists a unique structure of a simple vertex superalgebra on the $V^k(\mathfrak{h}) \otimes V^{k[n]}(\mathfrak{h})$-module

$$\mathcal{D}^k_{\mathbb{G}_m}[n] = \bigoplus_{\lambda \in \mathbb{Z}} \mathcal{F}^k_\lambda \otimes \mathcal{F}^{k[n]}_{-\lambda}$$

that is a conformal extension of $V^k(\mathfrak{h}) \otimes V^{k[n]}(\mathfrak{h})$. More precisely we have

$$\begin{aligned} &\mathcal{D}^k_{\mathbb{G}_m}[0] = \mathcal{D}^k_T \\ &\mathcal{D}^k_{\mathbb{G}_m}[n] = V^{k+k[n]}(\mathbb{C}) \otimes V_{\sqrt{n}\mathbb{Z}} \ \textit{if } n \neq 0 \end{aligned}$$

where $V_{\mathbb{Z}\sqrt{n}}$ is the lattice vertex algebra associated with the lattice $\mathbb{Z}\sqrt{n} \subset \mathbb{C}$. They also satisfy for all $m, n \in \mathbb{Z}$ the equality

$$\mathcal{D}^k_{\mathbb{G}_m}[m] \circ \mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n] = \mathcal{D}^k_{\mathbb{G}_m}[m+n]. \tag{10.3}$$

Before turning to the proof, to which the rest of this paragraph is devoted, a few remarks are in order.

Remark 10.1.2. *The most well-studied class of lattice vertex algebra is the case where $n \in 2\mathbb{Z}_{\geq 0}$, which corresponds to an even positive definite lattice. When allowing nonnegative odd values of n, we obtain certain vertex superalgebras that we* *have already encountered in the boson-fermion correspondence (Theorem 6.2.2). In general, when $n \in \mathbb{Z}$, the lattice vertex superalgebras $V_{\mathbb{Z}\sqrt{n}}$ still exist and are purely even if and only if $n \in 2\mathbb{Z}$ (see for instance §6 of [Xu98]).When n is negative, $V_{\mathbb{Z}\sqrt{n}}$ is $\frac{1}{2}\mathbb{Z}$-graded and is not lower bounded with every conformal degree space of infinite dimension (compare with the case of a simple group discussed in §9.3 and Remark 10.2.1).*

As in the case of a simple group the existence of the family enforces its unicity (Theorem 9.3.9). In fact, the exact same proof applies here, thanks to the unicity of $\mathcal{D}^k_{\mathbb{G}_m}[0] = \mathcal{D}^k_{\mathbb{G}_m}$ (Theorem 4.4.3) and Remarks 9.2.3, 9.3.3 and 9.3.8. Equality (10.3) follows from the same arguments as in the simple case (Proposition 9.3.1).

For the existence, first notice that we are done in the case $n = 0$ by the chiral Peter–Weyl Theorem 4.4.1. Assume that $n \neq 0$, then set

$$\mathcal{D}^k_{\mathbb{G}_m}[n] = V^{k+k[n]}(\mathbb{C}) \otimes V_{\mathbb{Z}\sqrt{n}}.$$

Since $n \neq 0$, $k + k[n] \neq 0$ and so $V^{k+k[n]}(\mathbb{C})$ is a nondegenerate Heisenberg vertex algebra and hence is simple. Since $V_{\mathbb{Z}\sqrt{n}}$ is also simple, so is $\mathcal{D}^k_{\mathbb{G}_m}[n]$. By definition, the lattice vertex superalgebra $V_{\mathbb{Z}\sqrt{n}}$ contains a Heisenberg vertex algebra at level 1 with basis element $\gamma \in V^1(\mathbb{C})$ such that

$$[\gamma_\lambda \gamma] = \lambda. \tag{10.4}$$

Let $\Gamma \in V^{k+k[n]}(\mathbb{C})$ be a generator such that

$$[\Gamma_\lambda \Gamma] = (k + k[n])\lambda. \tag{10.5}$$

By construction of the lattice vertex superalgebra, $\mathcal{D}^k_{\mathbb{G}_m}[n]$ is a conformal extension of $V^{k+k[n]}(\mathbb{C}) \otimes V^1(\mathbb{C})$. As a $V^{k+k[n]}(\mathbb{C}) \otimes V^1(\mathbb{C})$-module, we have the equality

$$\mathcal{D}^k_{\mathbb{G}_m}[n] = \bigoplus_{\lambda \in \mathbb{Z}} \mathcal{F}_0^{k+k[n]} \otimes \mathcal{F}^1_{\sqrt{n}\lambda}. \tag{10.6}$$

Now, set

$$\begin{pmatrix} \gamma_L \\ \gamma_R \end{pmatrix} = \begin{pmatrix} \frac{1}{nk[n]} & \frac{1}{\sqrt{n}} \\ \frac{1}{nk} & -\frac{1}{\sqrt{n}} \end{pmatrix} \begin{pmatrix} \Gamma \\ \gamma \end{pmatrix}, \tag{10.7}$$

and notice that by equation (10.2) the determinant of this matrix equals $-\frac{1}{\sqrt{n}} \neq 0$. In this new basis, it follows from equations (10.4) and (10.5) that the tensor product $V^{k+k[n]}(\mathbb{C}) \otimes V^1(\mathbb{C})$ is isomorphic (in obvious notations) to

$$V^k(\mathfrak{h}) \otimes V^{k[n]}(\mathfrak{h})$$

and that the natural conformal vectors are identified under this isomorphism. Finally, as a $V^k(\mathfrak{h}) \otimes V^{k[n]}(\mathfrak{h})$-module, one checks using the base change (10.7) that equation (10.6) becomes

$$\mathcal{D}^k_{\mathbb{G}_m}[n] = \bigoplus_{\lambda \in \mathbb{Z}} \mathcal{F}^k_\lambda \otimes \mathcal{F}^{k[n]}_{-\lambda}.$$

Remark 10.1.3. *The existence of this family of vertex algebras should naturally lead to another proof of Theorem 8.1.6. In fact, one can follow the strategy developed for the proof of Theorem 9.4.1 and reduce the statement to the well-studied representation theory of a lattice vertex algebra. This approach is more systematic than the ad hoc one we explained in Chapter 8. One major upshot is that we should be able to say additional things about the tensor structure on the category* $\mathcal{D}^{\kappa}_{T}\text{–Mod}^{1\times\mathcal{J}_\infty G}$ *using the results of §6.3 of [Cre+22] where the behaviour of intertwining operators is studied under the convolution operation.*

10.2 Associated scheme

We want to compute the associated scheme of the vertex superalgebras $\mathcal{D}^{k}_{\mathbb{G}_m}[n]$ introduced in Theorem 10.1.1.

In general, if V is a vertex superalgebra, we introduce, following [Zhu96], the following linear subspace

$$C_2(V) = \{x_{(-2)}y, x, y \in V\}.$$

It was shown (in *loc.cit.* in the purely even case but the proof carries to the super case, see also [Li21] for a discussion in the super case) that

$$R_V = V/C_2(V)$$

is a Poisson superalgebra whose structure is defined for all $x, y \in R$ by

$$\begin{aligned} \bar{x}\cdot\bar{y} &= \overline{x_{(-1)}y}, \\ \{\bar{x},\bar{y}\} &= \overline{x_{(0)}y}. \end{aligned}$$

Now, if V is a purely even vertex algebra then R_V is a commutative algebra and

$$\widetilde{X}_V = \mathrm{Spec}(R_V)$$

is called the associated scheme of V ([Ara12]) and its reduced scheme

$$X_V = (\widetilde{X}_V)_{red}$$

is called the associated variety.

Remark 10.2.1. *In the case where V is a vertex superalgebra, it is natural to define $\widetilde{X}_V$ to be the affine superscheme associated with R_V. The notion of reduced superscheme associated with a superscheme requires some care. If we proceed naively and kill nilpotent elements, all the information contained in the odd part of our superalgebra is lost in the process in characteristic $\neq 2$ as every odd element of a commutative superalgebra squares to zero. We refer to Chapters 3 and 10 of [CCF11] for a discussion of super algebraic geometry. For our purpose, all schemes being affine, and the category of affine superschemes being equivalent to that of commutative superalgebras, we will work in the latter setting.*

Let $n \in \mathbb{Z}$ and consider the lattice vertex superalgebra

$$V_{\mathbb{Z}\sqrt{n}} = \bigoplus_{\lambda \in \mathbb{Z}\sqrt{n}} \mathcal{F}^1_\lambda.$$

Let $b \in \mathcal{F}^1_0 = V^1(\mathbb{C})$ be a strong generator normalized by

$$[b_\lambda b] = \lambda.$$

The following material about lattice vertex algebras is standard (see for instance Theorem 5.5 of [Kac98] or Theorem 6.1.3 of [Xu98] for this level of generality). For all $\lambda \in \mathbb{Z}\sqrt{n}$, it is conventional to set $e^\lambda = |\lambda\rangle$. We have the equalities

$$b_{(0)} e^\lambda = \lambda e^\lambda, \tag{10.8}$$

$$\partial e^\lambda = \lambda b_{(-1)} e^\lambda, \tag{10.9}$$

$$Y(e^\lambda, z) = S_\lambda z^{\lambda b_{(0)}} \exp\left(\sum_{k<0} \lambda b_{(k)} \frac{z^{-k}}{-k}\right) \exp\left(\sum_{k>0} \lambda b_{(k)} \frac{z^{-k}}{-k}\right), \tag{10.10}$$

where S_λ is the endomorphism of $V_{\sqrt{n}\mathbb{Z}}$ uniquely determined by the conditions

$$[S_\lambda, b_{(n)}] = 0, \qquad S_\lambda(e^\mu) = e^{\lambda+\mu}$$

for all $n < 0$ and $\mu \in \mathbb{Z}\sqrt{n}$. It is a direct consequence of equation (10.10) that for $\lambda, \mu \in \mathbb{Z}\sqrt{n}$ we have the equality

$$e^\lambda_{(-\lambda\mu-1)} e^\mu = e^{\lambda+\mu} \tag{10.11}$$

The following was noticed by Li (see Proposition 3.12 of [Li05]), we recall the proof for the sake of completeness.

Proposition 10.2.2. *If n is a negative integer then $R_{V_{\mathbb{Z}\sqrt{n}}}$ is the zero ring.*

Proof. By equation (10.11) we have:

$$e^{\sqrt{n}}{}_{(n-1)} e^{-\sqrt{n}} = |0\rangle.$$

Since n is negative, this implies that $|0\rangle = 0$ in $R_{V_{\mathbb{Z}\sqrt{n}}}$. Therefore

$$1 = 0$$

in $R_{V_{\mathbb{Z}\sqrt{n}}}$ and the result follows. □

Remark 10.2.3. *The fact that $V = C_2(V)$ for a vertex algebra i.e., that $R_V = 0$ has the following striking consequence: there may not exist a lower truncated $\mathbb{Z}$-grading on V (Proposition 3.11 of [Li05]). Therefore this pathological behaviour of negatively shifted chiral differential operators (that one observes also in the simple case) is, at least for an algebraic torus, a genuine feature and does not simply mean we have made a wrong choice of conformal vector.*

Proposition 10.2.4 (See also Proposition 4.3 of [Li21]). *Assume that n is a nonnegative integer. Then $R_{V_{\mathbb{Z}\sqrt{n}}}$ is isomorphic to the polynomial superalgebra in 3 variables $e^{-\sqrt{n}}$, b and $e^{\sqrt{n}}$ (if n is odd then $e^{\pm\sqrt{n}}$ are odd variables) subject to the following relations*

$$b^{n+1} = 0, \tag{10.12}$$

$$(e^{-\sqrt{n}})^2 = (e^{\sqrt{n}})^2 = 0, \tag{10.13}$$

$$be^{-\sqrt{n}} = be^{\sqrt{n}} = 0, \tag{10.14}$$

$$e^{-\sqrt{n}}e^{\sqrt{n}} = \frac{(-\sqrt{n}b)^n}{n!}. \tag{10.15}$$

Its Poisson structure is uniquely determined by the following relations

$$\{b, e^{\sqrt{n}}\} = \sqrt{n}e^{\sqrt{n}}, \tag{10.16}$$

$$\{b, e^{-\sqrt{n}}\} = -\sqrt{n}e^{-\sqrt{n}}, \tag{10.17}$$

$$\{e^{-\sqrt{n}}, e^{\sqrt{n}}\} = \frac{(-\sqrt{n}b)^{n-1}}{(n-1)!}. \tag{10.18}$$

In particular, $R_{V_{\mathbb{Z}\sqrt{n}}}$ is a vector space of dimension $n+3$ with basis

$$\{e^{-\sqrt{n}}, 1, b, \ldots, b^n, e^{\sqrt{n}}\}.$$

Proof. First, since n is positive, for all $\lambda, \mu \in \mathbb{Z}\sqrt{n}$ of the same sign, we have

$$\lambda\mu \geqslant 0.$$

Therefore, it follows from equation (10.11) that $V_{\mathbb{Z}\sqrt{n}}$ is strongly generated by $e^{-\sqrt{n}}$, b and $e^{\sqrt{n}}$. It is a direct consequence of equation (10.11) that for all $\alpha > 1$, we have

$$e^{\pm\alpha\sqrt{n}} = 0$$

in $R_{V_{\mathbb{Z}\sqrt{n}}}$. In particular, only the $b_{(0)}$-weight spaces of weight $\{-\sqrt{n}, 0, \sqrt{n}\}$ survive in the quotient, which shows equality (10.13). Equation (10.9) readily implies equation (10.14). Equation (10.15) follows from a straightforward computation using formula (10.10). It is clear that equations (10.15) and (10.14) imply equation (10.12). Equations (10.17) and (10.16) follow from equation (10.8). Finally, equation (10.18) follows from formula (10.10).

We are left to see that this is all the relations. Let R be the superalgebra defined in the statement. It is of dimension $n+3$. We have just shown that we have a surjective map of superalgebras:

$$R \longrightarrow R_{V_{\mathbb{Z}\sqrt{n}}}.$$

So if we can show that $R_{V_{\mathbb{Z}\sqrt{n}}}$ is of dimension at least $n+3$, we are done. As a vector space, $R_{V_{\mathbb{Z}\sqrt{n}}}$ is the quotient of $V_{\mathbb{Z}\sqrt{n}}$ by $C_2(V_{\mathbb{Z}\sqrt{n}})$. According to Corollary 3.6

of [Li05] (whose proof carries over without change to the super case), $C_2(V_{\mathbb{Z}\sqrt{n}})$ is equal to the $\mathbb{C}$-span of elements of the form

$$e^{-\sqrt{n}}_{(-k_1-1)}\cdots e^{-\sqrt{n}}_{(-k_\alpha-1)}e^{\sqrt{n}}_{(-l_1-1)}\cdots e^{\sqrt{n}}_{(-l_\beta-1)}b_{(-m_1-1)}\cdots_{(-m_\gamma-1)}|0\rangle$$

where $k_1,\dots,k_\alpha,l_1,\dots,l_\beta,m_1,\dots,m_\gamma \geq 0$ such that at least one is nonzero. Elements of the previous shape have conformal degree

$$\sum_{i=1}^{\alpha}\left(\frac{n}{2}+k_i\right)+\sum_{i=1}^{\beta}\left(\frac{n}{2}+l_i\right)+\sum_{i=1}^{\gamma}(1+m_i)=(\alpha+\beta)\frac{n}{2}+\gamma+\sum_{i=1}^{\alpha}k_i+\sum_{i=1}^{\beta}l_i+\sum_{i=1}^{\gamma}m_i. \quad (10.19)$$

We claim that the span of the linearly independent family $\{e^{-\sqrt{n}},e^{\sqrt{n}},b^0,b^1,\dots,b^n\}$ (where $b^k=(b_{(-1)})^k|0\rangle$ for $k\geqslant 0$) of cardinality $n+3$ intersects $C_2(V)$ trivially. For conformal degree reasons, we are left to see that the same is true for the three families $\{e^{-\sqrt{n}}\}$, $\{e^{\sqrt{n}}\}$ and $\{b^0,\dots,b^n\}$. For the first two, this is easy to see. For the last family, notice that the only elements that may appear in a linear combination must satisfy $\alpha=\beta$ because of $b_{(0)}$-weights. But then if $\alpha=\beta>0$, by equation (10.19), we see that the conformal degree must be greater than $n+1$. This implies that $\alpha=\beta=0$ and we are done as we are reduced to the Heisenberg case. □

Proposition 10.2.5. *In the notations of Theorem 10.1.1, we have the following isomorphisms of Poisson superalgebras*

$$R_{\mathcal{D}^k_{\mathbb{G}_m}[n]}=0 \textit{ if } n<0,$$
$$R_{\mathcal{D}^k_{\mathbb{G}_m}[0]}=\mathcal{O}_{T^*\mathbb{G}_m},$$
$$R_{\mathcal{D}^k_{\mathbb{G}_m}[n]}=\mathbb{C}[\Gamma]\otimes R_{V_{\sqrt{n}}} \textit{ if } n>0.$$

Remark 10.2.6. *Note that the associated scheme does not depend on the level.*

10.3 The Moore–Tachikawa operation

Recall from equation (10.3) that for all $m,n\in\mathbb{Z}$ we have an isomorphism of vertex algebras

$$\mathcal{D}^k_{\mathbb{G}_m}[m]\circ\mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n]=\mathcal{D}^k_{\mathbb{G}_m}[m+n]$$

where $\circ$ denotes the convolution operation introduced in Section 9.2 (recall Remark 9.2.3). This convolution operation for vertex algebra has a classical analog for Poisson algebras (see [MT12]).

10.3.1 BRST cohomology for an abelian Lie algebra

Here, we spell out explicitly what is the BRST cohomology for a (1-dimensional) abelian Lie algebra in the super case following §7.1 of [AM25].

Let R be a commutative Poisson superalgebra equipped with an action of $\mathbb{G}_m$, that is, a $\mathbb{Z}$-graded commutative Poisson superalgebra. Assume that the action is Hamiltonian *i.e.*, there exists an even element $H \in R$ such that the grading is given by $\{H, \cdot\}$.

In this setting, the moment map is given by the unique $\mathbb{C}$-algebra map from $S(\mathfrak{h}) = \mathbb{C}[\gamma]$ to R that maps $\gamma \in \mathfrak{h}$ to $H \in R$ and the BRST complex is

$$\ldots \longrightarrow 0 \longrightarrow R \otimes \mathbb{C}\gamma \longrightarrow R \otimes \mathbb{C}1 \oplus R \otimes \mathbb{C}\gamma\gamma^* \longrightarrow R \otimes \mathbb{C}\gamma^* \longrightarrow 0 \longrightarrow \ldots$$

with differential

$$\mathrm{d} = \{H \otimes \gamma^*, \cdot\}.$$

Explicitly, if $r \in R$ is a parity homogeneous element and we set $|r| = 0$ (resp. 1) if r is even (resp. odd), we have

$$\begin{aligned}
\mathrm{d}(r \otimes \gamma) &= (-1)^{|r|}(-\{H, r\} \otimes \gamma\gamma^* + Hr \otimes 1),\\
\mathrm{d}(r \otimes 1) &= (-1)^{|r|}(\{H, r\} \otimes \gamma^*),\\
\mathrm{d}(r \otimes \gamma\gamma^*) &= (-1)^{|r|}(-Hr \otimes \gamma^*).
\end{aligned}$$

Write $R = \bigoplus_{n\in\mathbb{Z}} R_n$, then for all $n \in \mathbb{Z}$ and a parity homogeneous element $r_n \in R_n$, the previous equalities become:

$$\mathrm{d}(r_n \otimes \gamma) = (-1)^{|r|}(-nr_n \otimes \gamma\gamma^* - Hr_n \otimes 1), \tag{10.20}$$
$$\mathrm{d}(r_n \otimes 1) = (-1)^{|r|}(nr_n \otimes \gamma^*), \tag{10.21}$$
$$\mathrm{d}(r_n \otimes \gamma\gamma^*) = (-1)^{|r|}(-Hr_n \otimes \gamma^*). \tag{10.22}$$

In particular, since $H \in R_0$, we see that d preserves the grading. This means that the BRST complex is the direct sum of the subcomplexes

$$\ldots \longrightarrow 0 \longrightarrow R_n \otimes \mathbb{C}\gamma \longrightarrow R_n \otimes \mathbb{C}1 \oplus R_n \otimes \mathbb{C}\gamma\gamma^* \longrightarrow R_n \otimes \mathbb{C}\gamma^* \longrightarrow 0 \longrightarrow \ldots \tag{10.23}$$

Theorem 10.3.1. *Assume that H is not a zero divisor in R. Then we have*

$$\begin{aligned}
&H^{-1}_{BRST}(\mathfrak{h}, R) = 0,\\
&H^{0}_{BRST}(\mathfrak{h}, R) = H^{1}_{BRST}(\mathfrak{h}, R) = R_0/HR_0.
\end{aligned}$$

Proof. If $n \neq 0$, it is easy to see using (10.20), (10.21) and (10.22) that the subcomplex (10.23) is exact. Now consider the subcomplex

$$\ldots \longrightarrow 0 \longrightarrow R_0 \otimes \gamma \longrightarrow R_0 \otimes 1 \oplus R_0 \otimes \gamma\gamma^* \longrightarrow R_0 \otimes \gamma^* \longrightarrow 0 \longrightarrow \ldots$$

For an element $r+\bar{r} \in R_0$ (where r is even and $\bar{r}$ is odd) we have by equations (10.20), (10.21) and (10.22):

$$\begin{aligned}
\mathrm{d}((r+\bar{r})\otimes\gamma) &= H(-r+\bar{r})\otimes 1,\\
\mathrm{d}((r+\bar{r})\otimes 1) &= 0,\\
\mathrm{d}((r+\bar{r})\otimes\gamma\gamma^*) &= H(-r+\bar{r})\otimes\gamma^*.
\end{aligned}$$

Because H is not a zero divisor, the claim follows.

□

10.3.2 Poisson convolution operation

Let R, R' be two commutative Poisson superalgebras. Assume that R (resp. R') is acted on by $\mathbb{G}_m$ and the action is Hamiltonian with respect to a certain even element $H \in R$ (resp. $H' \in R'$).

The commutative Poisson superalgebra $R\otimes R'$ is acted on diagonally by $\mathbb{G}_m$ and the action is again Hamiltonian with respect to $H\otimes 1 + 1\otimes H'$. Now define

$$R\circ R' = H^0_{BRST}(\mathfrak{h}, R\otimes R')$$

and call $R\circ R'$ the convolution of R and R'. It is again a commutative Poisson superalgebra.

Let $m, n \geqslant 0$, we want to compare the Poisson superalgebras

$$R_{\mathcal{D}^k_{\mathbb{G}_m}[m]} \circ R_{\mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n]}$$

and

$$R_{\mathcal{D}^k_{\mathbb{G}_m}[m+n]}.$$

First we need to fix our choices of Hamiltonians. There are two $\mathbb{Z}$-gradings on

$$\mathcal{D}^k_G[m] = \bigoplus_{\lambda\in\mathbb{Z}} \mathcal{F}^k_\lambda \otimes \mathcal{F}^{k[m]}_{-\lambda}.$$

They are both Hamiltonian and are induced by the zero modes of γ_L and γ_R. The $\mathbb{G}_m$-action we consider is the right one, and so the Hamiltonian we choose is γ_R. The same story holds for $\mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n]$, with respect to the left $\mathbb{G}_m$-action. We choose, in obvious notations, the Hamiltonian δ_L.

We need to keep track precisely of the base change of Section §10.1. Recall that we set

$$\begin{aligned}
\begin{pmatrix}\Gamma\\ \gamma\end{pmatrix} &= \begin{pmatrix}1 & 1\\ \frac{1}{\sqrt{mk}} & -\frac{1}{\sqrt{mk[m]}}\end{pmatrix}\begin{pmatrix}\gamma_L\\ \gamma_R\end{pmatrix},\\
\begin{pmatrix}\Delta\\ \delta\end{pmatrix} &= \begin{pmatrix}1 & 1\\ -\frac{1}{\sqrt{nk[m]}} & -\frac{1}{\sqrt{nk[m+n]}}\end{pmatrix}\begin{pmatrix}\delta_L\\ \delta_R\end{pmatrix}.
\end{aligned}$$

Or, equivalently:

$$\begin{pmatrix}\gamma_L\\ \gamma_R\end{pmatrix} = \begin{pmatrix}\frac{1}{mk[m]} & \frac{1}{\sqrt{m}}\\ \frac{1}{mk} & -\frac{1}{\sqrt{m}}\end{pmatrix}\begin{pmatrix}\Gamma\\ \gamma\end{pmatrix}, \tag{10.24}$$

$$\begin{pmatrix}\delta_L\\ \delta_R\end{pmatrix} = \begin{pmatrix}\frac{1}{nk[m+n]} & \frac{1}{\sqrt{n}}\\ -\frac{1}{nk[m]} & -\frac{1}{\sqrt{n}}\end{pmatrix}\begin{pmatrix}\Delta\\ \delta\end{pmatrix}. \tag{10.25}$$

If we want to apply Theorem 10.3.1, the first step is to consider the tensor product

$$R_{\mathcal{D}^k_{\mathbb{G}_m}[m]} \otimes R_{\mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n]}$$

and check that the Hamiltonian is not a zero-divisor. By equations (10.24) and (10.25), the Hamiltonian equals

$$\gamma_R \otimes 1 + 1 \otimes \delta_L = \frac{1}{mk}\Gamma + \frac{1}{nk[m+n]}\Delta - \frac{1}{\sqrt{m}}\gamma + \frac{1}{\sqrt{n}}\delta \tag{10.26}$$

which is not a zero-divisor since Γ and Δ are free even variables. One checks using the relations (10.16), (10.17) and (10.18) that $(R_{\mathcal{D}^k_{\mathbb{G}_m}[m]} \otimes R_{\mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n]})_0$ is the subalgebra of $(R_{\mathcal{D}^k_{\mathbb{G}_m}[m]} \otimes R_{\mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n]})$ generated by $\Gamma, \Delta, e^{-\sqrt{m}}e^{-\sqrt{n}}, \gamma, \delta$ and $e^{\sqrt{m}}e^{\sqrt{n}}$. Now we set the Hamiltonian (10.26) to be zero, that is, add equation

$$\frac{1}{mk}\Gamma + \frac{1}{nk[m+n]}\Delta - \frac{1}{\sqrt{m}}\gamma + \frac{1}{\sqrt{n}}\delta = 0$$

to the relations of our subalgebra. We rewrite $\mathbb{C}[\Gamma, \Delta]$ as $\mathbb{C}[\Gamma+\Delta]\otimes\mathbb{C}[\frac{1}{mk}\Gamma + \frac{1}{nk[m+n]}\Delta]$, which is licit as the determinant of the matrix

$$\begin{pmatrix}1 & 1\\ \frac{1}{mk} & \frac{1}{nk[m+n]}\end{pmatrix}$$

is nonzero since $\frac{1}{k} \notin \mathbb{Z}$ by assumption.

We therefore see that there exists a surjection from the superalgebra on five variables $\Gamma + \Delta, e^{-\sqrt{m}}e^{-\sqrt{n}}, \gamma, \delta, e^{\sqrt{m}}e^{\sqrt{n}}$ where $\Gamma + \Delta$, γ and δ are even and $e^{\pm\sqrt{m}}e^{\pm\sqrt{n}}$ have the same parity as $m + n$, subject to the relations

$$(e^{-\sqrt{m}}e^{-\sqrt{n}})^2 = (e^{\sqrt{m}}e^{\sqrt{n}})^2 = 0, \tag{10.27}$$

$$\gamma^{m+1} = \delta^{n+1} = 0, \tag{10.28}$$

$$\gamma(e^{\pm\sqrt{m}}e^{\pm\sqrt{n}}) = \delta(e^{\pm\sqrt{m}}e^{\pm\sqrt{n}}) = 0, \tag{10.29}$$

$$(e^{-\sqrt{m}}e^{-\sqrt{n}})(e^{\sqrt{m}}e^{\sqrt{n}}) = (-1)^{mn}\frac{(-\sqrt{m}\gamma)^m}{m!}\frac{(-\sqrt{n}\delta)^n}{n!}. \tag{10.30}$$

to $R_{\mathcal{D}^k_{\mathbb{G}_m}[m]} \circ R_{\mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n]}$. This is clearly a map of graded vector spaces, where the grading on both sides is given by the powers of the free variable $\Gamma+\Delta$, each graded piece being of dimension $(m + 1)(n + 1) + 2$. Therefore, we are reduced to a simple dimension counting argument in the spirit of the proof of Proposition 10.2.4 that concludes this is an isomorphism.

Its Poisson structure is determined by the fact that $\Gamma + \Delta$ is in the Poisson center and the relations

$$\{\gamma, \delta\} = 0 \tag{10.31}$$

$$\{\gamma, e^{\pm\sqrt{m}}e^{\pm\sqrt{n}}\} = \pm\sqrt{m}e^{\pm\sqrt{m}}e^{\pm\sqrt{n}} \tag{10.32}$$

$$\{\delta, e^{\pm\sqrt{m}}e^{\pm\sqrt{n}}\} = \pm\sqrt{n}e^{\pm\sqrt{m}}e^{\pm\sqrt{n}} \tag{10.33}$$

$$\{e^{-\sqrt{m}}e^{-\sqrt{n}}, e^{\sqrt{m}}e^{\sqrt{n}}\} = (-1)^{mn}\left(\frac{(-\sqrt{m}\gamma)^{m-1}}{(m-1)!}\frac{(-\sqrt{n}\delta)^n}{n!} + \frac{(-\sqrt{m}\gamma)^m}{m!}\frac{(-\sqrt{n}\delta)^{n-1}}{(n-1)!}\right) \tag{10.34}$$

Proposition 10.3.2. *Let $m, n \geqslant 1$, there exists an injective morphism of Poisson superalgebras*

$$\iota : R_{\mathcal{D}^k_{\mathbb{G}_m}[m+n]} \longrightarrow R_{\mathcal{D}^k_{\mathbb{G}_m}[m]} \circ R_{\mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n]}$$

that is not surjective.

Proof. Let us think of $R_{\mathcal{D}^k_{\mathbb{G}_m}[m+n]}$ as being generated as a superalgebra by Λ, λ and $e^{\pm\sqrt{m+n}}$, in the notations of §10.2. Define ι to be the unique superalgebra morphism from $R_{\mathcal{D}^k_{\mathbb{G}_m}[m+n]}$ to $R_{\mathcal{D}^k_{\mathbb{G}_m}[m]} \circ R_{\mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n]}$ determined by

$$\begin{aligned}
\iota(\Lambda) &= \Gamma + \Delta, \\
\iota(e^{-\sqrt{m+n}}) &= (-1)^{mn}e^{-\sqrt{m}}e^{-\sqrt{n}}, \\
\iota(e^{\sqrt{m+n}}) &= e^{-\sqrt{m}}e^{-\sqrt{n}}, \\
\iota(\lambda) &= \frac{1}{\sqrt{m+n}}(\sqrt{m}\gamma + \sqrt{n}\delta).
\end{aligned}$$

A direct computation using equations (10.12)-(10.18) and equations (10.27)-(10.30) shows that this morphism is well-defined and preserves Poisson structures. This map is a map of graded vector spaces, the grading on the left-hand side (resp. right-hand side) being given by powers of Λ (resp. $\Gamma + \Delta$). It is easy to see this is an injection of each graded piece by a dimension counting argument, this proves the injectivity of ι. It is not surjective by the same dimension counting argument or since for instance, γ is clearly not in the image of ι. □

Remark 10.3.3. *For even values of m and n, there is no Poisson isomorphism between $R_{\mathcal{D}^k_{\mathbb{G}_m}[m+n]}$ and $R_{\mathcal{D}^k_{\mathbb{G}_m}[m]} \circ R_{\mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n]}$. In fact, notice that we are in the setting of the Zariski cancellation problem for Poisson algebras (see e.g., [GW20]). It follows from Corollary 5.4 of loc.cit. that there cannot exist any Poisson isomorphism between $R_{\mathcal{D}^k_{\mathbb{G}_m}[m+n]}$ and $R_{\mathcal{D}^k_{\mathbb{G}_m}[m]} \circ R_{\mathcal{D}^{-k[m]}_{\mathbb{G}_m}[n]}$. There is in this case a more elementary argument: the dimensions of the Zariski tangent spaces at any closed point do not match.*

Certain noncommutative analogs of the Zariski cancellation problem have been studied (see e.g. [BZ17]). A natural chiral quantization of the affine plane is the Heisenberg vertex algebra. It therefore seems natural to ask if a chiral analog of the Zariski cancellation problem holds. That is, if $\mathcal{H}$ is a Heisenberg vertex algebra, V, W are vertex algebras and $n \geqslant 1$ is an integer, does the existence of an isomorphism of vertex algebras

$$V \otimes \mathcal{H}^{\otimes n} \simeq W \otimes \mathcal{H}^{\otimes n}$$

enforces V and W to be isomorphic? Can one find conditions on V so that it holds, maybe for small values of n?

Chapter 11

Further perspectives

11.1 Beyond the principal case

Assume that G is a simple algebraic group and that $k \in \mathbb{C} \setminus \mathbb{Q}$. In this work we considered the quantum Hamiltonian reduction functor $H^*_{DS} = H^*_{DS,f_{prin}}$ associated with a principal nilpotent element f_{prin}. From there we defined the principal affine $\mathcal{W}$-algebra $W^k(\mathfrak{g}) = W^k(\mathfrak{g}, f_{prin})$ and the principal equivariant affine $\mathcal{W}$-algebra $\mathcal{W}^k_{G,f_{prin}}$ on G.

More generally, given any nilpotent element f there exists a quantum Hamiltonian reduction functor $H^*_{DS,f}$ associated with f by the work of [KRW03]. One may define in a similar fashion the affine $\mathcal{W}$-algebra associated with $(\mathfrak{g}, f)$ at level k to be

$$W^k(\mathfrak{g}, f) := H^*_{DS,f}(V^k(\mathfrak{g}))$$

and the equivariant affine $\mathcal{W}$-algebra associated with (G, f) at level k by

$$\mathcal{W}^k_{G,f} := H^*_{DS,f}(\mathcal{D}^k_G).$$

This vertex algebra comes equipped with two morphisms

$$W^k(\mathfrak{g}, f) \longrightarrow \mathcal{W}^k_{G,f} \longleftarrow V^{k^*}(\mathfrak{g})$$

so that we can define the category $\mathcal{W}^k_{G,f}\text{-Mod}^{\mathcal{J}_\infty G}$. It is natural to ask for a description of this category. Unlike in the principal case, there is to the best of the author's knowledge no clear quantum geometric Langlands interpretation of this category. An affine version of the Skryabin equivalence was studied in the principal case in [Ras21] and this was a motivation to state Conjecture 7.2.4. Since the Skryabin equivalence holds for any nilpotent element f ([Skr02]), it seems natural to expect an affine analogue involving $W^k(\mathfrak{g}, f)$.

Our construction still provides certain (possibly) interesting modules. Given a cocharacter $\gamma \in X_*(T)$ we can once again consider the family of $\mathcal{W}^k_{G,f}$-modules

$H^*_{DS,f}(\gamma \cdot \mathcal{D}^k_G)$. If the work of Arakawa and Frenkel [AF19] can be generalized beyond the principal case, one may carry over what we did in Chapter 7. For instance if one can show that for $\lambda \in P_+$ the $W^k(\mathfrak{g}, f)$-module

$$T^k_{\lambda,\gamma} := H^*_{DS,f}(\gamma \cdot V^k_\lambda)$$

is either zero or simple, then the proof of Theorem 7.2.5 applies to show that the modules $H^*_{DS,f}(\gamma \cdot \mathcal{D}^k_G)$ are either zero or simple. This would then allow for the formulation of a conjecture in the spirit of Conjecture 7.2.4. This might in turn be related to the work of Creutzig and Linshaw on the trialities of $\mathcal{W}$-algebra ([CL22a], [CL22b]). We plan to explore these directions in the future in collaboration with Thomas Creutzig.

The question of generalizing Arakawa–Frenkel duality beyond the principal case seems particularly interesting to us.

11.2 The chiral universal centralizer

We keep the previously introduced notations and consider a pair f, f' of nilpotent elements of $\mathfrak{g}$. Recall that we have morphisms of vertex algebras

$$V^k(\mathfrak{g}) \longrightarrow \mathcal{D}^k_G \longleftarrow V^{k^*}(\mathfrak{g}),$$
$$W^k(\mathfrak{g}, f) \longrightarrow \mathcal{W}^k_{G,f} \longleftarrow V^{k^*}(\mathfrak{g}).$$

We may further apply, following §7 of [Ara18], the functor $H^*_{DS,f'}$ with respect to the right embedding to obtain the chiral universal centralizer associated with (G, f, f') at level k

$$\mathcal{I}^k_{G,f,f'} := H^*_{DS,f'}(\mathcal{W}^k_{G,f}).$$

This vertex algebra comes equipped with two morphisms

$$W^k(\mathfrak{g}, f) \longrightarrow \mathcal{I}^k_{G,f,f'} \longleftarrow W^{k^*}(\mathfrak{g}, f').$$

Let us comment that our construction builds (possibly) interesting modules for $\mathcal{I}^k_{G,f,f'}$. Recall from Proposition 5.3.1 that

$$\mathcal{SF}(\mathcal{D}^\kappa_G) \simeq X_*(T) \times \check{P}.$$

We mostly exploited the subgroup $X_*(T) \times \{0\} \subset \mathcal{SF}(\mathcal{D}^k_G)$, because they are the only parameters that produce objects in $\mathcal{W}^k_{G,f}\text{–Mod}^{\mathcal{J}_\infty G}$. Let $(\gamma, x) \in X_*(T) \times \check{P}$, we can consider the $\mathcal{I}_{G,f,f'}$-module

$$H^*_{DS,f'}(H^*_{DS,f}((\gamma, x) \cdot \mathcal{D}^\kappa_G)).$$

For instance if $f = f' = f_{prin}$ are principal nilpotent elements, and $\mathcal{I}^k_G := \mathcal{I}_{G,f_{prin},f_{prin}}$ then the category $\mathcal{I}^k_G\text{–Mod}$ should be related to κ-twisted $\mathcal{D}$-modules on the loop group satisfying Whittaker equivariance conditions with respect to both the left and right actions of $\mathcal{L}G$ on itself. We are unaware of any published references mentioning conjectures on this category or its relevance to the quantum geometric Langlands program.

Note that when the level κ is critical, *i.e.*, $\kappa = \kappa^*$, and f, f' are principal or zero, then $\mathcal{I}_{G,f,f'}$ has been introduced and studied by Arakawa from a vertex algebraic point of view in his work on the class $\mathcal{S}$ ([Ara18]). This level is very special as it allows for the definition of an inverse quantum Hamiltonian procedure using the Feigin–Frenkel center (see §9 of *loc.cit.*).

For irrational values of k, in view of the chiral Peter–Weyl decomposition (Theorem 4.4.1), we have the equality as $V^k(\mathfrak{g}) \otimes V^{k^*}(\mathfrak{g})$-modules:

$$(\gamma, x) \cdot \mathcal{D}_G^\kappa = \bigoplus_{\lambda \in X^*(T)_+} ((\gamma + x) \cdot V_\lambda^k) \otimes (\omega_0^T(x) \cdot V_{-\omega_0\lambda}^{k^*}).$$

So that by taking the left and right quantum Hamiltonian reduction we obtain by Proposition 7.1.2 the equalities as $W^k(\mathfrak{g}) \otimes W^{k^*}(\mathfrak{g})$-modules:

$$\begin{aligned} &H_{DS}^*(H_{DS}^*((\gamma, x) \cdot \mathcal{D}_G^k)) = \bigoplus_{\lambda \in X^*(T)_+} T_{\lambda, \gamma+x}^k \otimes T_{-\omega_0\lambda, \omega_0^T(x)}^{k^*} && \text{if } \gamma + x, \omega_0^T(x) \in \check{P}_+, \\ &H_{DS}^*(H_{DS}^*((\gamma, x) \cdot \mathcal{D}_G^k)) = 0 && \text{else.} \end{aligned}$$

We do not know how to prove their simplicity but conjecture they are.

Remark 11.2.1. *The free field realization of the chiral universal centralizer given in [Fur25] might be useful in the study of its representation theory. We note that for $G = \mathrm{SL}_2$ and generic κ the chiral universal centralizer $\mathcal{I}_G^k$ was introduced by Frenkel and Styrkas in [FS06] and Frenkel and Zhu in [FZ12]. McRae and Yang have studied the representation theory of $\mathcal{I}_{SL_2}^k$ for generic values of k (see Remark 8.4 of [MY23]) and for $k = -1$ (see Theorem 8.2 of loc.cit.). In particular, in their result for $k = -1$, the Langlands dual group PSL_2 of SL_2 appears.*

Of course, a similar construction holds for arbitrary pairs of nilpotent elements (f, f') and the study of the representation theory of $\mathcal{I}_{G,f,f'}^k$ appear to us as a natural and widely open question.

11.3 Studying the vertex algebras of shifted chiral differential operators

The $\mathbb{Z}$-family of vertex algebras $\mathcal{D}_G^k[n]$ introduced in §9.1 and Chapter 10 played a central role in our work. Their existence is still conjectural in general and when they have been shown to exist, they are rather mysterious vertex algebras. For instance, their associated schemes do not seem to be known. In Chapter 10 we computed them in the case of a torus and found for instance that they are empty for negative values of n (Proposition 10.2.5). In general, we argued by an unicity argument (Theorem 9.3.9) that these vertex algebras behave well with respect to the convolution operation, *i.e.*, we have

$$\mathcal{D}_G^k[m] \circ \mathcal{D}_G^{-k[m]}[n] = \mathcal{D}_G^k[m+n] \tag{11.1}$$

for all $m, n \in \mathbb{Z}$. But at the level of the associated scheme, in the case $G = \mathbb{G}_m$, we have shown in §10.3.2 that equation (11.1) almost never holds. When n or m is zero, the result holds as the cotangent bundle of G behaves as a unit for convolution. If $mn < 0$ and $n + m > 0$, this does not hold for a somehow obvious since one of the two is the zero ring, and we expect the result to be nonzero. But even beyond this obstruction, that is when $n, m > 0$, we have shown this is wrong (Remark 10.3.3).

It would be interesting to have more examples beyond the multiplicative group case, for instance in the case of $G = \mathrm{PSL}_2$. More generally this raises the general question of understanding the interaction between the convolution operation and the operation of passing to the associated scheme.

11.4 Beyond irrational levels

Another natural question is that of rational levels. One first step is to understand what happens to the Peter–Weyl decomposition of $\mathcal{D}_G^\kappa$, and this has been studied for simple groups by Zhu ([Zhu11]). In the generic case, the theory of quantum groups was somehow hidden all along, it seems that for the rational case it should play a deep role (recall also Corollary 8.1.8 in which the combinatorics that appear seem to be related to the duality Lusztig introduces in §2.2.4 of [Lus93]). Some of the technology we used in our work, for instance the spectral flow group and the twisted quantum Hamiltonian reduction, make sense for all levels. It is unclear to us what the shifted algebras of chiral differential operators should be in this case and what role they should play. Understanding the interplay between twisted principal Hamiltonian reduction and the Feigin–Frenkel duality beyond irrational levels also seem interesting but subtle (see §4.4 of [AF19]).

The most general question our work raises is the following: given a reductive group G, a level κ and two nilpotent elements $f, f' \in \mathfrak{g}$, how does one define and study a suitable subcategory of $\mathcal{I}^\kappa_{G,f,f'}$−Mod ?

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